

BIOBUZZ lessons

Release

Catholic Central Spires Robotics

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How to use it

- **Start with the task you need, not with page one.** Every task lists what it needs finished first. If you missed a session, find the task you are up to and read its *Before you start*.
- **Do the step, then read the check.** The check tells you whether to go on. If what you see is not what the check describes, the *if it didn't* line under it is the first thing to try.

This guide walks you through making the robot do things, one task at a time. Each task page says what you will have when it is done, what has to be true before you start, and the steps to get there. Every step tells you what you will see when it worked and what to try when it did not.

How to use it

- **Start with the task you need, not with page one.** If you missed a session, ask a mentor which task you are up to.
- **Do the step, then read the check.** If what you see is not what the check says, ask a mentor before you go on.
- **The answers are in the back.** They are the real code from the solutions branch. Use them when you have tried and want a hint, not instead of trying.

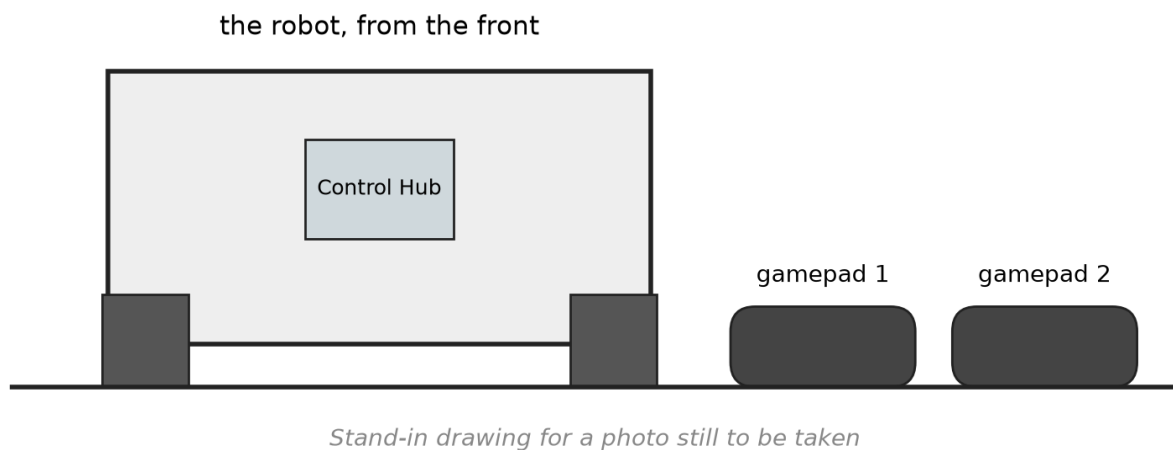


Figure 1. The robot from the front, on the floor, with both gamepads beside it, so a student can match what is in front of them to what the guide calls each part.

The tasks

3.1 L020: read the sticks

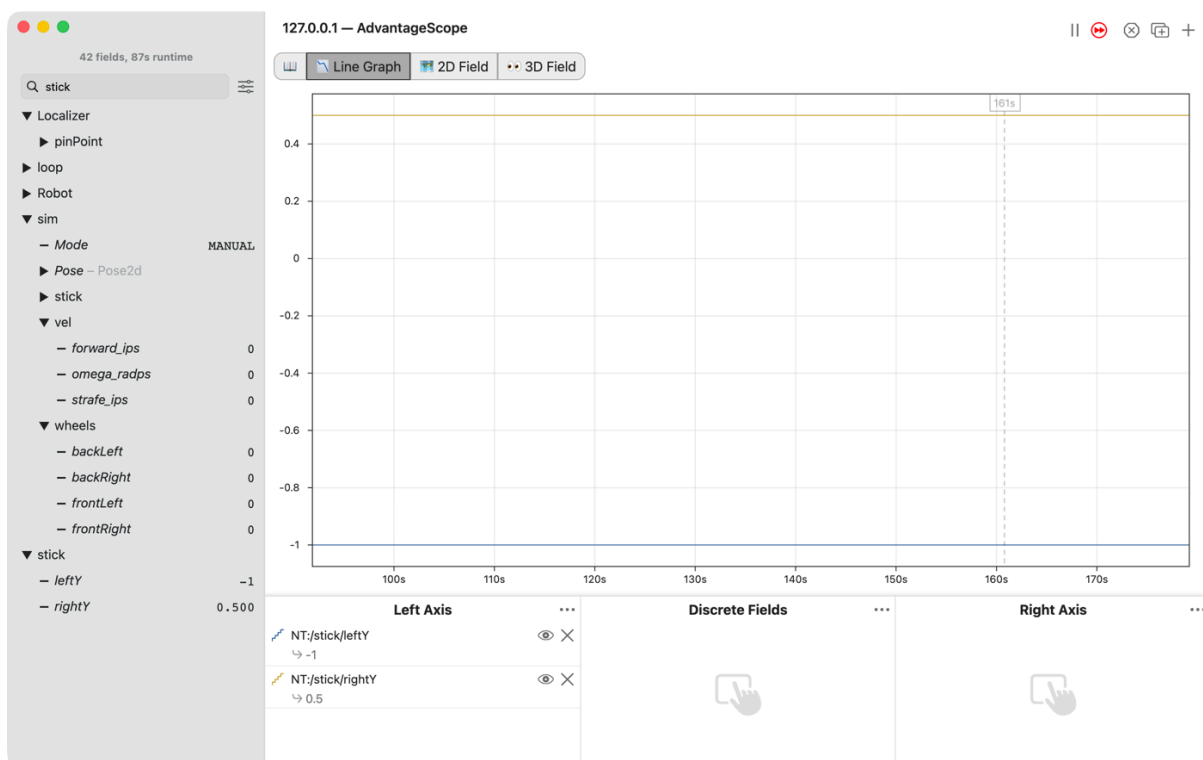


Figure 3.1. AdvantageScope’s Line Graph connected to the simulator, with NT:/stick/leftY drawn at -1 and NT:/stick/rightY drawn at 0.5, and both names in Left Axis.

Push the gamepad’s sticks and watch the numbers they send, live on the laptop. Nothing on the robot moves yet.

Before you start.

- Android Studio has the project open. Ask a mentor to get it there.
- AdvantageScope is installed on the same laptop.
- A gamepad is plugged into the laptop.
- No robot is needed.

The student writes in one folder, `mytry`, inside `TeamCode/java/org.firstinspires.ftc.teamcode`. It holds only a file called `.gitignore` until step L020S010 copies a starter into it. The `lessons` folder beside it holds the starting files.

3.1.1 The steps

L020S010: copy the starter into mytry

The `lessons` folder holds the starters: the file each lesson begins from. You copy one into your own folder, `mytry`, and change the copy.

Copy.

- Open `TeamCode/java/org.firstinspires.ftc.teamcode/lessons`.
- Click `L020SticksOpMode` once, and copy it: ⌘C on a Mac, Ctrl+C on Windows.
- Click the `mytry` folder once, and paste.
- A box called **Copy Class** opens with **Destination package** already ending in `mytry`. Click **OK**.
- If Android Studio asks whether to add the file to Git, click **Add**.

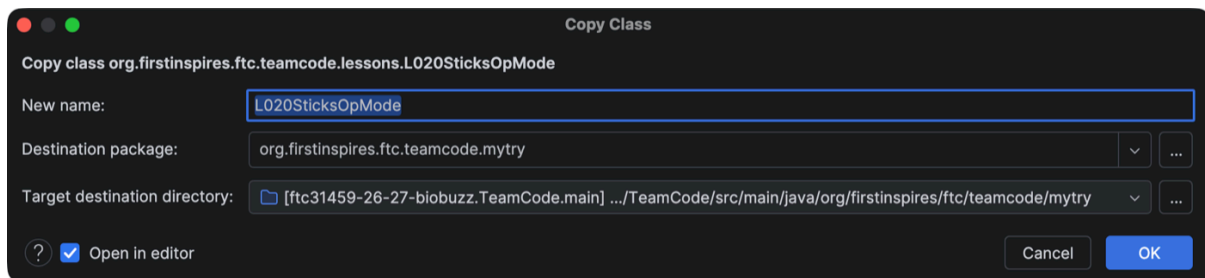


Figure 3.2. The Copy Class box with New name `L020SticksOpMode` and Destination package `org.firstinspires.ftc.teamcode.mytry`.

Check.

- `mytry` holds `L020SticksOpMode` beside `.gitignore`, the file is open, and its first line reads `package org.firstinspires.ftc.teamcode.mytry;`.

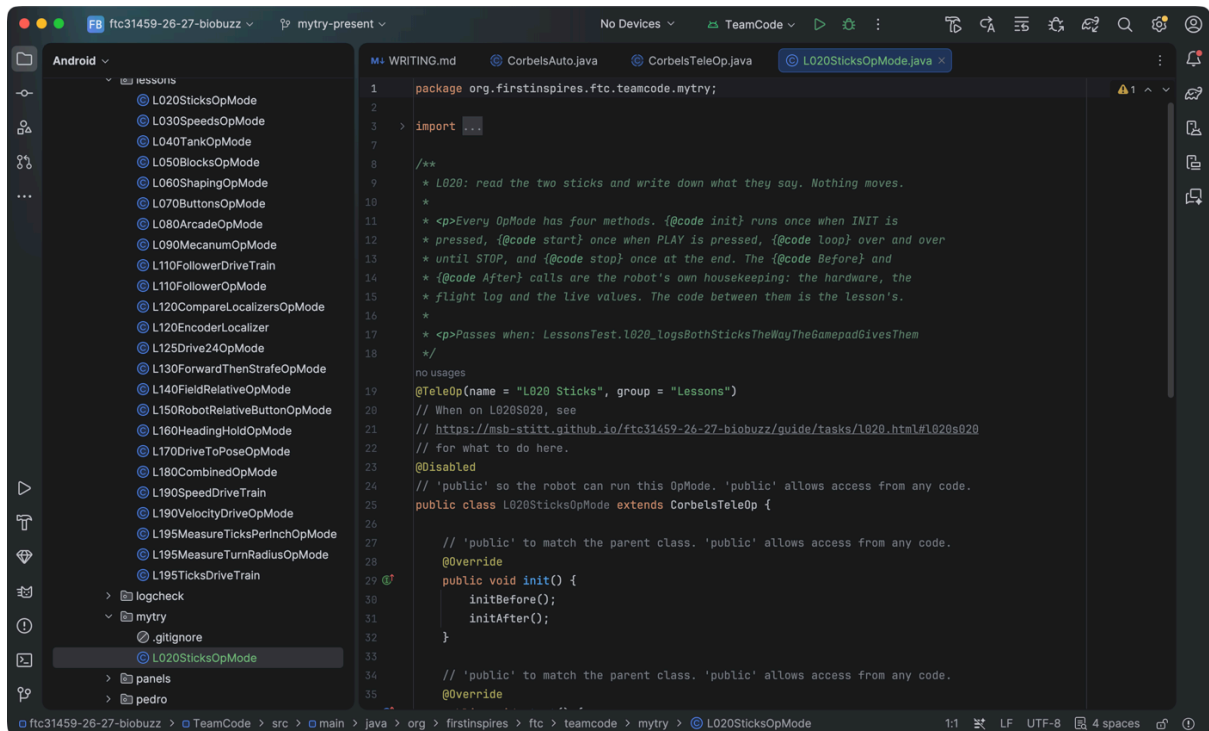


Figure 3.3. The Project window with `lessons` and `mytry` both open, `L020SticksOpMode` in each, and the copy open in the editor with its first line ending in `mytry`;

If it didn't, and the copy landed in `lessons` beside the starter, `lessons` was the folder clicked before pasting, so Destination package ended in `lessons`. Delete the copy, click `mytry`, and paste again.

L020S020: switch your copy on

Change. Find the line `@Disabled`, just under the comment that names this step. Click on it and delete the whole line: ⌘? on a Mac, Ctrl+Y on Windows.

`@Disabled` hides an OpMode from the robot's list. The starter in `lessons` keeps it, so the robot only ever lists your copy.

Check.

- Under the step's comment, the next lines read `// 'public' so the robot can run this OpMode. 'public' allows access from any code. and public class L020SticksOpMode.`

L020S030: read the left stick

The file is an **OpMode**: one program the robot can run. Pressing INIT runs its `init()` once. Pressing PLAY then runs its `loop()` over and over, many times a second, until STOP. The simulator presses all three for you.

Change.

- Scroll down to `loop()` and find the comment that names this step.
- Click at the end of its last line and press Enter. Type this much:

```
double leftStickY = gamepad1.
```

- A list opens with everything a gamepad has. Type `left_st` and the list shrinks to the left stick.
- Move the highlight to `left_stick_y` with the arrow keys, then press Tab.

- Type ; to end the line.

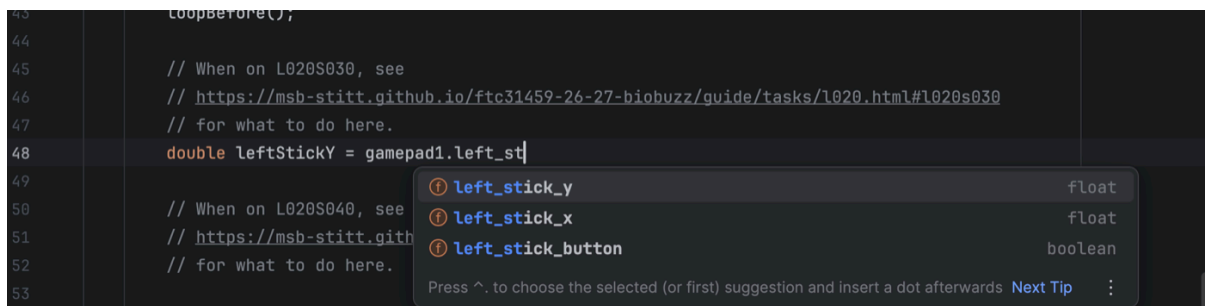


Figure 3.4. The list under `gamepad1` . after typing `left_st`, with `left_stick_y` highlighted and marked float.

```
// When on L020S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1020.html#1020s030
// for what to do here.
double leftStickY = gamepad1.left_stick_y;
```

`double` is Java's word for a number with a decimal point, like 0.5 or -1. `leftStickY` is a name you chose for one. The list said `float`, which is a smaller kind of decimal number. A `double` holds any float.

Now hold the pointer still over `left_stick_y`. A box opens and says what it is: *left analog stick vertical axis*. You can do this with any name you are not sure about.

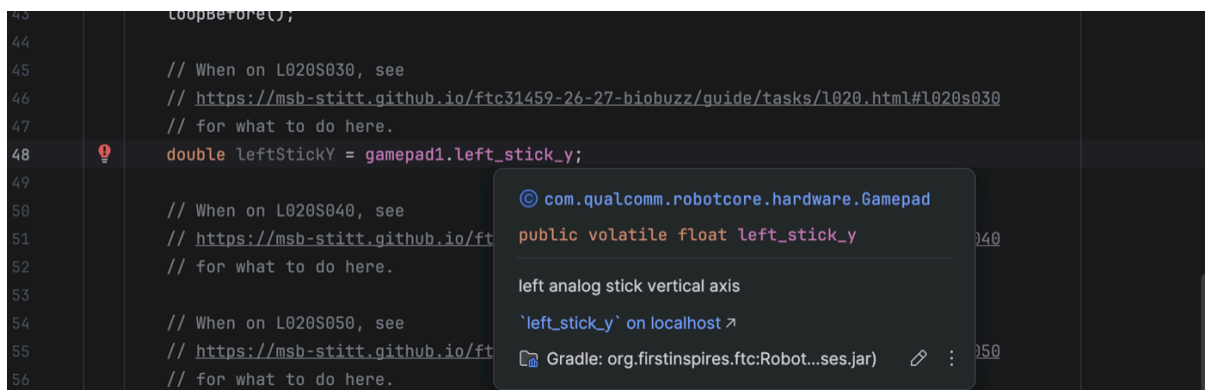
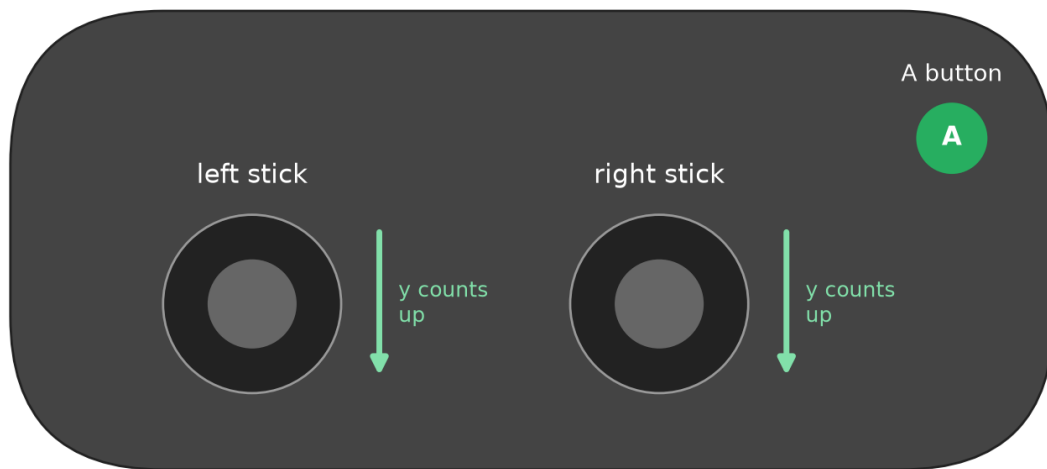


Figure 3.5. The box over `left_stick_y` reading `public volatile float left_stick_y` and *left analog stick vertical axis*.

a stick pushed away from the driver reads -1, pulled back +1



the driver holds it from this side

Stand-in drawing for a photo still to be taken

Figure 3.6. Gamepad 1 from above, with the left stick, the right stick and the A button labelled, and an arrow showing which way each y axis counts up.

Check.

- The line has no red underline.

If it **didn't**, and the line reads `left_stick_x`, the highlight was on the wrong name when Tab was pressed. Change the letter by hand.

L020S040: write the left stick down

Change. Under the comment that names this step, type `Track`. The list shows more than one Tracker. Pick the one from `org.firstinspires.ftc.teamcode.base` and press Tab. Then finish the line:

```
// When on L020S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1020.html#1020s040
// for what to do here.
Tracker.publish("stick/leftY", leftStickY);
```

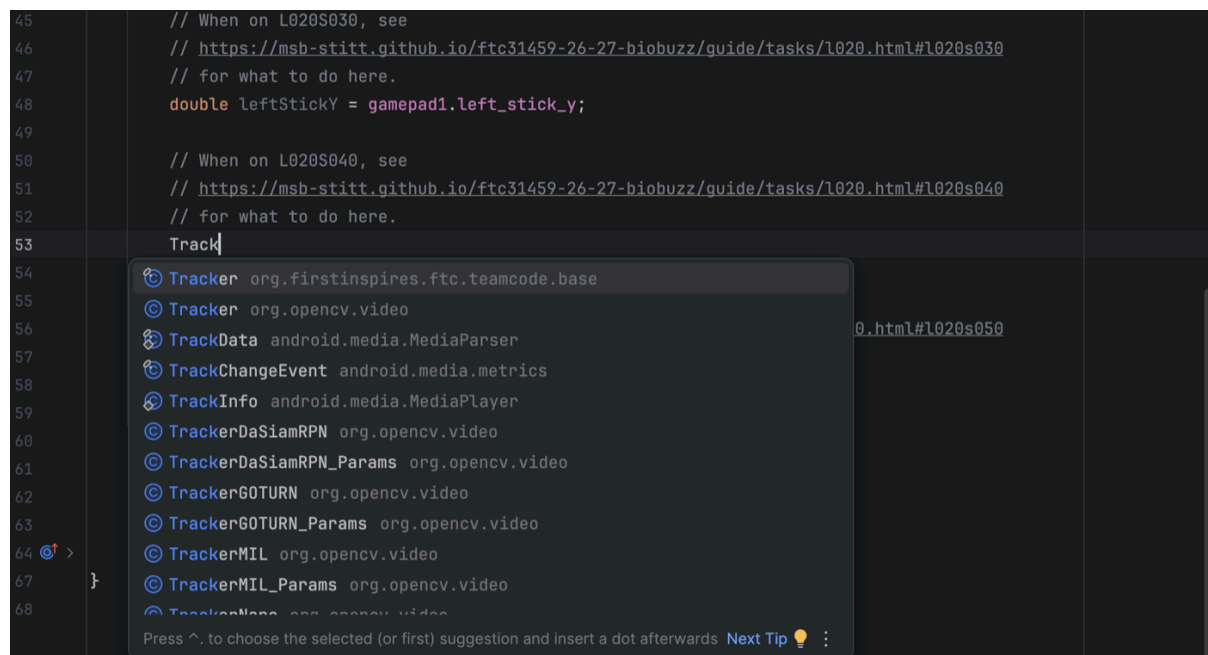


Figure 3.7. The list under `Tracker`, with `Tracker` from `org.firstinspires.ftc.teamcode.base` highlighted at the top and `Tracker` from `org.opencv.video` under it.

`Tracker.publish` takes a name and a value. It sends both to `AdvantageScope` while the code runs, and writes them in the flight log. Android Studio added an `import` line for `Tracker` near the top of the file.

Check.

- `Tracker` is not red, and there is an `import org.firstinspires.ftc.teamcode.base.Tracker;` line near the top.

The simulator is a robot that exists only in the laptop. It runs your `OpMode` just as the real robot would.

Simulator. Click the **Terminal** button near the bottom of the left edge. Type this and press Enter:

```
./gradlew :TeamCode:simRun --args="mytry.L020SticksOpMode --pad"
```

Check.

- It prints `Flight log:` and a folder, then `mytry.L020SticksOpMode running.`, and stays running.
- The folder is the project's top folder, the one that holds `README.md`.

`AdvantageScope` draws the numbers your code publishes, as they change.

`AdvantageScope`.

- Open **File > Connect to Simulator > NetworkTables 4**, and click **Line Graph**.
- Type `stick` in the search box at the top left, and click `NT:/stick/leftY`.
- Drag `leftY` from the `stick` folder down into **Left Axis**.

Before you push the stick, guess. When you push the left stick all the way away from you, will the line go to 1 or to -1? Now push it and see!

Check.

- The line moves when you push.
- All the way forward it reads -1, and all the way back it reads 1.

No gamepad. The simulator takes stick positions as words instead: `--args="mytry.L020SticksOpMode --left_stick_y=-1 --right_stick_y=0.5"` holds the left stick full forward and the right stick half back. The pictures on this page were taken that way.

The list in AdvantageScope has two stick folders. The one inside `sim` is the gamepad as the simulator read it; the one on its own is what the student's code published. In L020 they read the same, because the code passes the stick on unchanged.

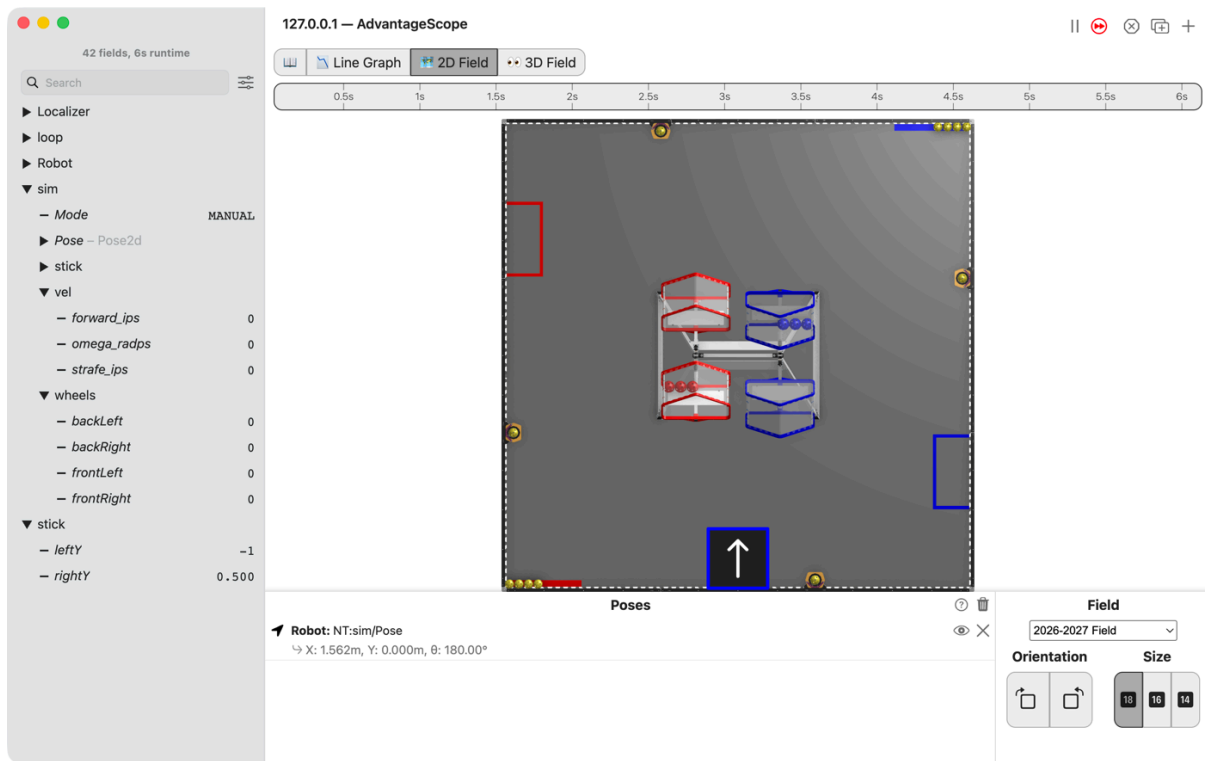


Figure 3.8. AdvantageScope's 2D Field with the robot `NT:sim/Pose` at its start. The list shows the four `sim/wheels` at 0, `stick/leftY` at -1 and `stick/rightY` at 0.5.

The simulator also sends where the robot is and the four wheel powers, as above. They stay still in L020, since nothing drives the wheels.

L020S050: the right stick too

Change. Click in the Terminal and press Ctrl+C to stop the simulator. That is Ctrl+C on a Mac too, not ⌘C. Under the comment that names this step, read and write down the right stick the same way:

```
// When on L020S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1020.html#1020s050
// for what to do here.
double rightStickY = gamepad1.right_stick_y;
Tracker.publish("stick/rightY", rightStickY);
```

Simulator. Run the same command again.

AdvantageScope. Drag `rightY` from the `stick` folder into **Left Axis** beside `leftY`.

Check.

- Each line follows its own stick.

- Push each stick all the way both ways. The numbers never go past 1 or below -1, and forward is -1 on both.

Log.

- Press Ctrl+C in the Terminal to stop the simulator.
- In AdvantageScope, open **File > Open Log(s)...** Go to the project's top folder. Open the newest file whose name starts L020SticksOpMode.
- Drag leftY and rightY from stick into **Left Axis**.

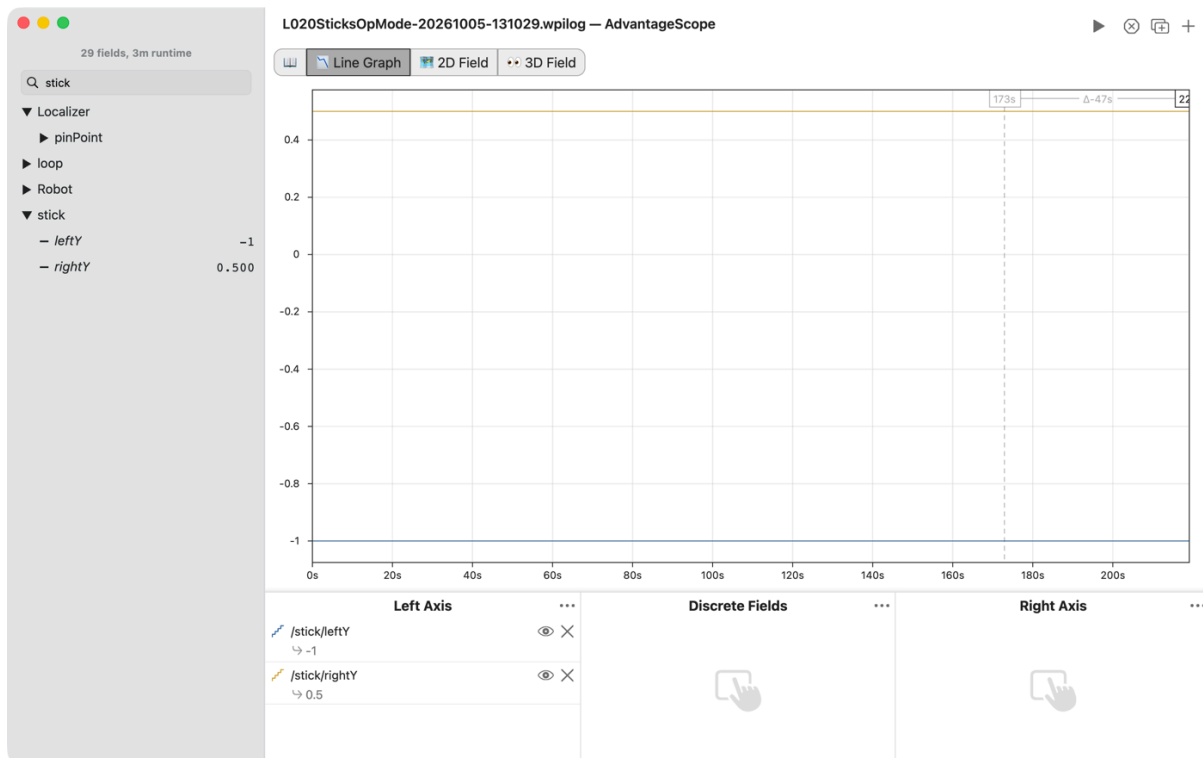


Figure 3.9. AdvantageScope's Line Graph of the flight log L020SticksOpMode-....wpilog, with /stick/leftY at -1 and /stick/rightY at 0.5 for the whole run.

Check.

- The lines show what the sticks did during the run.
- It is the same stick/leftY and stick/rightY you watched live, kept in a file you can open again later.

What you just did. loop runs over and over, from PLAY until STOP, and each pass is one look at the gamepad. The code read both sticks into doubles and wrote each one down under a name. AdvantageScope showed the values live, and the flight log kept them to look at after the run. A stick reads from -1 to 1, and forward is -1, which is backwards from the way anybody thinks about driving. L030 turns it round.

Where next. L030 puts a minus sign in front, so forward is positive.

3.2 L030: turn the sticks into speeds

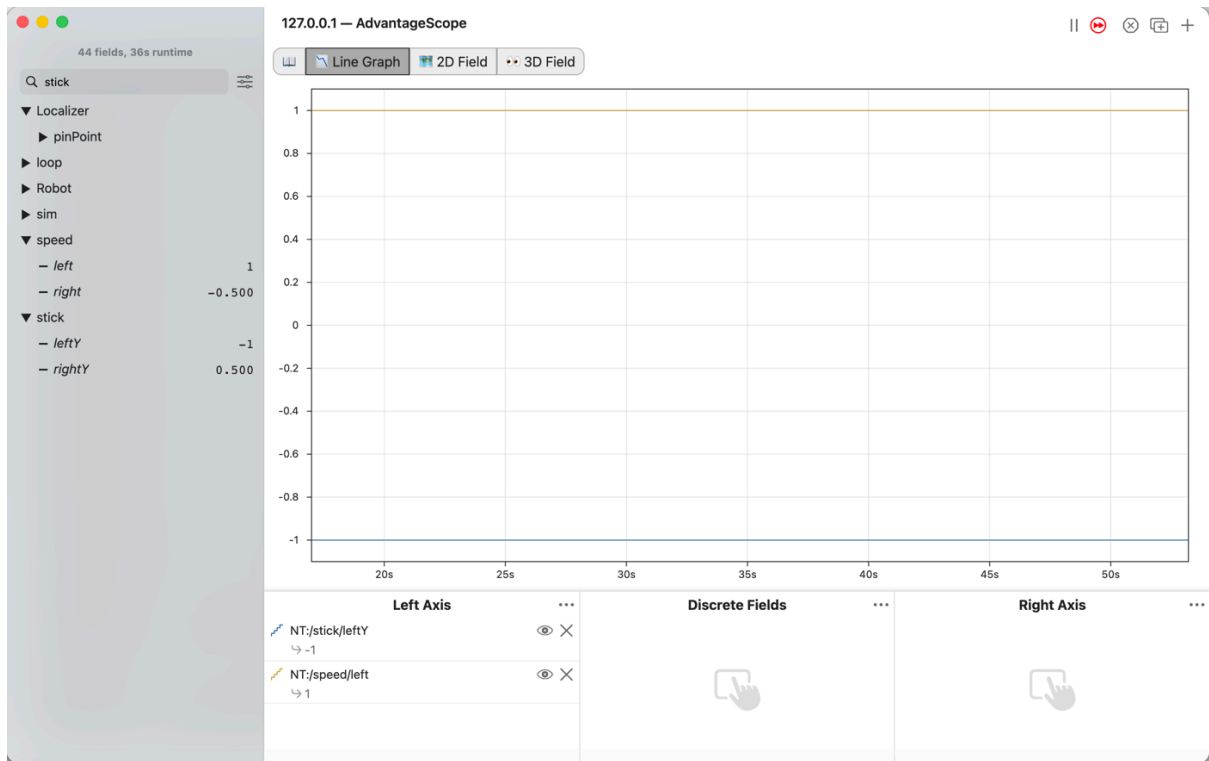


Figure 3.10. AdvantageScope's Line Graph connected to the simulator, with NT:/stick/leftY drawn at -1 and NT:/speed/left drawn at 1, and speed/right reading -0.500 beside stick/rightY at 0.500 in the list.

Turn each stick into a wheel speed, where forward is positive. Nothing on the robot moves yet.

Before you start.

- L020 is done, so mytry holds L020SticksOpMode.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- No robot is needed.

3.2.1 The steps

L030S010: copy the starter into mytry

Copy.

- Open TeamCode/java/org.firstinspires.ftc.teamcode/lessons.
- Click L030SpeedsOpMode once, and copy it: ⌘C on a Mac, Ctrl+C on Windows.
- Click the mytry folder once, and paste.
- The **Copy Class** box opens with **Destination package** already ending in mytry. Click **OK**.
- If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- mytry holds L020SticksOpMode and L030SpeedsOpMode, and the new file is open.

L030S020: switch your copy on

Change. Find the line `@Disabled`, just under the comment that names this step, and delete the whole line, the way you did in L020.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. 'public' allows access from any code.

L030S030: the left speed

We are working toward tank drive, where the left stick drives the left wheels and the right stick drives the right wheels. Push a stick away from you and its wheels should turn forward. We call forward speeds positive and backward speeds negative. But in L020 a stick pushed away read -1, so its number has to be turned round to make a speed.

The starter already has the four lines you wrote in L020. They read the sticks and write them down.

Change. Under the comment that names this step, type `double leftSpeed = -leftS`. The list offers `leftStickY`, the name from the lines above. Press Tab, then finish the two lines:

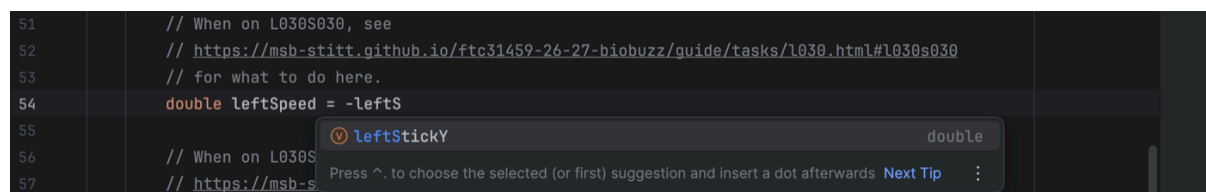


Figure 3.11. The list under `-leftS`, offering the variable `leftStickY`, marked `double`.

```
// When on L030S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1030.html#1030s030
// for what to do here.
double leftSpeed = -leftStickY;
Tracker.publish("speed/left", leftSpeed);
```

The minus sign turns the number round: -1 becomes 1, and 0.5 becomes -0.5. `leftStickY` is still what the stick says, and `leftSpeed` is how fast the wheels should go. Keeping two names for the two ideas means each name always says what it holds.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L030SpeedsOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Type `stick` in the search box. Drag `leftY` from the `stick` folder, and `left` from the `speed` folder, into **Left Axis**.

Check.

- Push the left stick away from you. `stick/leftY` goes to -1, and `speed/left` goes to 1.
- The two lines are always mirror images of each other.

L030S040: the right speed

Change. Press Ctrl+C in the Terminal to stop the simulator. Under the comment that names this step, do the same for the right stick:

```
// When on L030S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1030.html#1030s040
```

```
// for what to do here.
double rightSpeed = -rightStickY;
Tracker.publish("speed/right", rightSpeed);
```

Simulator. Run the same command again.

AdvantageScope. Drag rightY from stick and right from speed into **Left Axis**.

Check.

- Push both sticks away from you. Both speeds read 1.
- Pull both back, and both read -1.

No gamepad. `--args="mytry.L030SpeedsOpMode --left_stick_y=-1 --right_stick_y=0.5"` holds the left stick full forward and the right stick half back. The picture at the top was taken that way: speed/left reads 1 and speed/right reads -0.500.

What you just did. Each speed is its stick with a minus sign in front, so forward is positive from here on. The stick and the speed have separate names because they are separate things: one is what the driver's hand does, the other is what the wheels should do. A name like `leftStick = -gamepad1.left_stick_y` would hide that.

Where next. L040 sends leftSpeed and rightSpeed to the wheels and drives the robot in the simulator.

3.3 L040: drive like a tank

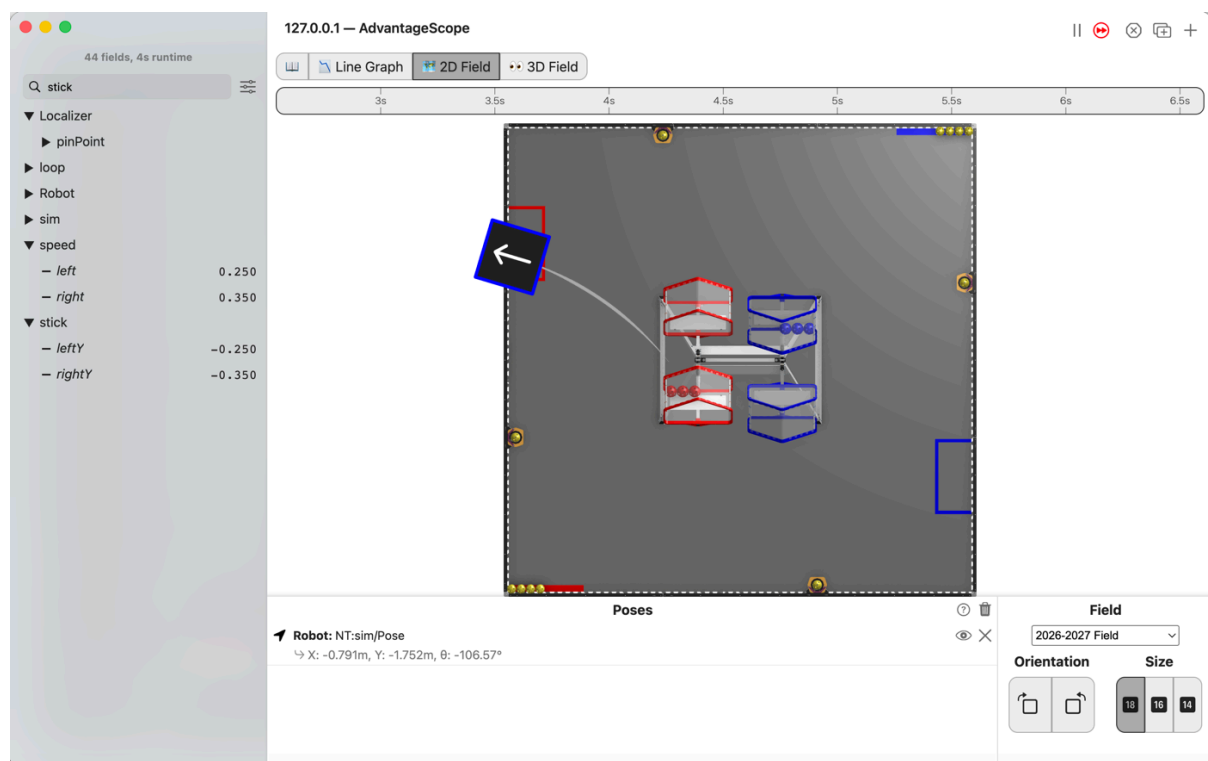


Figure 3.12. AdvantageScope's 2D Field connected to the simulator, with the robot NT:sim/Pose near the left wall after curving left from its start, speed/left reading 0.250 and speed/right 0.350.

The two speeds drive the wheels, and the robot drives in the simulator: tank drive!

Before you start.

- L030 is done, so `mytry` holds `L030SpeedsOpMode`.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- No robot is needed. The robot drives in the simulator only; L050 puts it on blocks.

3.3.1 The steps

L040S010: copy the starter into mytry

Copy. In `lessons`, click `L040TankOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L040TankOpMode`, and the new file is open.

L040S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public' so the robot can run this OpMode`. `'public'` allows access from any code.

L040S030: the left wheels

The starter already reads the sticks and turns them into `leftSpeed` and `rightSpeed`, as you did in L030.

Change. The robot has four motors, one on each wheel. `hardware` holds all four. Under the comment that names this step, type `hardware.front`. The list offers `frontLeft` and `frontRight`. Pick `frontLeft` and press **Tab**.

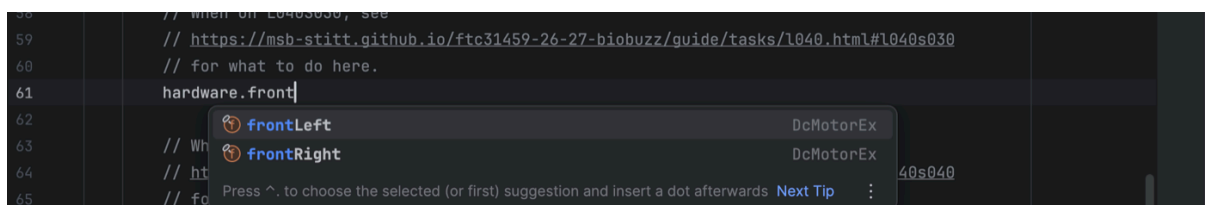


Figure 3.13. The list under `hardware.front`, offering `frontLeft` and `frontRight`, each marked `DcMotorEx`.

Type `.setP`. The list shows what a motor can be told to do. The one we want is `setPower` (double power).

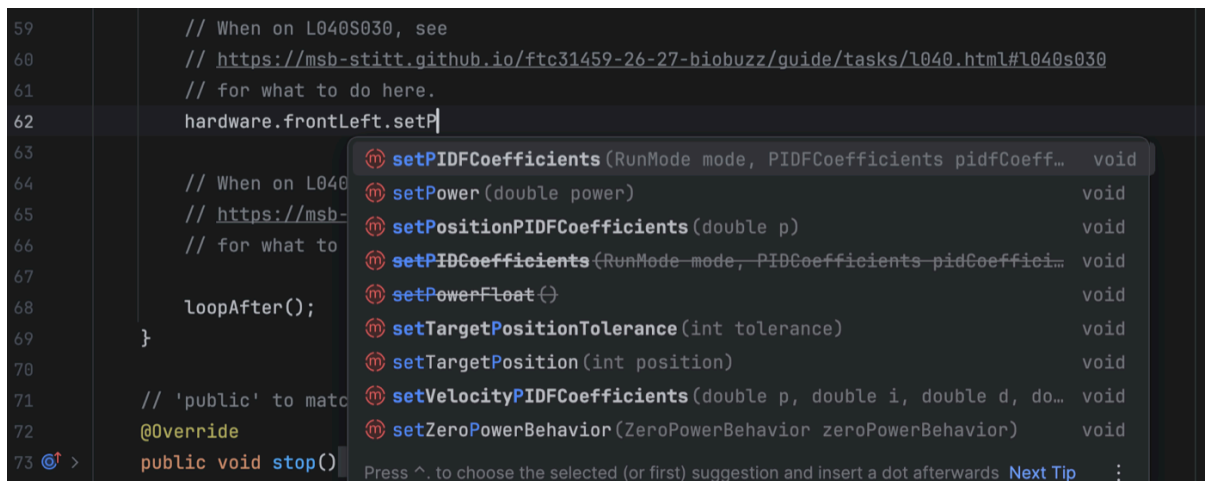


Figure 3.14. The list under `hardware.frontLeft.setP`, with `setPower (double power)` second.

Finish the line, and add the same for the back left wheel:

```
// When on L040S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1040.html#1040s030
// for what to do here.
hardware.frontLeft.setPower(leftSpeed);
hardware.backLeft.setPower(leftSpeed);
```

A power is a number from -1 to 1. It is how hard the motor pushes: 1 is full forward, -1 is full backward, and 0 is stop.

Check.

- Neither line has a red underline.

L040S040: the right wheels

Change. Under the comment that names this step, do the same for the right side:

```
// When on L040S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1040.html#1040s040
// for what to do here.
hardware.frontRight.setPower(rightSpeed);
hardware.backRight.setPower(rightSpeed);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L040TankOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, and click **2D Field**. Open the `sim` folder and drag `Pose` into **Poses**. The robot appears near the bottom edge of the field. A thin line runs to it from the middle of the field. That is where the simulator put the robot before the OpMode moved it to its start, so you can ignore it.

Check.

- Push both sticks gently forward, and the robot drives forward.
- Pull both back, and it comes back.
- Push one forward and pull the other back, and it spins in place.
- Push the right stick further than the left, and it curves to the left.

No `gamepad`. `--args="mytry.L040TankOpMode --left_stick_y=-0.25 --right_stick_y=-0.35"` holds both sticks gently forward, the right a little further. The picture at the top was taken that way, 4 s after connecting. The simulator has no walls, so a robot left running drives off the field.

If it didn't, and the robot creeps with both sticks let go, the sticks do not read exactly 0. Nothing in L040 tidies that up; L060 does.

What you just did. Each pass through `loop` reads the sticks, turns them into speeds, and hands each speed to the two motors on its side. That is tank drive. In the simulator every motor turns its wheel forward for a positive power. On the robot, a motor mounted the other way round turns its wheel backward, and L050 is where that is found and fixed.

Where next. L050 runs this on the robot, on blocks.

3.4 L050: drive the robot on blocks

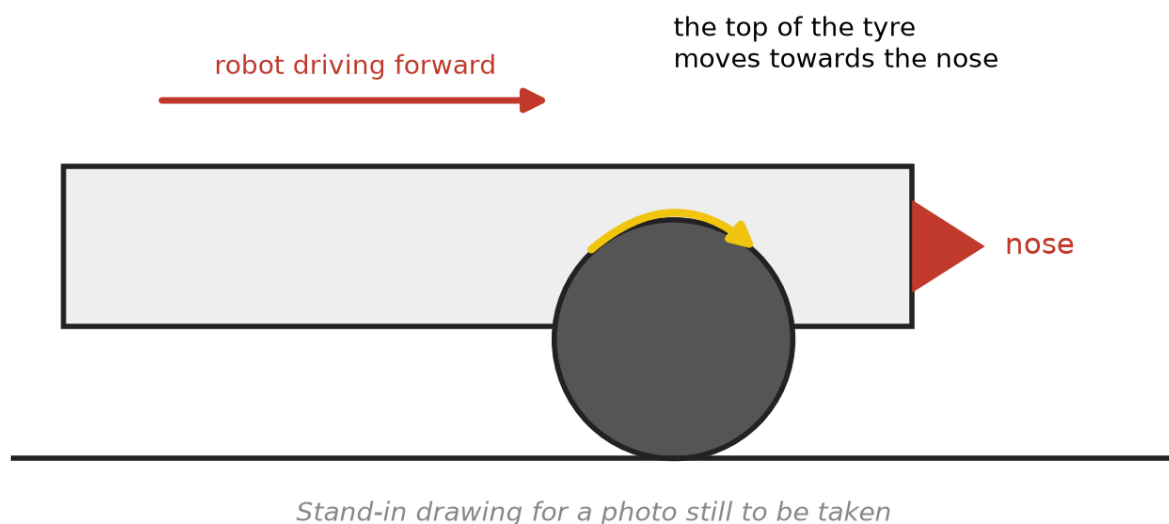


Figure 3.15. One wheel from the side, with an arrow on the top of the tyre showing which way it travels when that wheel is driving the robot forward.

Tell each motor which way it is mounted. Then run tank drive on the robot, up on blocks, and check that every wheel turns the way its stick says.

Before you start.

- L040 is done, so `mytry` holds `L040TankOpMode`.
- A robot, a Driver Station with a gamepad, and a USB cable from the laptop to the Control Hub.
- Two blocks to rest the robot on, so no wheel touches the ground, and about 2 m of clear floor for the last step.

3.4.1 The steps

L050S010: copy the starter into `mytry`

Copy. In `lessons`, click `L050BlocksOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- mytry holds L050BlocksOpMode, and the new file is open.

L050S020: switch your copy on

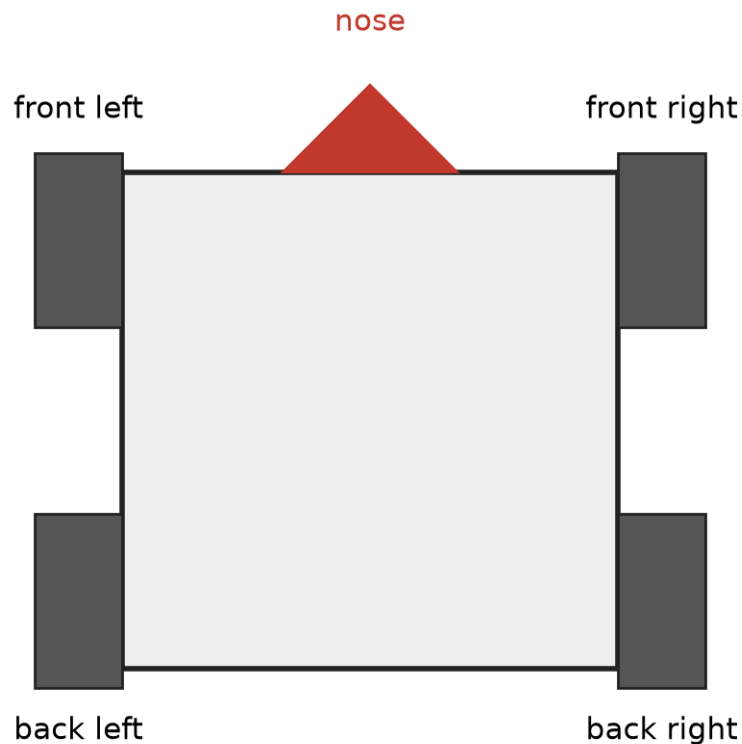
Change. Delete the line @Disabled, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public' so the robot can run this OpMode.` 'public' allows access from any code.

L050S030: label the robot

Label. Put a strip of tape on the robot's front, and mark it FRONT. Mark the left side LEFT and the right side RIGHT, as the robot sees them when it drives forward.



Stand-in drawing for a photo still to be taken

Figure 3.16. The robot from above with its nose marked, and each of the four wheels labelled front left, front right, back left and back right.

Check.

- Stand behind the robot, facing the way FRONT faces. LEFT is on your left.

L050S040: tell each motor which way it is mounted

The two sides of the robot face opposite ways. So the same power turns the left motors one way and the right motors the other. In L040 the simulator did not care. A real robot does.

`Constants.java` already says which way each motor is mounted, and

`hardware.mecanumConfig` holds what it says.

Change. Scroll up to `init()` and find the comment that names this step. Type `hardware.frontLeft.setD`, and pick `setDirection`. Inside the brackets, type `hardware.mecanumConfig.frontLeftD`. The list offers `frontLeftDirection.get()`, marked `Direction`. Pick it.

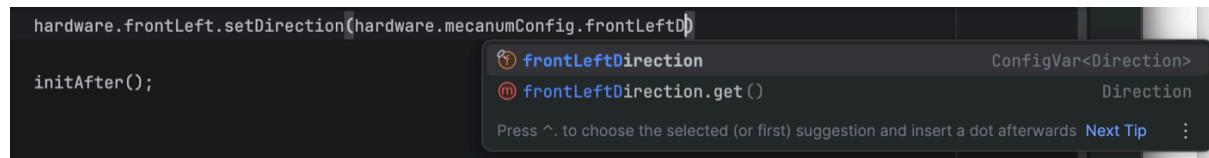


Figure 3.17. The list under `hardware.mecanumConfig.frontLeftD`, offering `frontLeftDirection` and `frontLeftDirection.get()`, the second marked `Direction`.

Finish the line, and do the same for the other three motors:

```
// When on L050S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1050.html#1050s040
// for what to do here.

hardware.frontLeft.setDirection(hardware.mecanumConfig.frontLeftDirection.get());
hardware.frontRight.setDirection(hardware.mecanumConfig.frontRightDirection.get());
hardware.backLeft.setDirection(hardware.mecanumConfig.backLeftDirection.get());
hardware.backRight.setDirection(hardware.mecanumConfig.backRightDirection.get());
```

`init` runs once, when `INIT` is pressed, so the motors know their directions before `PLAY`.

Check.

- Each line ends `.get() ;`, and none has a red underline.

L050S050: send the code to the robot

The **Control Hub** is the robot's computer: the box the motors plug into. The **Driver Station** is the screen the drivers hold. It picks an `OpMode`, has the `INIT`, `PLAY` and `STOP` buttons, and passes the gamepad to the robot.

Robot.

- Rest the robot on the blocks, so all four wheels are off the ground.
- Turn it on, and plug the laptop into the Control Hub with the USB cable.

Send.

- In Android Studio, the device list left of **TeamCode** reads **No Devices** until the robot is plugged in. Pick the Control Hub from it.
- Click the green triangle right of **TeamCode**.

Check.

- Android Studio says the install worked, and the Driver Station shows the robot.

L050S060: watch the wheels

Gamepad. Plug the gamepad into the Driver Station. On the gamepad, hold `Start` and press `A`. That makes it gamepad 1.

Run. On the Driver Station, pick `L050 Blocks` from the `Lessons` group. Press `INIT`.

AdvantageScope.

- Join the laptop to the robot's Wi-Fi. Ask a mentor for its name and password the first time.
- Open **AdvantageScope > Settings...** on a Mac. On Windows it is **App > Show Preferences...**. Type `192.168.43.1` in **Robot Address**, and click the `?`.
- Open **File > Connect to Robot > NetworkTables 4**.

The first time AdvantageScope talks to the robot, the operating system might ask for your permission. Please give it.

Check.

- The title bar reads `192.168.43.1`, and the list on the left shows `Localizer`.

Run. Press `PLAY`.

Check.

- Push the left stick a little way forward. In AdvantageScope, `speed/left` goes above 0.
- Both left wheels turn, with the top of each tyre moving toward `FRONT`.
- Let go, and do the same with the right stick and the right wheels.

Write down any wheel that turns the other way, or that turns for the other side's stick.

L050S070: set each wheel right

Turns backward. A wheel that turns backward needs its direction swapped in `TeamCode/src/main/java/org/firstinspires/ftc/teamcode/pedro/Constants.java`. Find its line, such as `c.frontLeftDirection.set(DcMotorSimple.Direction.FORWARD);`, and swap `FORWARD` for `REVERSE`, or `REVERSE` for `FORWARD`.

Turns for the other stick. A wheel that turns for the other side's stick is plugged into a port the Robot Configuration gives a different name. Ask a mentor to check the names on the Driver Station.

Send. Send the code again, as in L050S050, and repeat L050S060.

Check.

- All four wheels pass L050S060.

L050S080: drive it on the floor

Drive. Press `STOP`. Lift the robot off the blocks and put it on the floor, with room in front of it. Press `INIT`, then `PLAY`.

Check.

- Push both sticks gently forward, and the robot drives forward!
- Pull both back, and it comes back.
- Push one forward and pull the other back, and it spins in place.

Live on the robot is for practice. An `OpMode` in `lessons` or `mytry` turns on the robot's `NetworkTables` server at `INIT`, which is what AdvantageScope connects to. FTC rule R704 forbids third-party telemetry over Wi-Fi at competitions, so a competition `OpMode` lives in another folder, where it stays off.

On a Mac, the **Settings...** window is titled **Preferences**.

If it didn't, and the robot creeps with both sticks let go, the sticks do not read exactly 0. L060 fixes that.

What you just did. Which way a motor turns is set by how it is bolted on and wired, not by the code. `setDirection` tells each motor which way it is mounted, so from then on a positive power turns every wheel forward. The directions live in one place, `Constants.java`. The code reads them from there, and so will the drivetrains in later lessons and AutoTune.

Where next. L060 stops the creep, and makes a gently pushed stick mean gently.

3.5 L060: shape the sticks

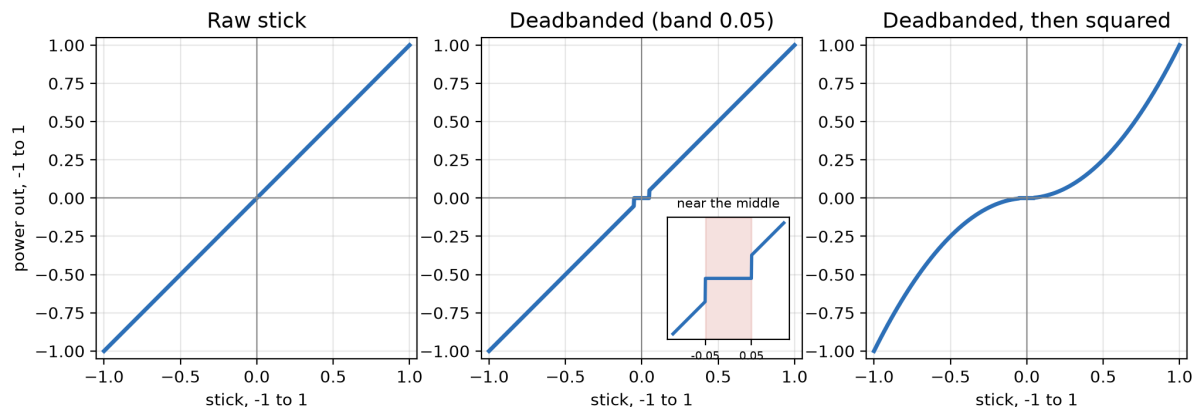


Figure 3.18. Three graphs side by side, each with the stick from -1 to 1 along the bottom and the power out from -1 to 1 up the side: the raw stick as a straight line, the deadbanded stick as a line with a flat step at the middle, and the squared stick as an S through the middle.

Make the robot sit still when the sticks are let go, and make a gentle push mean gently.

Before you start.

- L050 is done, so `mytry` holds `L050BlocksOpMode`, and the robot's wheels all turn the right way.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot and the Driver Station for the last step.

3.5.1 The steps

L060S010: copy the starter into mytry

Copy. In `lessons`, click `L060ShapingOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L060ShapingOpMode`, and the new file is open.

L060S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public' so the robot can run this OpMode.` 'public' allows access from any code.

L060S030: write a deadband

A stick that has been let go does not always read 0. It can read a few hundredths, and a few hundredths of power is enough to make a robot crawl across the floor on its own. A **deadband** fixes that: any reading closer to the middle than the band counts as let go.

Change. Scroll to the bottom of the file and find the comment that names this step. Under it, write a method of your own:

```
// When on L060S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1060.html#1060s030
// for what to do here.
double deadband(double value, double band) {
    if (Math.abs(value) < band) {
        return 0;
    }
    return value;
}
```

A **method** is a named piece of code you can use again. This one takes two numbers, value and band, and gives back one number. return is the number it gives back.

if runs the lines inside its { } only when what is in its brackets is true. Here, return 0 runs only for a stick close to the middle.

Math.abs takes the minus sign off: Math.abs(-0.03) is 0.03. deadband is given the speed, so a stick pulled halfway back gives -0.5, and -0.5 is less than 0.05. But it is far from the middle, so it has to be compared without its minus sign.

Check.

- The method has no red underline.
- deadband is grey, with *no usages* above it. Nothing uses it yet.

L060S040: use the deadband

Change. In loop(), under the comment that names this step, make a power for each side:

```
// When on L060S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1060.html#1060s040
// for what to do here.
double band = 0.05;
double leftPower = deadband(leftSpeed, band);
Tracker.publish("power/left", leftPower);
double rightPower = deadband(rightSpeed, band);
Tracker.publish("power/right", rightPower);
```

Then change the four setPower lines below, so each one gets the power and not the speed:

```
hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);

hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L060ShapingOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Drag left from the speed folder and left from the power folder into **Left Axis**.

Check.

- Let go of the sticks. `speed/left` may sit a little off 0, and `power/left` reads 0.
- The wheels in `sim/wheels` all read 0.

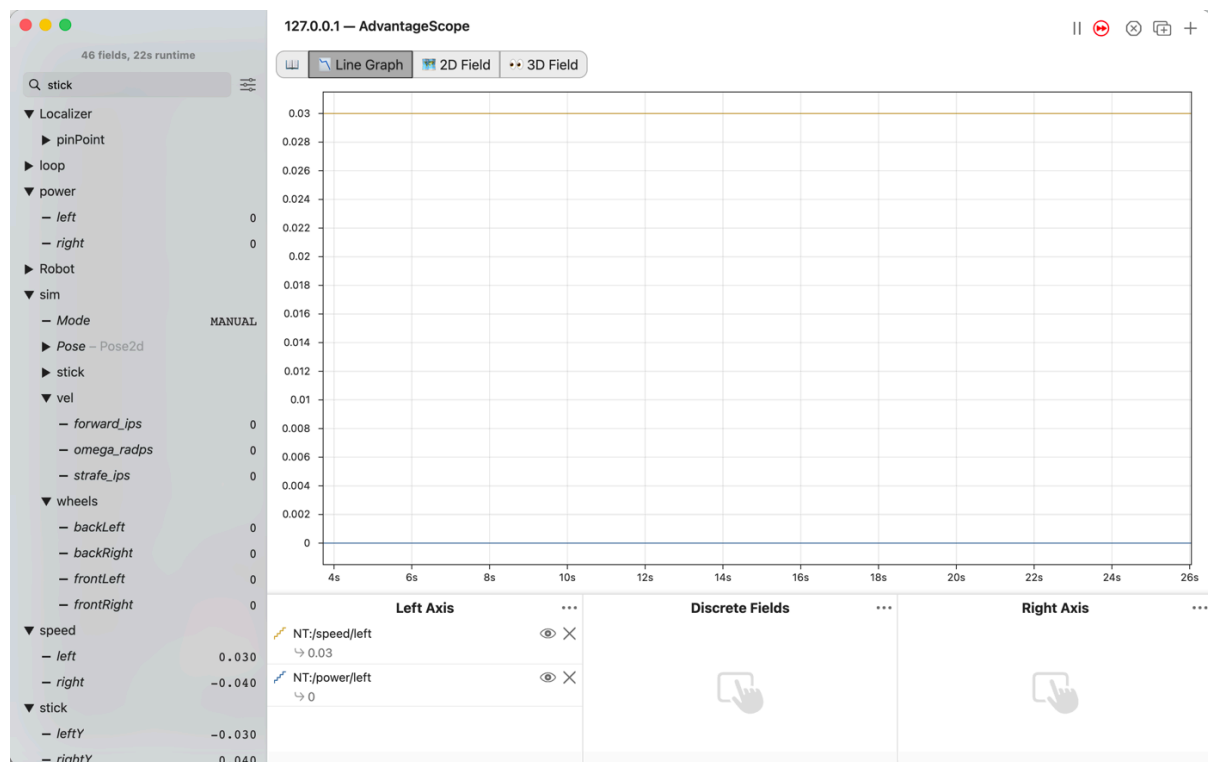


Figure 3.19. AdvantageScope's Line Graph with `NT:/speed/left` at 0.03 and `NT:/power/left` at 0, and the list showing `power/left` and `power/right` at 0, all four `sim/wheels` at 0, and `speed/right` at -0.040.

L060S050: write squaring

With the deadband in, a tenth of a push is still a tenth of the power. Near the middle that is more robot than a driver wants. **Squaring** fixes it: half a push becomes a quarter of the power, and a full push is still full power.

Change. Press `Ctrl+C` in the Terminal to stop the simulator. Find the comment that names this step. Under it, write a second method:

```
// When on L060S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1060.html#l060s050
// for what to do here.
double squared(double value) {
    return value * Math.abs(value);
}
```

`value * value` would lose the minus sign: -0.5 times -0.5 is 0.25, and backward would turn into forward. `value * Math.abs(value)` keeps it: -0.5 times 0.5 is -0.25.

Check.

- The method has no red underline.

L060S060: use squaring

Change. Go back to the two deadband lines from L060S040. Put `squared(` in front of `deadband`, and one more `)` before the `;`:

```
double leftPower = squared(deadband(leftSpeed, band));
```

```
double rightPower = squared(deadband(rightSpeed, band));
```

squared works on what deadband gives back.

Simulator. Run the same command again, and connect AdvantageScope.

Check.

- Push the left stick halfway forward. speed/left reads 0.5, and power/left reads 0.25.
- Push it all the way, and both read 1.
- Pull it back, and the power goes below 0.

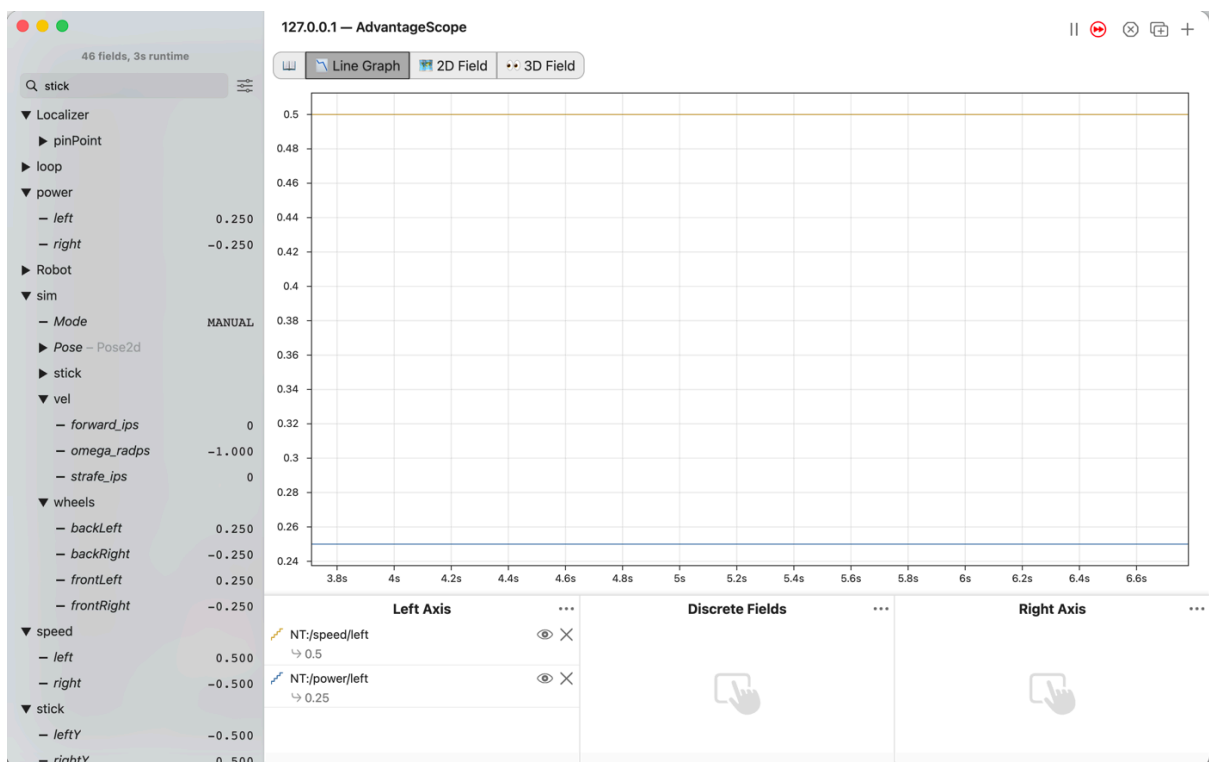


Figure 3.20. AdvantageScope's Line Graph with NT: /speed/left at 0.5 and NT: /power/left at 0.25, and the list showing power/right at -0.250 beside speed/right at -0.500.

L060S070: drive it on the robot

Send.

- Press Ctrl+C in the Terminal to stop the simulator.
- Plug the laptop into the Control Hub with the USB cable.
- Pick the Control Hub from the device list, and click the green triangle right of **TeamCode**.

Check.

- Android Studio says the install worked.

Run. On the Driver Station, pick L060 Shaping from the Lessons group. Press INIT.

AdvantageScope.

- Join the laptop to the robot's Wi-Fi.
- Open **File > Connect to Robot > NetworkTables 4**. AdvantageScope still has the robot's address from L050.

Check.

- The title bar reads 192.168.43.1.

Run. Press PLAY. Drag `left` from the speed folder and `left` from the power folder into **Left Axis**.

Check.

- Let go of both sticks, and the robot sits still, with `power/left` at 0.
- Push a stick about a third of the way, and the robot creeps along under control.
- Push it all the way, and the robot goes full speed.

No `gamepad.` `--args="mytry.L060ShapingOpMode --left_stick_y=-0.03 --right_stick_y=0.04"` holds both sticks just off centre; the first picture was taken that way. `--left_stick_y=-0.5 --right_stick_y=0.5` holds them half forward and half back for the second.

If it didn't, and the robot still creeps on the floor, the band is smaller than what this gamepad reads when it is let go. Raise it a little: 0.05 to 0.1 is the usual range. A band much bigger than that throws away real driving.

If it didn't, and the robot goes forward but will not back up, a sign was lost. Check `squared` uses `Math.abs`, and `deadband` too.

What you just did. Both methods sit between the speed and the wheels. Each one takes a number and gives one back, and neither knows what a wheel is. `speed/left` is still what the stick asks for, and `power/left` is what the wheels get, so AdvantageScope shows both side by side.

Where next. L070 reads the buttons and the sideways sticks.

3.6 L070: read the buttons

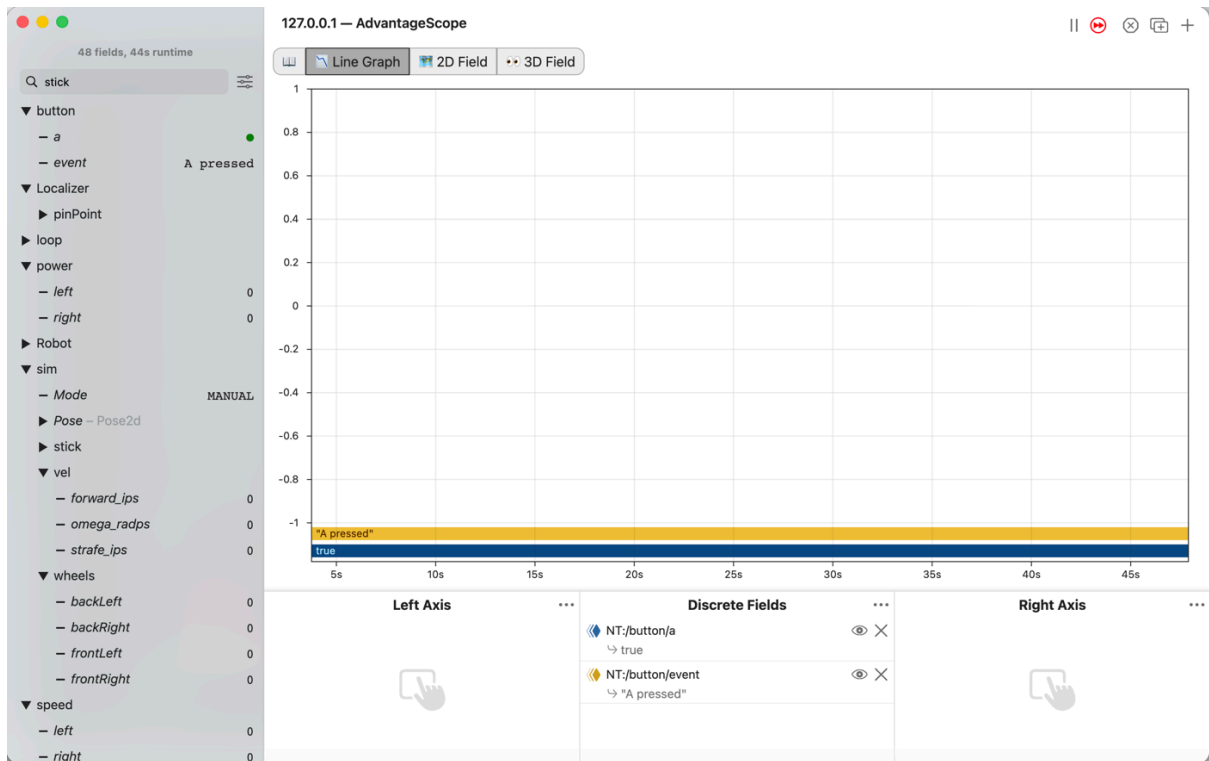


Figure 3.21. AdvantageScope’s Line Graph with NT : /button/a at true and NT : /button/event at “A pressed” in **Discrete Fields**, drawn as two bars along the bottom of the graph, and the list showing button/a as a green dot and button/event as A pressed.

Read the A button, say the moment it goes down and the moment it comes up, and read both sticks sideways. The robot drives as it did in L060 the whole time.

Before you start.

- L060 is done, so mytry holds L060ShapingOpMode.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot and the Driver Station for the last step.

3.6.1 The steps

L070S010: copy the starter into mytry

Copy. In `lessons`, click `L070ButtonsOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- mytry holds L070ButtonsOpMode, and the new file is open.

L070S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. `'public'` allows access from any code.

L070S030: read the A button

A stick is a number. A button is only down or up, so Java keeps it as a **boolean**: `true` when it is down, `false` when it is up.

Change. In `loop()`, find the comment that names this step, below the four `setPower` lines. Under it, type `boolean buttonA = gamepad1.a`. The list offers `a`, marked `boolean`.

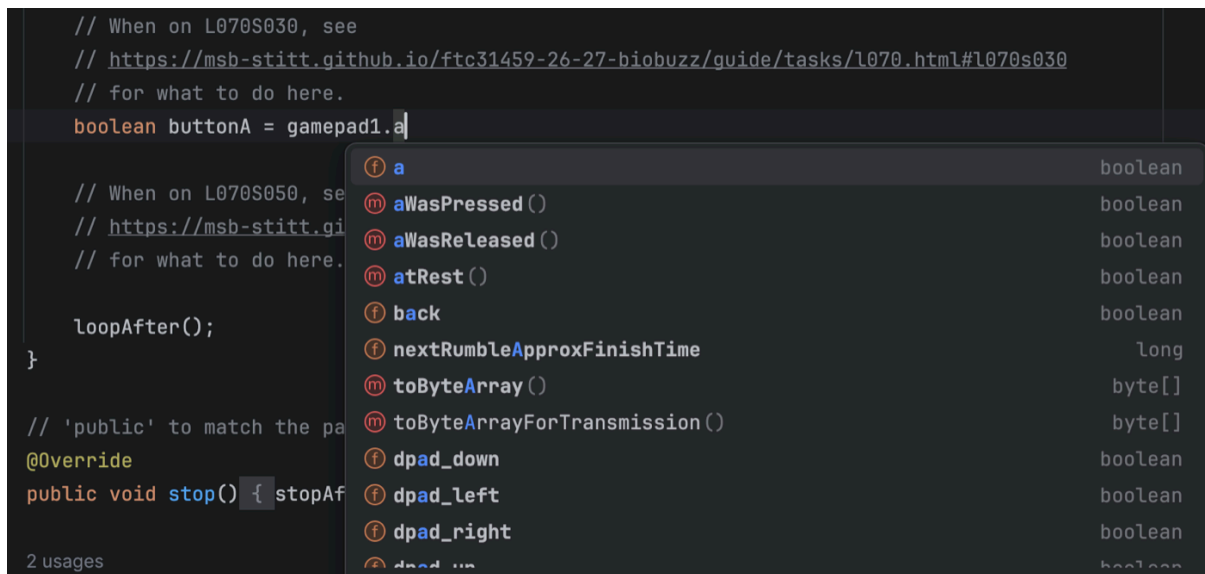


Figure 3.22. The list under `gamepad1.a`, offering a marked `boolean` first, then `aWasPressed()` and `aWasReleased()`.

Pick `a`, and finish the lines:

```
// When on L070S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /1070.html#1070s030
// for what to do here.
boolean buttonA = gamepad1.a;
Tracker.publish("button/a", buttonA);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L070ButtonsOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Open the button folder, and drag `a` into **Discrete Fields**.

Check.

- `button/a` is a red dot.
- Press A and hold it, and the dot turns green.
- Let go, and it turns red again.

L070S040: remember last loop's button

`buttonA` says whether A is down in this loop. To know the moment it changed, the code also needs to know whether it was down in the loop before. A variable made inside `loop()` is gone when the loop ends, so this one goes outside it.

Change. Press Ctrl+C in the Terminal to stop the simulator. Scroll to the top of the class, and find the comment that names this step. Under it, type:

```
// When on L070S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1070.html#1070s040
// for what to do here.
boolean previousButtonA;
```

A variable made here is a **field**. It keeps its value from one loop to the next. A boolean field starts out false.

Check.

- The line has no red underline.

L070S050: say when the button changes

Change. Back in `loop()`, find the comment that names this step. Under it, type:

```
// When on L070S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1070.html#1070s050
// for what to do here.
if (buttonA != previousButtonA) {
    if (buttonA) {
        Tracker.publish("button/event", "A pressed");
    } else {
        Tracker.publish("button/event", "A released");
    }
}
previousButtonA = buttonA;
```

`!=` means “is not the same as”. The first `if` is true only on the loop where the button changed. The second one says which way it changed. `else` is what runs when the `if` is false. The last line saves this loop’s button for the next loop.

The text in quotes is a **String**. Tracker writes it down the same way it writes a number.

Simulator. Run the same command again, and connect AdvantageScope.

AdvantageScope. Drag event from the button folder into **Discrete Fields**. Until A has been pressed once, event is crossed out, because nothing has been written to it yet.

Holding A down does not write A pressed again. event only changes on the loop the button changed.

Check.

- Press A and hold it. button/a turns into a green dot, and event reads A pressed.
- Let go. The dot turns red, and event reads A released.

L070S060: read the sticks sideways

Each stick also moves left and right. Pushed right, it reads above 0. Pushed left, it reads below 0. Tank drive does not use these yet. Later lessons drive sideways with them.

Change. Press Ctrl+C in the Terminal to stop the simulator. Near the top of `loop()`, find the comment that names this step, under the lines that read the sticks up and down. Under it, type:

```
// When on L070S060, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/1070.html#1070s060
// for what to do here.
double leftStickX = gamepad1.left_stick_x;
```

```
Tracker.publish("stick/leftX", leftStickX);
double rightStickX = gamepad1.right_stick_x;
Tracker.publish("stick/rightX", rightStickX);
```

Simulator. Run the same command again, and connect AdvantageScope.

Check.

- Push the left stick to the right, and `stick/leftX` goes above 0.
- Push the right stick to the left, and `stick/rightX` goes below 0.
- speed and power stay at 0, and the robot does not move.

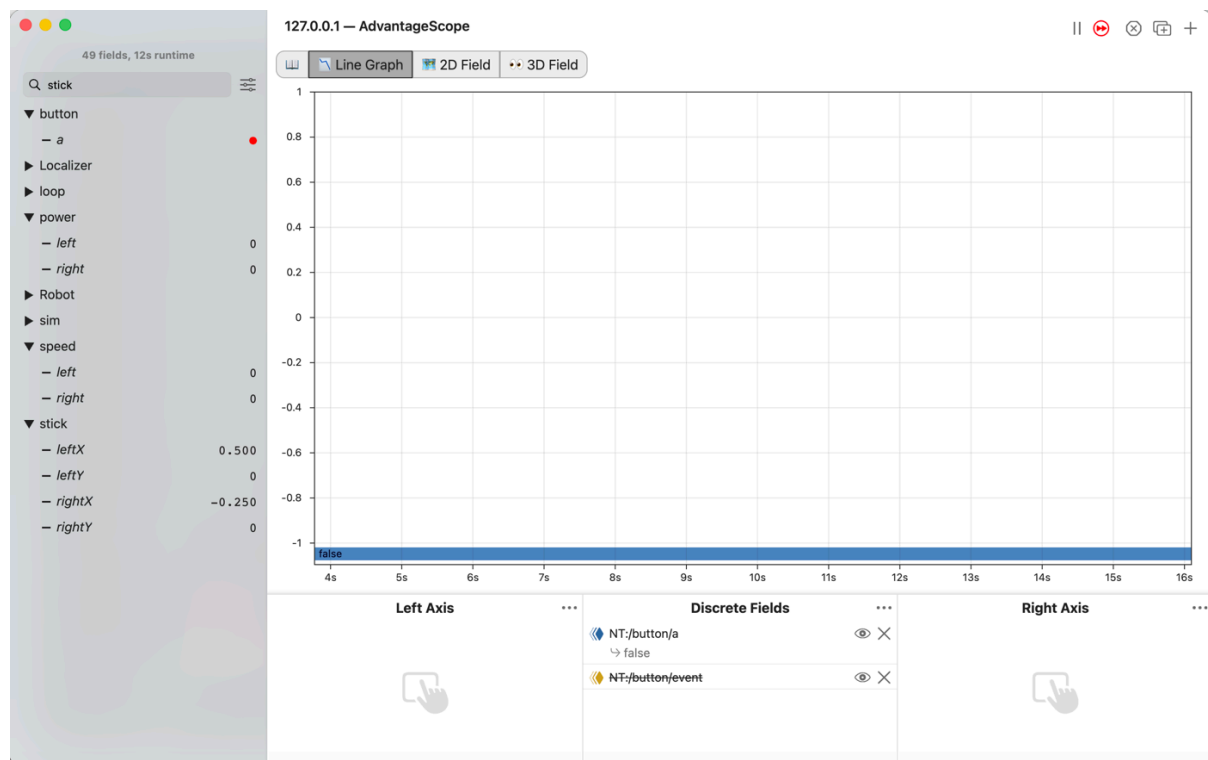


Figure 3.23. AdvantageScope's list with the `stick` folder open, showing `leftX` at 0.500 and `rightX` at -0.250, `leftY` and `rightY` at 0, and speed and power at 0.

L070S070: watch the buttons on the robot

Robot. Stop the simulator. Send the code, run L070 Buttons from the Driver Station, and watch it in AdvantageScope.

Check.

- Press A, and in AdvantageScope `button/event` reads A pressed.
- Let go, and it reads A released.
- The robot still drives with the sticks as it did in L060.

No gamepad. `--args="mytry.L070ButtonsOpMode --a=true"` holds A down for the whole run, so event reads A pressed and never changes; the picture at the top was taken that way. A release needs a gamepad. `--left_stick_x=0.5 --right_stick_x=-0.25` holds the sticks sideways for the last picture. The tests press and let go of A without a gamepad.

If it didn't, and event reads A pressed but never A released, the last line, `previousButtonA = buttonA;`, is missing. The test fails with letting go says so.

If it didn't, and event reads A pressed when A is let go, the two `Tracker.publish` lines are swapped.

What you just did. A boolean is either `true` or `false`. A field keeps a value between loops, and comparing this loop with the last one finds the moment something changed. That is how a button press can do one thing once, rather than once every loop it is held.

Where next. L080 drives the robot with one stick for going and one for turning.

3.7 L080: drive like an arcade game



Figure 3.24. AdvantageScope's Line Graph with `NT:/power/left` at 1 and `NT:/power/right` at 0.6 on **Left Axis**, and the list showing `arcade/forward` at 1 and `arcade/turn` at -0.250.

Drive with one stick for going and one for turning: arcade drive. When the sticks ask for more power than a motor has, scale every wheel down together, so the robot still drives the curve the driver asked for.

Before you start.

- L070 is done, so `mytry` holds `L070ButtonsOpMode`.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot, the Driver Station, and about 3 m of clear floor for the last step.

3.7.1 The steps

L080S010: copy the starter into mytry

Copy. In `lessons`, click `L080ArcadeOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- mytry holds L080ArcadeOpMode, and the new file is open.

L080S020: switch your copy on

Change. Delete the line @Disabled, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this OpMode. 'public' allows access from any code.

L080S030: forward and turn

Tank drive asks the driver to think in left wheels and right wheels. Arcade drive asks for forward and turn instead, which is how a driver already thinks. The code works out the two sides.

The starter has the sticks, the buttons, deadband and squared from the lessons before. It does not drive the wheels yet.

Change. In `loop()`, find the comment that names this step. Under it, type:

```
// When on L080S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /1080.html#l080s030
// for what to do here.
double band = 0.05;
double forward = squared(deadband(-leftStickY, band));
Tracker.publish("arcade/forward", forward);
double turn = squared(deadband(-rightStickX, band));
Tracker.publish("arcade/turn", turn);
```

forward comes from the left stick, up and down. turn comes from the right stick, sideways. Both get a minus sign. For forward, that makes pushing away from you positive, as in L030. For turn, it makes a turn to the left positive. That is the way robots measure turning: left, or counter-clockwise, is positive.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L080ArcadeOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, and open the arcade folder.

Check.

- Push the left stick halfway forward, and forward reads 0.25.
- Push the right stick halfway right, and turn reads -0.25.
- The robot does not move yet.

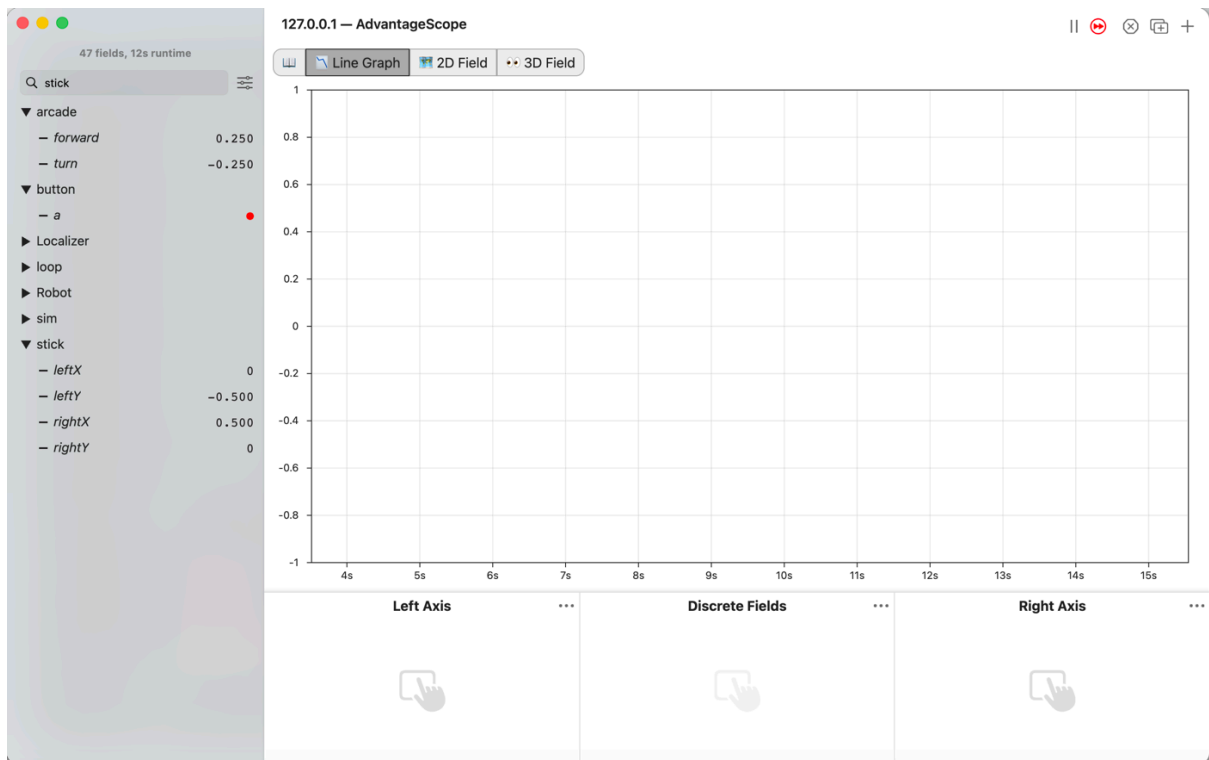


Figure 3.25. AdvantageScope's list with the arcade folder open, showing forward at 0.250 and turn at -0.250, beside stick/leftY at -0.500 and stick/rightX at 0.500.

L080S040: mix them into two sides

To turn right, the left side goes faster and the right side goes slower. A turn to the right is below 0, so taking turn away from the left side speeds it up, and adding it to the right side slows that side down.

Change. Press Ctrl+C in the Terminal to stop the simulator. Find the comment that names this step. Under it, type:

```
// When on L080S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l080.html#l080s040
// for what to do here.
double leftPower = forward - turn;
double rightPower = forward + turn;
Tracker.publish("power/left", leftPower);
Tracker.publish("power/right", rightPower);

hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);
hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

Simulator. Run the same command again, and connect AdvantageScope. Click **2D Field**. The robot from L040 is still in **Poses**.

Check.

- Push the left stick forward, and the robot drives forward.
- Push the right stick right, and it turns right.
- Push both, and it drives a curve.

Now push the left stick all the way forward and the right stick halfway right. Click **Line Graph**, and drag left and right from the power folder into **Left Axis**. power/left reads 1.25.

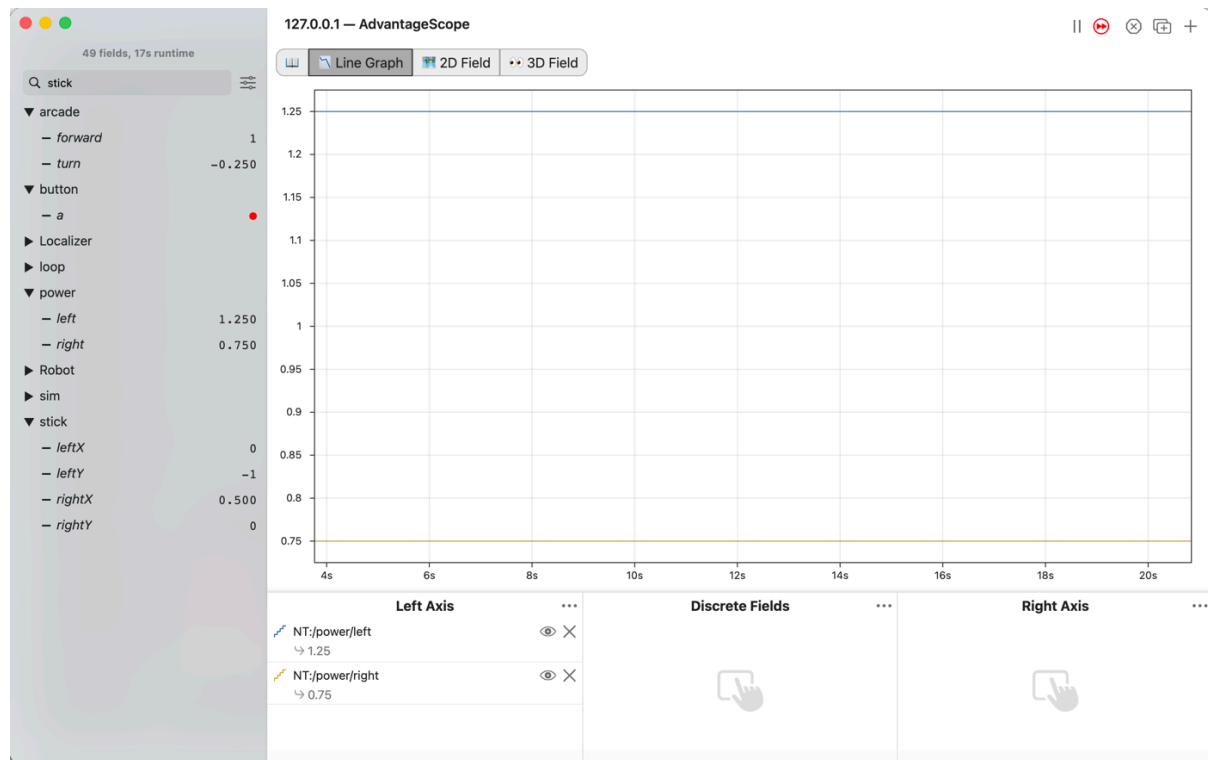


Figure 3.26. AdvantageScope's Line Graph with NT:/power/left at 1.25 and NT:/power/right at 0.75 on **Left Axis**, and the list showing arcade/forward at 1 and arcade/turn at -0.250.

A motor's power only goes from -1 to 1. On the robot, the Control Hub cuts 1.25 down to 1 and leaves 0.75 alone. The left side loses some of what it asked for and the right side loses none, so the robot drives a wider curve than the sticks asked for. The simulator does not cut it, so the curve there looks right. The number gives it away.

L080S050: scale, don't cut

The fix is to divide both sides by the bigger one. 1.25 and 0.75 become 1 and 0.6. The robot goes a little slower, and the curve stays the one the driver asked for.

Change. Press Ctrl+C to stop the simulator. Go back to the two lines from L080S040 that make leftPower and rightPower. Under them, type:

```
double leftPower = forward - turn;
double rightPower = forward + turn;
double biggest = Math.max(1, Math.max(Math.abs(leftPower),
Math.abs(rightPower)));
leftPower = leftPower / biggest;
rightPower = rightPower / biggest;
```

Math.max gives back the bigger of two numbers. The inner one finds the bigger power, with its minus sign taken off. The outer one makes sure biggest is never less than 1. Without it, a gentle 0.25 would be divided by 0.25 and become full power.

The last two lines have no double in front. leftPower already exists, so they change it rather than making a new one.

Simulator. Run the same command again, and connect AdvantageScope.

Check.

- Push the left stick all the way forward and the right stick halfway right. `power/left` reads 1, and `power/right` reads 0.6.
- Push the left stick halfway forward alone, and both read 0.25, the same as before this step.

L080S060: drive it on the robot

Robot. Stop the simulator. Send the code, and run L080 Arcade from the Driver Station.

Check.

- Push the left stick halfway forward and the right stick a quarter right, and the robot drives a circle to the right.
- Push both sticks twice as far, and it drives a circle the same size, only faster.

No `gamepad`. `--args="mytry.L080ArcadeOpMode --left_stick_y=-0.5 --right_stick_x=0.5"` holds the sticks for the first picture. `--left_stick_y=-1 --right_stick_x=0.5` holds them for the other two, before and after L080S050. The simulator has no walls, so a robot left running drives off the field.

If it didn't, and the robot turns left when the right stick is pushed right, the minus sign is missing from `-rightStickX`, or `-` and `+` are swapped in the two mix lines.

If it didn't, and the robot goes full speed with a gentle push, `biggest` starts below 1. Check the 1 in the outer `Math.max`.

What you just did. Arcade drive mixes two things the driver wants into two things the wheels do. When the mix asks for more than a motor has, dividing keeps the shape of the request and gives up some speed. Cutting at 1 keeps the speed and changes the shape. The Control Hub cuts; it is the code's job to scale first.

Where next. L090 drives a mecanum robot, which adds a third thing to do with the sticks: going sideways.

3.8 L090: slide it sideways

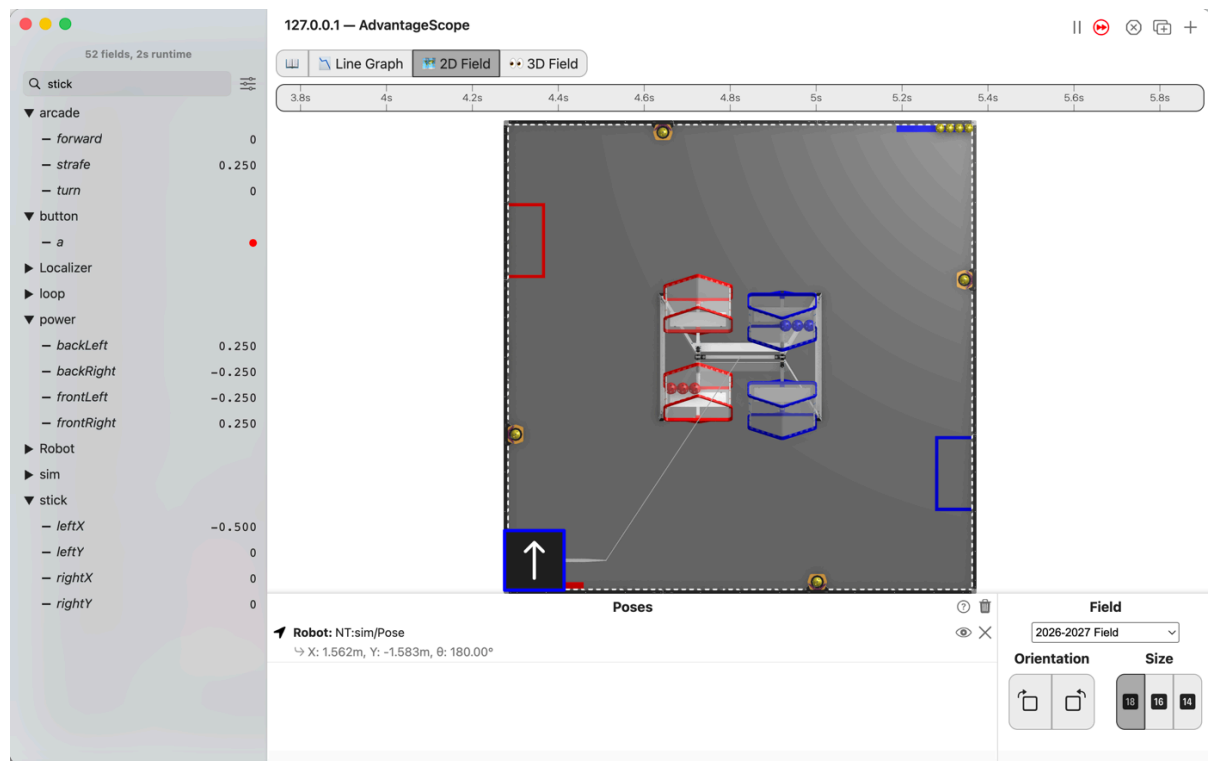


Figure 3.27. AdvantageScope's 2D Field with the robot `NT:sim/Pose` slid to its own left along the bottom edge, its nose still pointing the same way at 180.00° , and the list showing `arcade/strafe` at 0.250 and the four wheel powers at -0.250, 0.250, 0.250 and -0.250.

Drive a mecanum robot: the left stick drives and slides, the right stick turns, and all three can happen at once. Each wheel gets its own power.

Before you start.

- L080 is done, so `mytry` holds `L080ArcadeOpMode`.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot on blocks, as in L050, the Driver Station, and about 2 m by 2 m of clear floor for the last step.

3.8.1 The steps

L090S010: copy the starter into mytry

Copy. In `lessons`, click `L090MecanumOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L090MecanumOpMode`, and the new file is open.

L090S020: switch your copy on

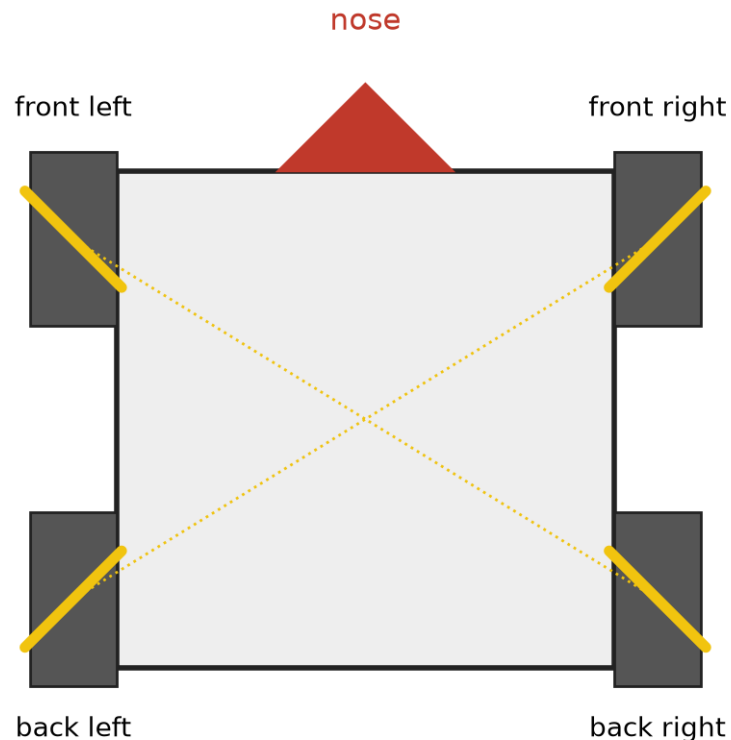
Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. 'public' allows access from any code.

L090S030: read the slide

A mecanum wheel has small rollers around it, set at 45 degrees. Spinning the wheel pushes the robot along and sideways at the same time. With four of them, the robot can drive, slide and turn.



the top roller of each wheel, at 45 degrees: together they make an X

Figure 3.28. The robot from above with its nose marked, and the top roller of each of the four wheels drawn as a line at 45 degrees, so the four lines make an X across the robot.

Seen from above, the top rollers make an X. Game Manual 0 shows it on [this page](#). A mecanum wheel is a left wheel or a right wheel, and the two are mirror images. goBILDA's page shows both: <https://www.gobilda.com/96mm-mecanum-wheel-set-70a-durometer-bearing-supported-rollers/>

The starter has forward and turn from L080. The slide is the third number, and it comes from the left stick, sideways.

Change. In `loop()`, find the comment that names this step. Under it, type:

```
// When on L090S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1090.html#1090s030
// for what to do here.
double strafe = squared(deadband(-leftStickX, band));
Tracker.publish("arcade/strafe", strafe);
```

Strafe is the robot word for sliding sideways. The stick reads above 0 to the right, and the minus sign makes a slide to the left positive, the same way a turn to the left is.

Check.

- Neither line has a red underline.

L090S040: give each wheel its power

Each wheel adds up the three numbers its own way:

Wheel	forward	strafe	turn
front left	+	-	-
front right	+	+	+
back left	+	+	-
back right	+	-	+

Change. Find the comment that names this step. Under it, type:

```
// When on L090S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1090.html#1090s040
// for what to do here.
double frontLeftPower = forward - strafe - turn;
double frontRightPower = forward + strafe + turn;
double backLeftPower = forward + strafe - turn;
double backRightPower = forward - strafe + turn;
Tracker.publish("power/frontLeft", frontLeftPower);
Tracker.publish("power/frontRight", frontRightPower);
Tracker.publish("power/backLeft", backLeftPower);
Tracker.publish("power/backRight", backRightPower);

hardware.frontLeft.setPower(frontLeftPower);
hardware.frontRight.setPower(frontRightPower);
hardware.backLeft.setPower(backLeftPower);
hardware.backRight.setPower(backRightPower);
```

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L090MecanumOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, and click **2D Field**. Open the arcade and power folders.

Check.

- Push the left stick halfway left. `strafe` reads 0.25.
- The front right and back left wheels read 0.25, and the other two read -0.25.
- The robot slides to its own left, and its nose keeps pointing the same way!
- Forward and turning work as they did in L080.

L090S050: scale all four

Push the left stick into a corner, all the way forward and all the way left. `forward` and `strafe` are both 1, so the front right and back left wheels ask for 2. That is twice what a motor has.

Change. Press Ctrl+C in the Terminal to stop the simulator. Go back to the four lines that make the wheel powers. Under them, type:

```
double frontLeftPower = forward - strafe - turn;
double frontRightPower = forward + strafe + turn;
```

```

double backLeftPower = forward + strafe - turn;
double backRightPower = forward - strafe + turn;
double biggest = Math.max(1, Math.max(Math.max(Math.abs(frontLeftPower),
Math.abs(frontRightPower)),
      Math.max(Math.abs(backLeftPower), Math.abs(backRightPower))));
frontLeftPower = frontLeftPower / biggest;
frontRightPower = frontRightPower / biggest;
backLeftPower = backLeftPower / biggest;
backRightPower = backRightPower / biggest;

```

This is L080's scaling, for four wheels. The two inner `Math.max` calls find the bigger of each pair, and the next one finds the biggest of all four. Type it as two lines, pressing Enter where the first one ends, and Android Studio moves the second one in for you.

Simulator. Run the same command again, and connect AdvantageScope.

Check.

- Push the left stick into the forward-left corner. The front right and back left wheels read 1, and the other two read 0.
- The robot drives diagonally, forward and to its left, with its nose still pointing forward.

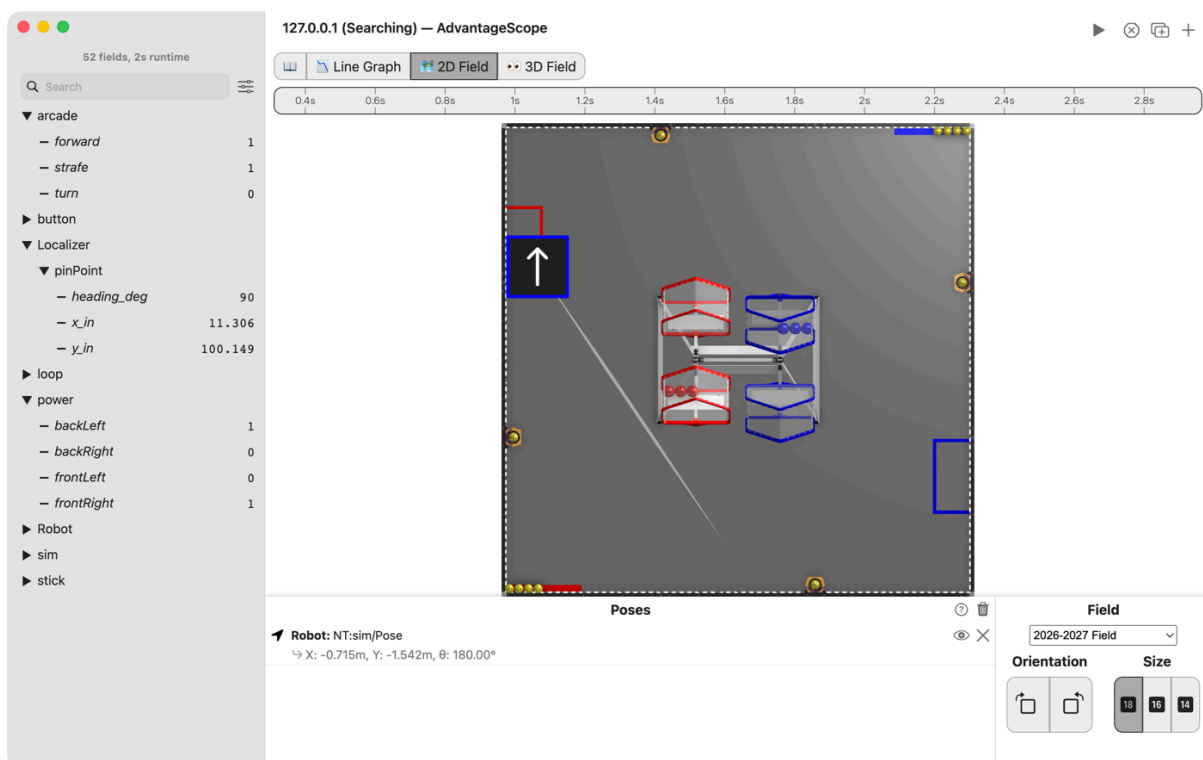


Figure 3.29. AdvantageScope's 2D Field and list with the left stick in the forward-left corner: arcade/forward and arcade/strafe at 1, power/frontRight and power/backLeft at 1, power/frontLeft and power/backRight at 0, and the robot's heading still 180.00°.

L090S060: check every wheel on blocks

A mecanum wheel spinning forward pushes the robot forward and sideways at once. The front left and back right wheels push forward and to the right. The front right and back left wheels push forward and to the left. Spinning backward, each pushes the opposite way. The robot moves the way the four pushes add up.

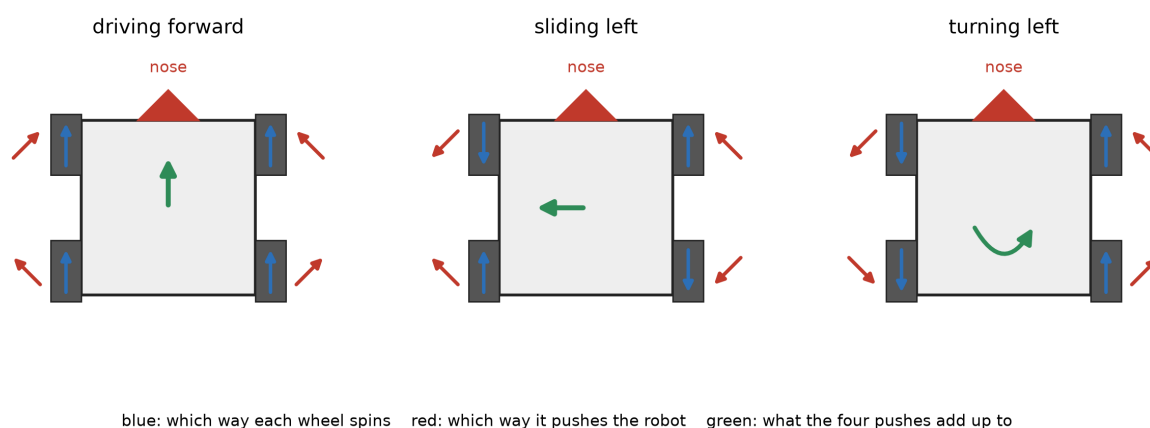


Figure 3.30. The robot from above, three times: driving forward, sliding left and turning left. On each wheel, an arrow for the way it spins and an arrow at 45 degrees for the way it pushes the robot, and in the middle one arrow for where the four pushes add up to.

Robot. Stop the simulator. Rest the robot on the blocks, so all four wheels are off the ground. Send the code, and run `L090 Mecanum` from the Driver Station.

Check.

- Push each way gently and watch the top of each tyre:

Push	front left	front right	back left	back right
left stick forward	forward	forward	forward	forward
left stick left	backward	forward	forward	backward
right stick left	backward	forward	backward	forward

Left stick forward: all four push forward, and their sideways pushes cancel. Left stick left: the forward and backward pushes cancel, and all four push left. Right stick left: the left side pushes backward and the right side forward, so the robot turns left.

Write down any wheel that turns the other way. A wheel that turns backward for every push needs its direction swapped, as in L050S070. A wheel that is right for forward but backward for a slide is on the other corner. Check the X from above.

L090S070: drive it on the floor

Drive. Press STOP. Lift the robot off the blocks and put it on the floor. Press INIT, then PLAY.

Check.

- Push the left stick left, and the robot slides to its own left with its nose still pointing the same way.
- Push it into a corner, and the robot drives diagonally.
- Push the right stick while sliding, and it turns as it slides.

No gamepad. `--args="mytry.L090MecanumOpMode --left_stick_x=-0.5"` slides it left for the first picture. `--left_stick_y=-1 --left_stick_x=-1` holds the corner for the second.

If it didn't, and the robot slides right when the stick goes left, the minus sign is missing from `-leftStickX`, or the strafe signs are swapped in the four lines.

If it didn't, and the robot turns or grinds instead of sliding, on the floor, a wheel is on the wrong corner. Forward still works with a wheel in the wrong place, because all four still push forward. It is the sideways pushes that stop cancelling. This is what the arithmetic says. No robot here has been driven with a wheel on the wrong corner.

What you just did. Three numbers, four wheels, one line each. Those four lines are the whole of mecanum drive, and the path follower in later lessons uses the same four. The scaling is L080's, for four wheels instead of two.

Where next. L110 hands the wheels to the path follower.

3.9 Watch the numbers while it drives

3.9.1 What you will have when this is done

The robot's own numbers on a screen while somebody drives it. You get a table of every value the lesson sends, a graph of any two of them, and a picture of where the robot thinks it is. After this, "it went crooked" is a graph you can point at. It stops being an argument.

3.9.2 Before you start

- L060 is finished, because step 3 graphs the two numbers L060's squaring makes.
- A robot with its battery in, running a lesson.
- A laptop or a phone that can join the robot's Wi-Fi.

3.9.3 The steps

Step 1: open Panels

Join the robot's Wi-Fi, then open `http://192.168.43.1:8001`.

Panels is a web page the Robot Controller app serves. Nothing in the lesson starts it and nothing turns it on.

You'll know it worked when the page loads and shows its panels, even before an OpMode runs.

If it didn't, and the page never loads, the laptop is on the wrong Wi-Fi. Check which network it joined.

If it didn't, and you get a different page full of tuning buttons, that is AutoTune on port 10158. It is a second server in the same app, and it is not this.

Step 2: find the numbers your lesson sends

Run L060 Shaping, the copy finished in L060, and press PLAY. The Telemetry panel fills with one line per value, each shaped `name: number`.

The Graph panel has a list to pick from, and a name only joins that list **after it has been sent once**. So start the OpMode first, then go looking. The list is flat, in the order the names first arrived.

You'll know it worked when `speed/left` and `power/left` are both in the list, and both numbers move when you move the left stick. The [cheat sheet](#) lists every name each lesson sends.

If it didn't, and a name you expected is missing, nothing sent it. Either that lesson does not publish it, or the line that publishes it is not typed yet.

Step 3: graph the two against each other

Pick `speed/left` and `power/left` and push the stick slowly from nothing up to full.

You'll know it worked when the two lines leave zero together and the shaped one stays under the raw one the whole way up, until they meet at full stick. That is L060 drawn: squaring a number below 1 makes it smaller.

If it didn't, and the two lines lie exactly on top of each other, nothing is being shaped. That is L060S060 not typed yet.

If it didn't, and both lines look like a staircase rather than a curve, that is the sampling and it is

normal. The robot sends to Panels every 75 ms and drops the loops in between, so with a loop of about 5 ms, roughly one loop in 15 reaches the graph. For the same reason one slow loop will not show in `loop/period_ms`. Graph `loop/max_period_ms` instead: it only ever steps up, and a step means some loop blocked.

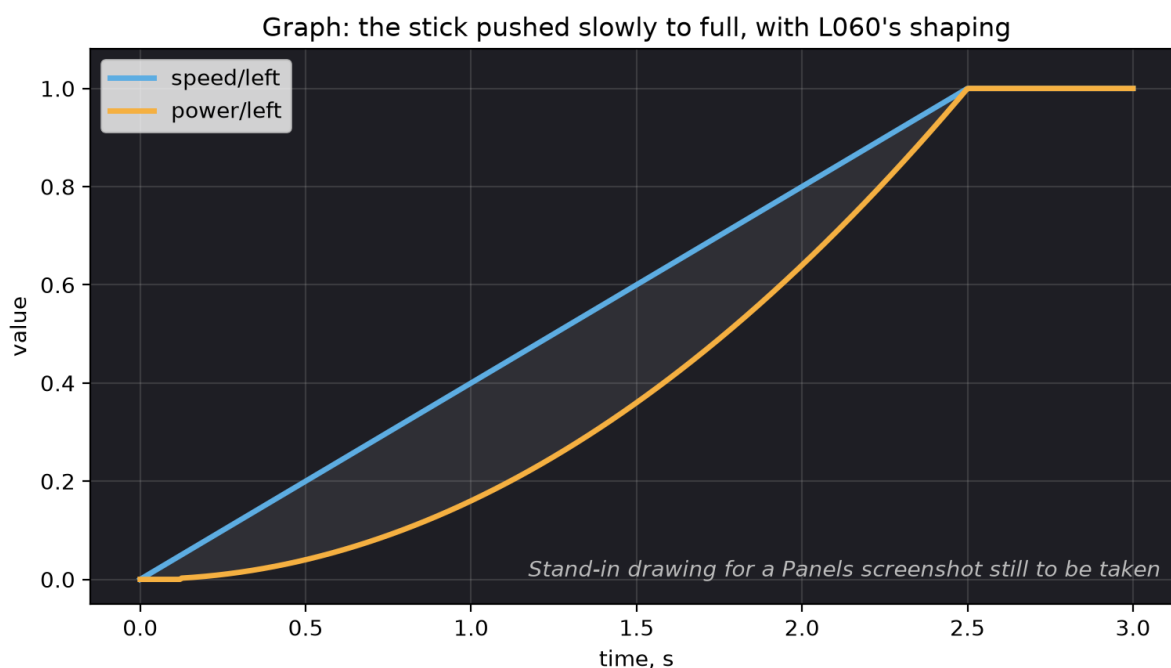


Figure 3.31. The Panels Graph panel with `speed/left` and `power/left` both plotted while the stick is pushed slowly to full, so the gap between the two curves is the thing a student is looking for.

Step 4: look at where it thinks it is

Open the Field panel. There is nothing to set up: the lesson sends the field's coordinates with every frame.

A blue square 18 in by 18 in is the robot, and the line out of it is the way the robot faces. A grey line behind it is where it has been since the OpMode started. In a lesson that follows a path you also get an orange line for the piece of path being followed, and a green square for the nearest point on that path at the heading the path wants there.

You'll know it worked when the blue square moves as the robot moves. Turn the robot, and the line out of the square swings round with it.

If it didn't, and the robot draws in a corner with most of the square off the field, nothing is wrong. Pedro puts (0, 0) in a corner and the field runs 0 in to 144 in, so a lesson that never sets a pose starts there.

If it didn't, and there is no green square or orange line, you are in a teleop lesson. There is no path to draw. You get the robot and its trail.

3.9.4 What you just did

Panels has no separate graph call. It reads the telemetry text, and any line shaped `name: number` becomes a table row and a graph series at once. That is why one `Tracker.publish` call gets you both, and why a name may hold anything except a colon.

The same call also writes the number into the flight log on the robot. Panels is the live view and the log is the recording, and they carry the same names, so a graph you found while driving is a graph you can find again afterwards.

The gap between the blue square and the green one is worth knowing. The green square is the nearest

point on the path, not where the robot should have got to by now. So the gap says how far off the line the robot is and how crooked it is sitting. It does not say whether the robot is running late.

3.9.5 Where next

- [task.bench](#) is the other half of this. It runs the same lessons on a laptop, with no robot at all.

3.10 Check your work without a robot

3.10.1 What you will have when this is done

Two ways to try a lesson with no robot in front of you. One runs every test and tells you which lessons are finished. The other runs a single lesson on the laptop and draws the robot moving, so you can see what your code does before it is your turn at the field.

3.10.2 Before you start

- Android Studio opens the project, and you can type a Gradle command into its terminal. Ask a mentor to get a laptop that far.
- [L040](#) is written, or step 2 has no lesson to show you.
- AdvantageScope installed on the same laptop, for step 2 and step 3.

3.10.3 The steps

Step 1: run the tests and read the report

```
./gradlew :TeamCode:test
```

That runs the tests for everything the lessons are built on, and leaves the lesson tests out. It says so while it runs:

```
Skipping org.firstinspires.ftc.teamcode.lessons.*. Run them with
:TeamCode:testLessons.
```

You'll know it worked when it ends with BUILD SUCCESSFUL. Nothing underneath the lessons is broken, so whatever fails next is yours.

If it didn't, and it says SDK location not found, the laptop has no `local.properties` file yet. That file is not kept in git, and Android Studio writes it the first time it opens the project. Open the project once, then run the command again.

Now the tests for your own work:

```
./gradlew :TeamCode:testLessons
```

On a fresh copy, where nobody has written a lesson yet, the report counts 44 tests and 44 failures. Forty-four failures is not a broken copy. Each one is a lesson nobody has written, and the count drops as you write them.

You'll know it worked when the last lines name a report:

```
> There were failing tests. See the report at: file:///.../index.html
```

Open that link in a browser, click the class and then the test, and you get the message the test wrote. The console does not give you that. All the console gives you is this:

```
L120EncoderLocalizerTest > countsStraightAhead FAILED
    java.lang.AssertionError at L120EncoderLocalizerTest.java:61
```

A file and a line, and no word about what went wrong. The report for that same run also holds robot relative while held expected:<1.0> but was:<5.551115123125783E-17>, which is L150's starter copied and not yet changed. Every *if it didn't* line in this guide quotes a message out of the report, so every one of them starts here.

Step 2: watch one lesson move

A test says where the robot ended up. It does not show you the robot. For that, run one lesson on its own:

```
./gradlew :TeamCode:simRun --args="lessons.L040TankOpMode --left_stick_y=-1
--right_stick_y=-1"
```

The first word inside the quotes is the lesson's class name, with the package it sits in before it. Each option after it holds one gamepad control at one value for the whole run, named the way the gamepad names it. So `--left_stick_y=-1` is the left stick pushed fully forward, and `--a=true` is the A button held down. Ask for `--args="--help"` and it lists every control you can set.

To push the sticks yourself, plug a gamepad into the laptop and add `--pad`:

```
./gradlew :TeamCode:simRun --args="lessons.L040TankOpMode --pad"
```

One gamepad becomes `gamepad1` on its own. With two plugged in, hold Start and press A on the one you want to drive with.

Plug it in before the run or during it, whichever suits. Pull it out and the sticks go back to rest, so the robot stops. Plug it back in and it is yours again.

Use `--pad`, or set the controls yourself, but not both. Together they are an error, and the run names the setting that clashed.

It prints where the flight log goes, then runs:

```
Flight log: <the repository's top folder>
lessons.L040TankOpMode running. Connect AdvantageScope to 127.0.0.1 as
NetworkTables 4, and Ctrl-C to stop.
```

Open AdvantageScope and connect to 127.0.0.1 as NetworkTables 4. The topics arrive under `sim/`. Add a 2D field and give it `sim/Pose`.

You'll know it worked when the robot moves. `sim/wheels/frontLeft` and its three neighbours say what reached each motor, and `sim/vel/forward_ips` says how fast the robot is going.

If it didn't, and the robot sits still with all four wheels reading 0, that lesson is still blank. Watched with the old L2b lesson's blanks open and both sticks forward: the pose never changed for the whole run. Nothing is wrong with the simulator. It ran a lesson that writes nothing to the motors.

If it didn't, and no topics arrive at all, nothing connected. The address is 127.0.0.1, and the kind is NetworkTables 4 rather than a log file.

Ctrl-C stops it. The flight log is in the project's top folder, which the first line it prints names.

Which way round the field gets drawn

The corner the arrow starts in and the way it points do not agree with the numbers going out. Trust that the robot moves, and how far it moves, rather than which way it faces on the screen.

Step 3: open the log a failing test left

Some test classes hand you their flight log when they fail. Watched, by asking the simulator's own test for 400 in when the robot drives 54.75 in:

```
java.lang.AssertionError: drove forward, and got a fair way: 54.75097300967889
AdvantageScope can open what ran:
```

```
/var/folders/.../Tank-20260928-065400.wpilog (37502 bytes)
```

You'll know it worked when that file opens in AdvantageScope as a log, not as a connection, and holds the run up to the moment the test gave up. There is more in it than `simRun` sends out: `/Robot/Pose`, `/Robot/Speeds`, `/Robot/Twist`, all four wheels, and the loop timing.

If it didn't, and the failure names no log, that class never asked for one. The lesson tests are all like that today. So `l2b_theSticksMoveTheSimulatedRobot` fails with `drove forward`, and got a `fair way: 0.0` and no path to open. A log does get written, and it holds 0 bytes, because nothing closed it. Watched. Use step 2 instead, and tell a mentor.

3.10.4 What you just did

Both halves run the same model of the robot, and it is a small one. Each loop, the four wheel powers are mixed back into forward, sideways and turn. Each of those three chases what was asked of it with a 0.15 s lag. The answer is added up into a pose.

Axis	Flat out
Forward	64.4 in/s
Sideways	43.6 in/s
Turn	4.0 rad/s

So the model has no slip, no scrub, no sagging battery, no floor, no field wall, no motor wired backwards and no Pinpoint. It will drive straight through the perimeter and never mention it, and it cannot tell you a path is too fast for the wheels. When the simulator and the robot disagree, the robot is right.

That is why a green test run is not a finished lesson. These two catch the mistakes that live in the arithmetic, which are most of them. What is left over is the kind of mistake only a robot on a floor will show you, and that is why every lesson's last step is a robot on a floor.

3.10.5 Where next

- [L110](#) hands the wheels to the path follower. From here on you can try each lesson on the laptop before you take it to the field.

3.11 L110: hand the wheels to the path follower

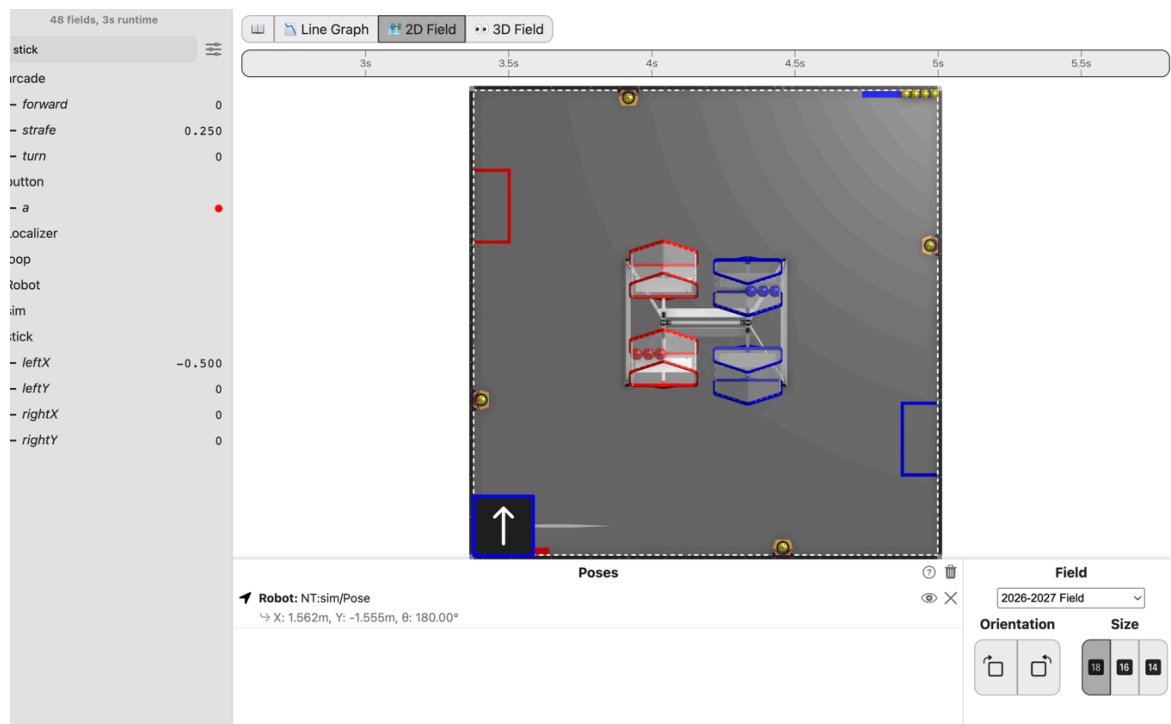


Figure 3.32. AdvantageScope's 2D Field with the robot `NT:sim/Pose` slid to its own left along the bottom edge, its nose still pointing the same way at 180.00° , and the list showing `arcade/strafe` at 0.250, as in L090, with the path follower holding the drivetrain.

The robot drives on the sticks as it did in L090, and the path follower is holding the drivetrain. The sticks still decide where the robot goes. The path follower is Pedro Pathing's code that drives the robot along a path; from L125 on, it drives the robot itself.

Before you start.

- L090 is done, so `mytry` holds `L090MecanumOpMode`.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot, the Driver Station, and about 2 m by 2 m of clear floor for the last step.

A follower holding the drivetrain knows where the robot is. It can hold a heading, and it can drive a path. Up to L090 the follower was handed a stand-in that ignored it, so the lesson had the motors to itself. L120 onwards need the follower to hold the real one.

3.11.1 The steps

L110S010: copy the two starters into mytry

This lesson has two files. One is the `OpMode`. The other is the drivetrain: the code that turns three numbers into four wheel powers.

Copy. In `lessons`, click `L110FollowerDriveTrain` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**. Do the same for `L110FollowerOpMode`.

Check.

- mytry holds L110FollowerDriveTrain and L110FollowerOpMode.

L110S020: switch your copy on

Change. In L110FollowerOpMode, delete the line @Disabled, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this OpMode. `'public'` allows access from any code.

L110S030: write the mix

Open L110FollowerDriveTrain. It extends CorbelsDriveTrain, which means it gets everything CorbelsDriveTrain already has. That class sets each motor's direction, scales the powers as L090 did, and sends them to the motors. The one part it leaves to you is mix.

wheels is an **array**: four numbers in a row, in one variable. wheels[FL] is the front left wheel's slot, and FR, BL and BR name the other three.

Change. In mix, find the comment that names this step. Under it, type L090's four lines, each into its own slot:

```
// When on L110S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l110.html#l110s030
// for what to do here.
wheels[FL] = forward - strafe - turn;
wheels[FR] = forward + strafe + turn;
wheels[BL] = forward + strafe - turn;
wheels[BR] = forward - strafe + turn;
```

Check.

- No line has a red underline, and FL, FR, BL and BR turn purple.

L110S040: make room for the drivetrain

Change. Open L110FollowerOpMode. Near the top of the class, find the comment that names this step. Under it, type:

```
// When on L110S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l110.html#l110s040
// for what to do here.
L110FollowerDriveTrain drivetrain;
```

This is a field, like previousButtonA in L070. It holds the drivetrain from init until the OpMode stops.

Check.

- The line has no red underline.

L110S050: hand the drivetrain to the follower

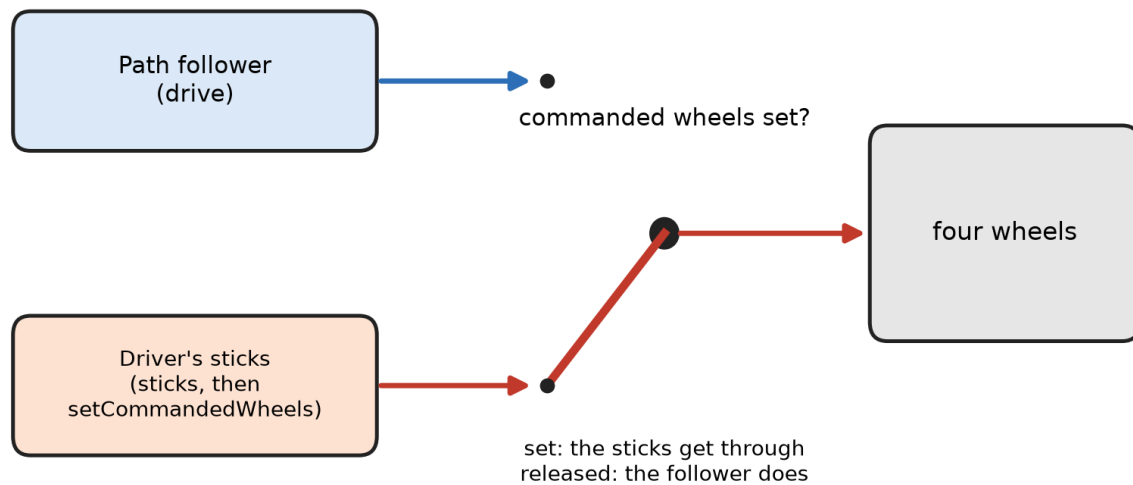


Figure 3.33. Two arrows reaching the same four wheels, one from the path follower and one from the driver's sticks, with the commanded-wheels switch between them showing which one gets through.

Change. In `init()`, find the comment that names this step. Under it, type the first line below. Then change `initAfter();` to `initAfter(drivetrain);`:

```

// When on L110S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l110.html#l110s050
// for what to do here.
drivetrain = new L110FollowerDriveTrain(hardware);

initAfter(drivetrain);

```

`new` makes the drivetrain. `initAfter` makes the path follower, and the follower holds what it is handed. Up to L090, `initAfter()` was handed nothing, because the lesson drove its own wheels.

The four `setDirection` lines from L050 are gone from `init`. The drivetrain sets each motor's direction itself, from the same config.

Check.

- Neither line has a red underline.

L110S060: drive with the sticks

Change. In `loop()`, find the comment that names this step, under the `strafe` lines. Under it, type:

```

// When on L110S060, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l110.html#l110s060
// for what to do here.
drivetrain.sticks(forward, strafe, turn);

```

`sticks` sends the three numbers through your mix. Then it scales them and saves them. At the end of the loop, `loopAfter()` updates the follower, and the follower drives the wheels.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L110FollowerOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `arcade` folder.

Check.

- Push the left stick halfway left. `strafe` reads 0.25, and the robot slides to its own left with its nose pointing the same way.
- Forward, turning and the corners all work as they did in L090.

L110S070: drive it on the robot

Robot. Stop the simulator. Send the code, and run `L110 Follower` from the Driver Station.

Check.

- It drives the way L090 did: forward, sideways, turning, and the diagonals.

No gamepad. `--args="mytry.L110FollowerOpMode --left_stick_x=-0.5"` slides it left for the picture at the top. The simulator has no walls, so a robot left running slides off the field.

If it didn't, and the robot does not move at all, `initAfter` is still `initAfter()`. The sticks command the drivetrain, but nothing drives it, because the follower is holding a stand-in. `LessonsTest.l110_theFollowerDrivesTheSticks` fails with the follower slides it left: `front left expected:<-0.25> but was:<0.0>`.

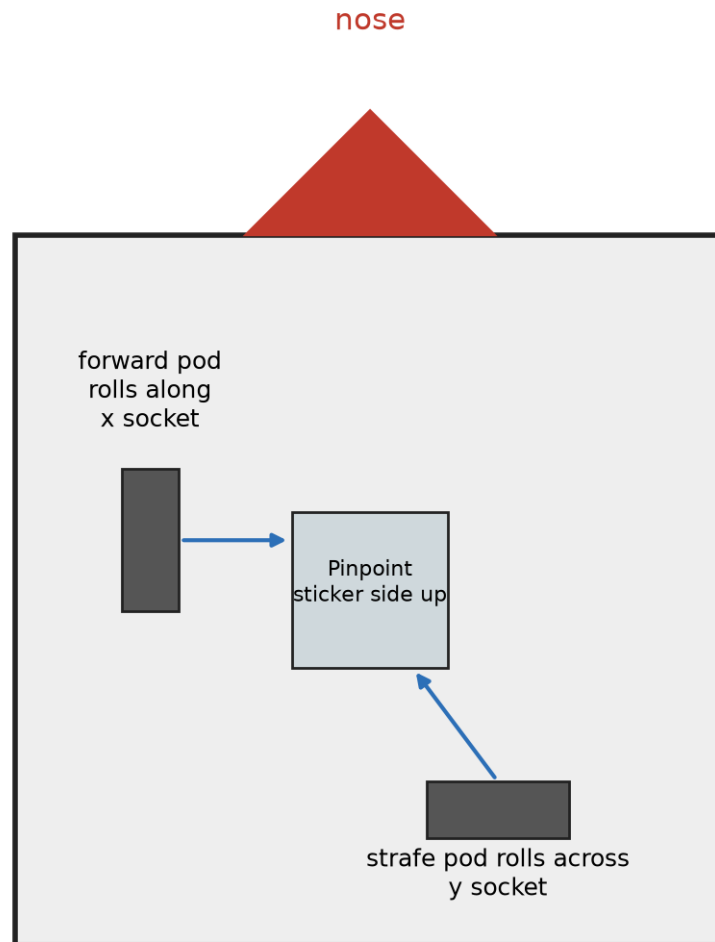
If it didn't, and it slides the wrong way or turns when it should slide, a sign in `mix` is wrong. `LessonsTest.l110_mixGivesEachWheelItsOwnSum` names the wheel and the move, such as `strafe left: back left`.

What you just did. The drivetrain asks one question every time the follower updates it: has the lesson commanded the wheels? If so, it uses what the lesson commanded. If not, it mixes what the follower asked for. Either way the three numbers go through the same `mix`, so the sticks and the follower agree about which wheel does what.

That door is ours, not Pedro's. Pedro's own mecanum drivetrain does only what the follower says. Ours lets the driver reach past it to the wheels, which the teleop lessons after L120 need. The rest of `CorbelsDriveTrain`, which answers everything else Pedro's follower asks a drivetrain, is written out with a comment above each method.

Where next. L115 mounts the Pinpoint, which the follower uses to know where the robot is.

3.12 L115: mount the Pinpoint and plug it in



Stand-in drawing for a photo still to be taken

Figure 3.34. The Pinpoint on the robot from above, with the forward pod and the strafe pod drawn at right angles to each other, each labelled with the socket it plugs into, and the sticker side marked facing up.

The Pinpoint is bolted on, its two pods are in the right sockets, and the Control Hub finds it when you press INIT. The robot does not yet know how far the pods sit from its middle. L117 measures that.

Before you start.

- L110 is done, so the follower holds the drivetrain. The follower is what builds the Pinpoint.
- The robot, the two odometry pods, and whatever driver the robot's own screws take.
- The I2C cable the Pinpoint came with, and the Driver Station.

The three rules in L115S010 and L115S020 are Pedro's, from [its Pinpoint page](#).

3.12.1 The steps

L115S010: put one pod along the robot and one across it

The Pinpoint counts two wheels that nothing drives. One rolls along the length of the robot, and that one is the **forward** pod. The other rolls across it, and that one is the **strafe** pod.

Both have to touch the floor the whole time, and neither may be able to lift. A pod that skips loses count, and nothing after it can tell that it did.

Mount. Bolt the two pods on, one along the robot and one across it. Plug the forward pod into the Pinpoint's x socket and the strafe pod into its y socket.

Check.

- Push the robot forward by hand. The forward pod's wheel turns.
- The strafe pod's wheel drags sideways.

L115S020: plug it into an I2C port that is not port 0

The Pinpoint talks to the Control Hub over I2C. Port 0 already has the Control Hub's own IMU on it, so the Pinpoint goes on another port.

Mount. Bolt the Pinpoint on with its sticker side, the side the ports are on, facing **up**. Plug its cable into I2C port 1, 2 or 3 on the Control Hub.

Configure. On the Driver Station, open the robot's configuration. Add the Pinpoint on that port, and name it `pinpoint`, spelled exactly that way. That is the name the code asks for:

```
c.name.set("pinpoint");
```

Check.

- Pick L110 Follower from the Lessons group and press INIT.
- INIT finishes with no error.

INIT looks for the Pinpoint by its name, so if INIT finishes, it was found.

L115S030: hold the robot still while it inits

Pressing INIT builds the follower, and the follower builds the Pinpoint. The Pinpoint then resets its own gyro, which takes about a quarter of a second. The robot has to sit still for it.

Robot. Put the robot on the floor, step back, safety glasses on! Pick L110 Follower on the Driver Station and press INIT, but not PLAY yet. Watch it in AdvantageScope.

Check.

- Before you press PLAY, `Localizer/pinPoint/heading_deg` holds steady while the robot sits still.

Drive. Press PLAY and drive a slow square.

Check.

- The heading comes back to where it started.

If it didn't, and a pod wheel does not turn at all in L115S010, it is not reaching the floor. Most mounts have a slot for that.

If it didn't, and the Driver Station shows this at INIT, the name in the configuration is not the name in the code:

```
Unable to find a hardware device with name "pinpoint" and type
```

GoBildaPinpointDriver

A still robot cannot show whether the numbers are live; the square shows that.

If it didn't, and the heading crawls one way while the robot sits still, INIT happened while the robot was moving. Press STOP and INIT again without touching it.

If it didn't, and forward and sideways come out swapped once L117 has tuned the pods, the two pods are in each other's sockets. L115S010 is where that gets fixed, and no amount of tuning works around it.

Where the rest comes from. The missing-device message is the one `HardwareMap.get` throws in the SDK. The reset at INIT is `PinpointConfig's` default `resetMode`, which `PinpointLocalizer's` constructor acts on. The still robot and the quarter second are from the goBILDA driver's own javadoc.

What you just did. Every lesson from L120 on believes the Pinpoint. L120 puts a second way of tracking the robot beside it and shows the drive encoders drifting away from it over a run. L195 measures the robot against it. A pod in the wrong socket makes all of that quietly wrong instead of loudly broken.

The Pinpoint needs nothing from the drive motors to know where the robot is. Its two wheels are not driven, so a mecanum roller cannot scrub them sideways and a slipping wheel cannot spin them.

Where next. L117 lets AutoTune tune the robot.

3.13 L117: let AutoTune tune the robot

The robot measures itself. It finds which way each motor turns and where the pods sit. It finds how fast it can drive and how hard it can brake. Pedro Pathing's AutoTune does the measuring. You paste what it finds into `Constants.java`.

Before you start.

- L115 is done, so the Pinpoint and its two pods are mounted and plugged in, and L050's motor directions are set.
- The robot, the Driver Station, a laptop that can join the robot's Wi-Fi, and blocks to lift the robot's wheels off the floor.
- Clear floor for L117S040: at least 48 in ahead of the robot, 48 in to its left and 36 in to its right, plus a tile of room all round.

The step names and the messages quoted are from the tuner code in `pedro/procedures/`, and the address from Pedro's [Mecanum page](#).

3.13.1 The steps

L117S010: open AutoTune

AutoTune is a web page the robot serves. It is not in the Driver Station's list of OpModes. It starts its own OpMode, called `Pedro Tuning`, when a step needs the robot to move.

Send. Send the code to the robot. Join the laptop to the robot's Wi-Fi. In a browser, open `http://192.168.43.1:10158`.

The first time the browser talks to the robot, the operating system might ask for your permission. Please give it.

Check.

- The page lists four tuners: Mecanum Tuner, Tests, Pinpoint Tuner and Foresight Tuner.

L117S020: find which way each motor turns

Put the robot on blocks, so its wheels spin freely.

Robot. Pick `Mecanum Tuner`. Type the four motor names as the Robot Configuration spells them: `Front Left`, `Front Right`, `Back Left` and `Back Right`. Each motor then spins on its own, at half power, beside a picture of which wheel it is. Say whether it spun, and whether it spun forward or reversed.

Change. At the end, AutoTune shows Java that starts `public static MecanumConfig drivetrainConfig`. Open `Constants`, in the `pedro` folder. Find the eight lines that set the names and directions. Put the eight AutoTune shows in their place. Keep `manualBrakeMode` and `powerThreshold`. AutoTune does not print them.

Check.

- The directions match what L050 found on blocks.

L117S030: measure the pods

The Pinpoint needs four more numbers: which way each pod counts, and how far each one sits from the middle of the robot. The `x` offset is how far sideways the forward pod is from the point the robot turns about, and left counts up. The `y` offset is how far forward the strafe pod is, and forward counts up. This robot's are about 1.06 in and 3.36 in.

Put the robot on the floor.

Robot. Pick `Pinpoint Tuner`. Keep the name `pinpoint` and the pod type it offers. Then do the three pushes it asks for, stopping each one when you are done:

- Push the robot forward.
- Push the robot to the left.
- Spin the robot half a turn on the spot, counterclockwise.

Change. Replace `localizerConfig` in `Constants` with the `PinpointConfig` AutoTune shows.

Check.

- In `Tests`, pick `Localization Test` and drive with the gamepad.
- Driving forward raises `x`, and strafing left raises `y`.

L117S040: let the robot find its speed and its brakes

This one drives itself. Clear the floor first.

Robot. Pick `Foresight Tuner`. Keep the distances and the speed it offers: 48 in, 30 in/s and 36 in. It drives forward and left as fast as it can, coasts to a stop, turns back and forth, spins, brakes, and steps 12 in to 24 in each way.

Change. Replace `foresightConfig` in `Constants` with the `ForesightConfig` AutoTune shows. Keep the five lines it does not print, from `translationalConstraint` to `holdPointHeadingScaling`.

Three of its numbers are kept twice in `Constants`, for the lessons' own drivetrain. Copy each one up into its own line:

- the number in `c.coast.set(...)` goes in `powerPerInchPerSecond`;
- `maxAchievableForwardVelocity` goes in `maxForwardInchesPerSecond`;
- `maxAchievableStrafeVelocity` goes in `maxStrafeInchesPerSecond`.

Check.

- Each of the three pairs holds the same number.

L117S050: watch it drive a line

Send. Send the code again, so the new `Constants` is on the robot.

Robot. In `Tests`, pick `Line Test`, with the distance at 48.

Check.

- The robot drives 48 in and stops on the line.

No gamepad, except in L117S030's check. `AutoTune` runs its own `OpMode`, and nothing here runs in the simulator.

If it didn't, and the Mecanum Tuner stops with this, a motor other than the one named `spun`, or none did:

The wrong motor spun. Check that your motors are plugged into the correct ports, **and** that they are configured correctly. Then, **try** again.

It is the message for an answer of No to “Did the front left motor spin?”, read from `MecanumTuner.java`. Swap two motor names and each of those motors is given the other's direction, so both wheels turn backwards. L050's checks on blocks cannot tell that from a backwards wheel. This one can.

If it didn't, and `x` and `y` swap over in the Localization Test, the two pods are in each other's sockets. L115S010 is where that gets fixed, and no tuning works around it.

If it didn't, and the robot brakes late or never holds still after L117S040, check that the five hold lines are still in `foresightConfig`.

What you just did. Every number in `Constants.java` either came off this robot or is waiting to. `AutoTune` measures the ones that need the robot driving itself. L195 measures the two it does not: ticks per inch and the turn radius.

Where next. L120 compares the Pinpoint with the drive encoders and logs both.

3.14 L120: work out where the robot is from the drive wheels

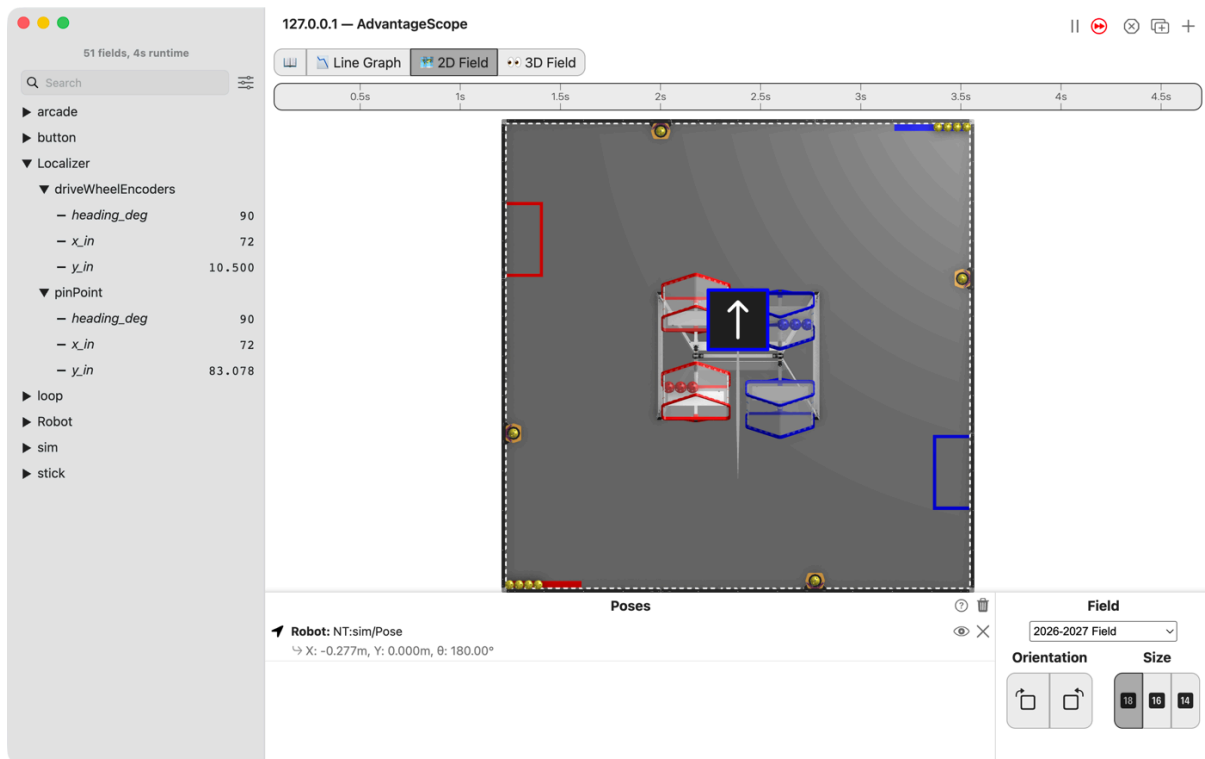


Figure 3.35. AdvantageScope's 2D Field and list with the Localizer folder open: pinPoint/y_in at 83.078 while the robot drives forward, and driveWheelEncoders still at x 72, y 10.5 and heading 90, where it started.

Your own localizer runs beside the robot's own, and both are published so they can be graphed against each other. Yours counts how far the four drive wheels roll. The robot's own counts the Pinpoint's two pods. Neither is the truth. A tape measure is.

Before you start.

- L110 is done, so the follower holds a real drivetrain, and L115 is done, so the Pinpoint is on the robot.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot on blocks, as in L050, the Driver Station, a tape measure, and about 3 m of clear floor for the last step.

The simulator turns each wheel's encoder by how far the wheel rolls, at `Constants.ticksPerInch`, and turns the IMU with the robot. It has no slip, so a finished localizer follows the Pinpoint there more closely than it will on the robot. The tests check the arithmetic step by step.

3.14.1 The steps

L120S010: copy the two starters into mytry

This lesson has two files. One is the OpMode. The other is the **localizer**: the code that works out where the robot is.

Copy. In lessons, click `L120CompareLocalizersOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click

Add. Do the same for `L120EncoderLocalizer`.

Check.

- `mytry` holds `L120CompareLocalizersOpMode` and `L120EncoderLocalizer`.

L120S020: switch your copy on

Change. In `L120CompareLocalizersOpMode`, delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. `'public'` allows access from any code.

L120S030: run your localizer beside the robot's own

The robot's own localizer reads the Pinpoint. A **shadow** localizer runs beside it. It is updated and published every loop, but it never steers the robot.

Change. In `L120CompareLocalizersOpMode`, find `shadowLocalizers()` and the comment that names this step. Under it, type:

```
// When on L120S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
/l120.html#l120s030
// for what to do here.
shadowLocalizers.add("driveWheelEncoders",
    new L120EncoderLocalizer(new HardwareWheelSource(hardware)));
```

`HardwareWheelSource` turns red, because this file does not know where it lives yet.

- Click `HardwareWheelSource`.
- Press Option+Enter on a Mac, Alt+Enter on Windows, and pick **Import class...**
- Android Studio adds an `import` line at the top of the file, and the red goes away.

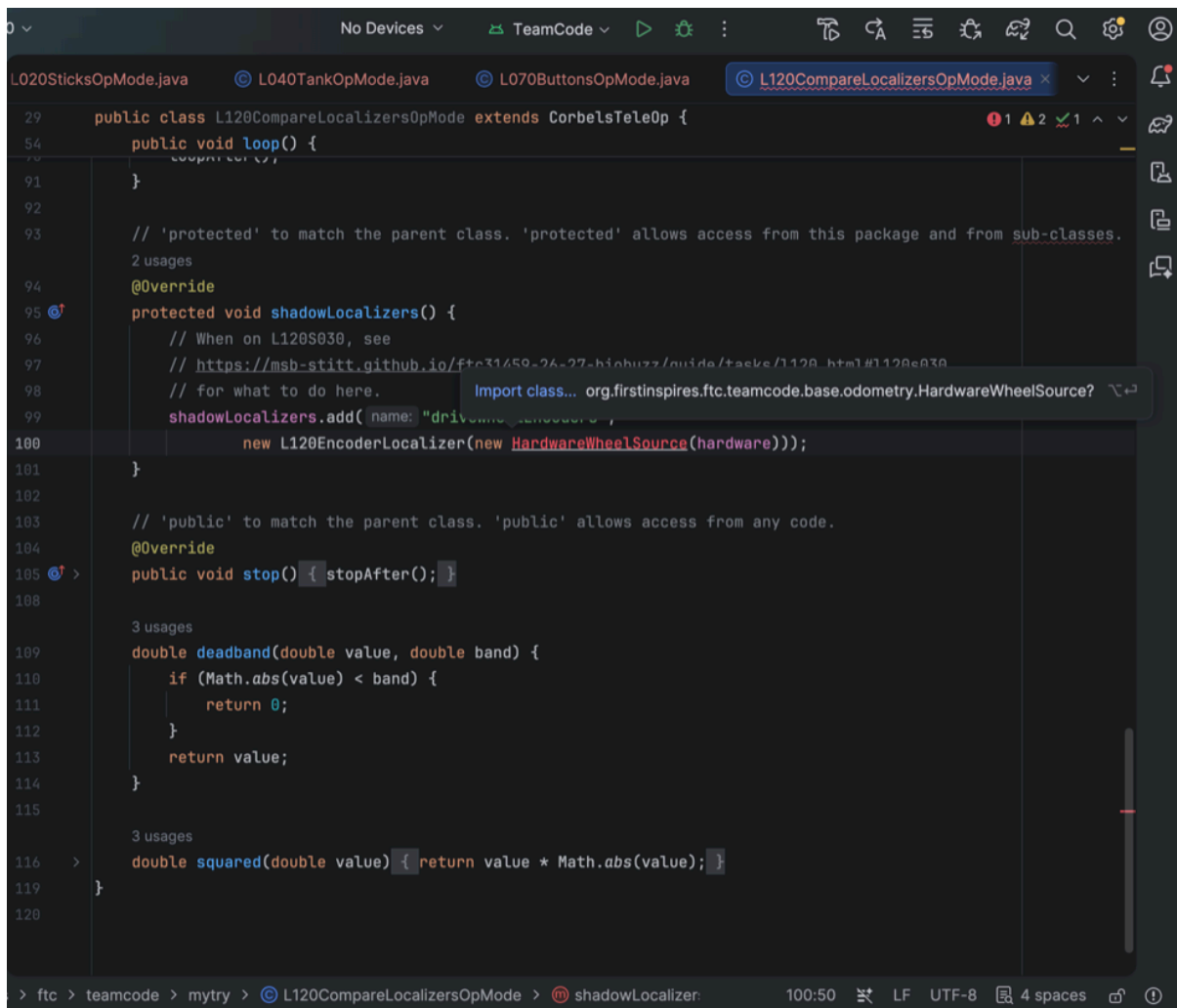


Figure 3.36. Android Studio with `HardwareWheelSource` underlined in red and the popup offering `Import class... org.firstinspires.ftc.teamcode.base.odometry.HardwareWheelSource?` with `\n↵`.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L120CompareLocalizersOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020. Open the `Localizer` folder, then `driveWheelEncoders` and `pinPoint` inside it.

Check.

- Both start at `x_in 72`, `y_in 10.5` and `heading_deg 90`.
- Push the left stick forward. `pinPoint/y_in` climbs, and `driveWheelEncoders` stays where it started. Yours does not count anything yet.

`x_in` is how many inches the robot is from the left edge of the 2D Field, and `y_in` how many from its bottom edge. `heading_deg` is the way the robot's nose points, and 90 is straight up the field. Since L040, the `Poses` list has called that same direction 180°. Both mean up the field.

L120S040: add up how far the wheels rolled

`update` runs once a loop. Each time, it can only see how far each wheel has rolled in total, so it remembers last loop's numbers and takes them away.

The first loop has nothing to take away from, so it only remembers.

Change. Press Ctrl+C in the Terminal to stop the simulator. Open L120EncoderLocalizer. In update, find the comment that names this step. Under it, type:

```
// When on L120S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l120.html#l120s040
// for what to do here.
if (lastWheels == null) {
    lastWheels = wheels.clone();
    lastRawHeading = rawHeading;
    return;
}

double frontLeft = wheels[0] - lastWheels[0];
double frontRight = wheels[1] - lastWheels[1];
double backLeft = wheels[2] - lastWheels[2];
double backRight = wheels[3] - lastWheels[3];

double forward = (frontLeft + frontRight + backLeft + backRight) / 4;
double strafe = (-frontLeft + frontRight + backLeft - backRight) / 4;

x += forward;
y += strafe;
```

Android Studio types the } for you when you press Enter after the {. Type the rest below it.

null means “nothing yet”. .clone() makes a copy, so next loop’s numbers do not change it. return leaves update early, until next loop.

The two long lines are L090’s table read backwards. There, three numbers made four wheel powers. Here, four wheel moves make forward and strafe again, in inches.

Check.

- In the Terminal, run this. It prints 8 tests completed, 2 failed, then BUILD FAILED and more red lines. That is what failing tests look like here, and two are meant to fail now.

```
./gradlew :TeamCode:testDebugUnitTest --tests '*L120EncoderLocalizerTest*'
```

The two that fail are appliesTheHeadingSoSidewaysIsRelativeToTheRobot and setPoseMovesTheEstimateWithoutMovingTheRobot. Both are about which way the robot faces. That is the next step.

L120S050: turn the step to face the way the robot faces

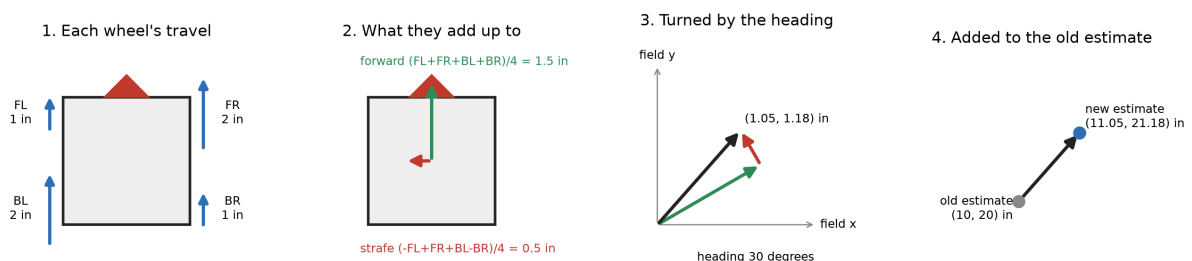


Figure 3.37. One loop of dead reckoning: the four wheel travels since the last loop, the forward and left arrows those add up to, the same pair turned to point along the robot’s heading, and the old estimate with that pair added on the end.

forward is forward for the robot, not for the field. The robot starts facing along the field’s y, so its forward is the field’s y. L120S040 added it to x.

Change. Delete the two lines `x += forward;` and `y += strafe;`. Find the comment that names this step. Under it, type:

```
// When on L120S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l120.html#l120s050
// for what to do here.
double heading = rawHeading + headingOffset;
x += forward * Math.cos(heading) - strafe * Math.sin(heading);
y += forward * Math.sin(heading) + strafe * Math.cos(heading);
```

`Math.cos` and `Math.sin` share the step out between `x` and `y`. Facing along `x`, `cos` is 1 and `sin` is 0, so forward all goes on `x`. Facing along `y`, `cos` is 0 and `sin` is 1, so forward all goes on `y`. In between, each gets a part.

heading counts in **radians**, not degrees. One whole turn is about 6.28 radians. `Math.cos` and `Math.sin` use radians.

Check.

- Run the same test command again. It ends with `BUILD SUCCESSFUL`.

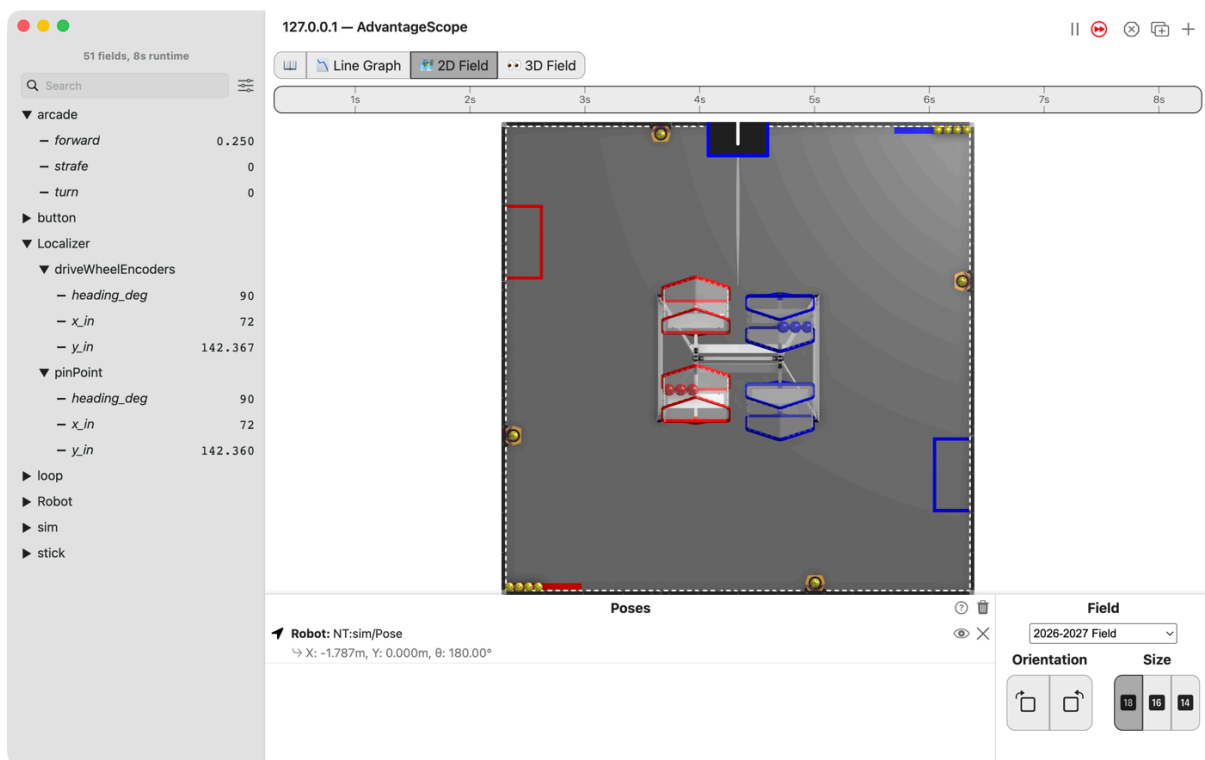


Figure 3.38. AdvantageScope's 2D Field with the robot at the top of the field, its arrow pointing up, and the list with the Localizer folder open: `driveWheelEncoders` and `pinPoint` both at `x 72` and heading `90`, and `y_in` at `142.367` and `142.360`.

Simulator. Run the command from L120S030 again, and connect AdvantageScope.

Check.

- Push the left stick forward. This time `driveWheelEncoders/y_in` climbs with `pinPoint /y_in`, and the two stay within a tenth of an inch.

The simulator turns each wheel's encoder by how far it rolls, so your localizer has something to count.

L120S060: drive it, then measure it

Robot. Stop the simulator. Send the code, run `L120 Compare Localizers` from the Driver Station, and watch it in AdvantageScope.

Check.

- Drive about 80 in straight ahead and stop. Both `y_in` numbers moved, and they do not quite agree.
- Measure how far the robot really went with the tape measure.

Then put the robot on the blocks and push the left stick forward. The wheels spin, and the robot does not go anywhere. `driveWheelEncoders` walks off across the field. `pinPoint` stays put, because its pods are off the floor too.

No gamepad. `--args="mytry.L120CompareLocalizersOpMode --left_stick_y=-0.5"` drives forward for the picture at the top, and for L120S050's.

If it didn't, and `driveWheelEncoders` is missing from the Localizer folder in L120S030, the two lines are not in `shadowLocalizers()`. `LessonsTest.l120_theEncoderLocalizerRunsAlongsideAndIsLogged` fails with the shadow localizer is logged: and the list of names that were published.

If it didn't, and `countsStrafingLeft` fails with `left` is `+y` when facing `+x`, a sign in the strafe line is wrong. With the sign in front of `frontRight` flipped, it reads `expected:<12.0>` but was:`<6.0>`, and three other tests fail with it.

If it didn't, and all 8 tests fail with `Cannot load from double array` because `"this.lastWheels"` is null, the first-loop if is missing.

What you just did. That is **dead reckoning**: believe the wheels, and add up every step. It has no other source, so a wheel that spins without carrying the robot moves the estimate. The test `aWheelThatSlipsIsSimplyBelieved` shows it on a laptop: one wheel rolling 24 in on its own moves the estimate 6 in. The Pinpoint's pods only measure and never push, so they slip less. That is the case for having them.

The heading here is the one at the end of each step. While driving and turning at once, the middle of the step would be closer. Taking the end needs no subtraction of two headings, so a turn through the back of the circle needs no care.

The localizer reports a speed of zero. Only `x_in`, `y_in` and `heading_deg` are published for a shadow, so nothing reads its speed.

Where next. L125 drives 24 inches on its own.

3.15 L125: the first autonomous

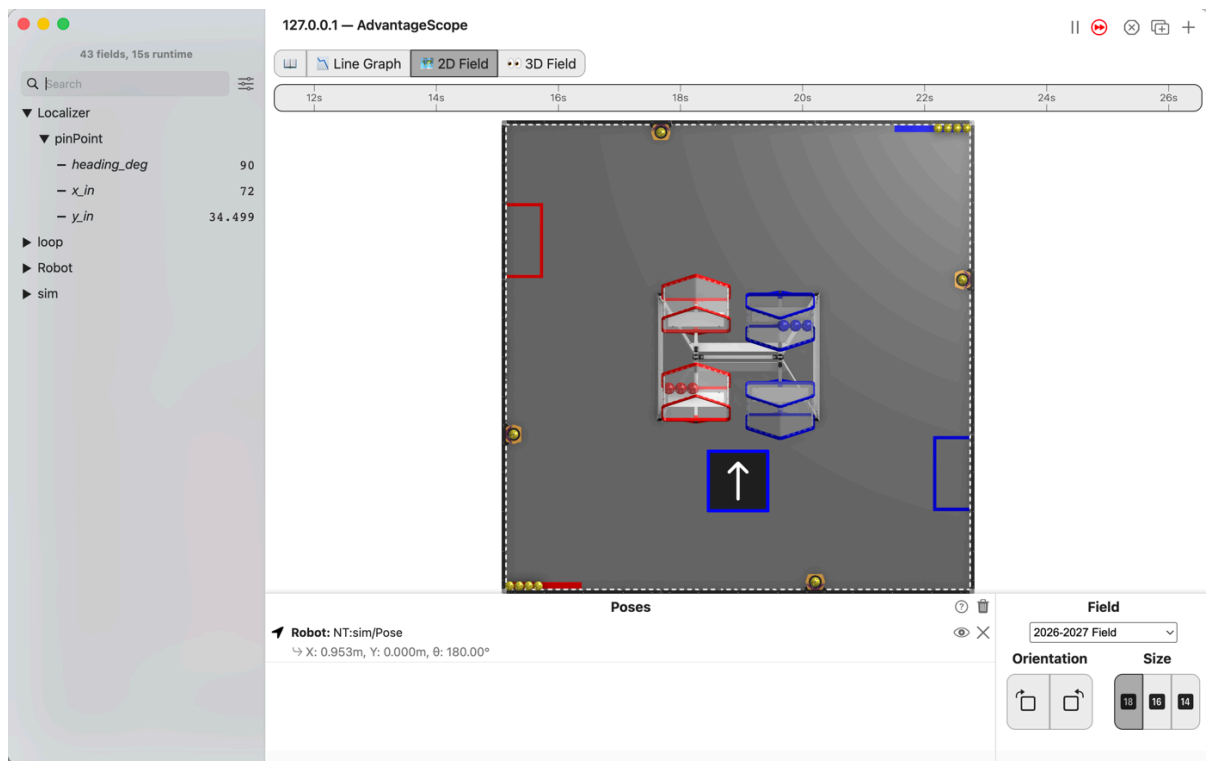


Figure 3.39. AdvantageScope's 2D Field with the robot stopped 24 inches up the field from where it started, its arrow pointing up the field, and the list showing pinPoint/y_in at 34.499, x_in at 72 and heading_deg at 90.

The robot drives 24 inches forward on its own and stops there. Nobody touches a stick. The path follower does the driving, and you write one line to tell it where to go.

Before you start.

- L120 is done. Nothing here uses L120's code; the follower uses the robot's own localizer.
- AdvantageScope is installed on the same laptop. No gamepad is needed.
- The robot, the Driver Station, a piece of tape, a tape measure, and about 1 m of clear floor in front of the robot for the last step.

3.15.1 The steps

L125S010: copy the starter into mytry

Copy. In `lessons`, click `L125Drive24OpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L125Drive24OpMode`, and the new file is open.

L125S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. 'public' allows access from any code.

L125S030: say where to go

An autonomous has no sticks to read. It has a `routine`, and the robot runs it after `PLAY`. The routine hands back a **command**: a list of things to do, in order.

`start` and `end` are already near the top of the class, 24 inches apart along the field's `y`. The robot starts facing along `y`, so the line from one to the other is straight ahead. `startPose()` tells the follower where the robot is at the start.

Change. In `routine()`, find the comment that names this step. Delete the line `return Command.NOOP`; under it, and type:

```
// When on L125S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l125.html#l125s030
// for what to do here.
return sequential(follow(follower, line(start, end).constant(start)));
```

Read it from the inside out. `line(start, end)` is the path. `.constant(start)` holds the heading `start` has the whole way. `follow` drives the path. `sequential` makes the list, and this list has one thing in it.

`sequential`, `follow` and `line` turn red. For each one:

- Click it.
- Press `Option+Enter` on a Mac, `Alt+Enter` on Windows. This time the popup says **Import static method...**, because each is a method, not a class. Pick it.
- Android Studio adds an `import static` line at the top of the file.

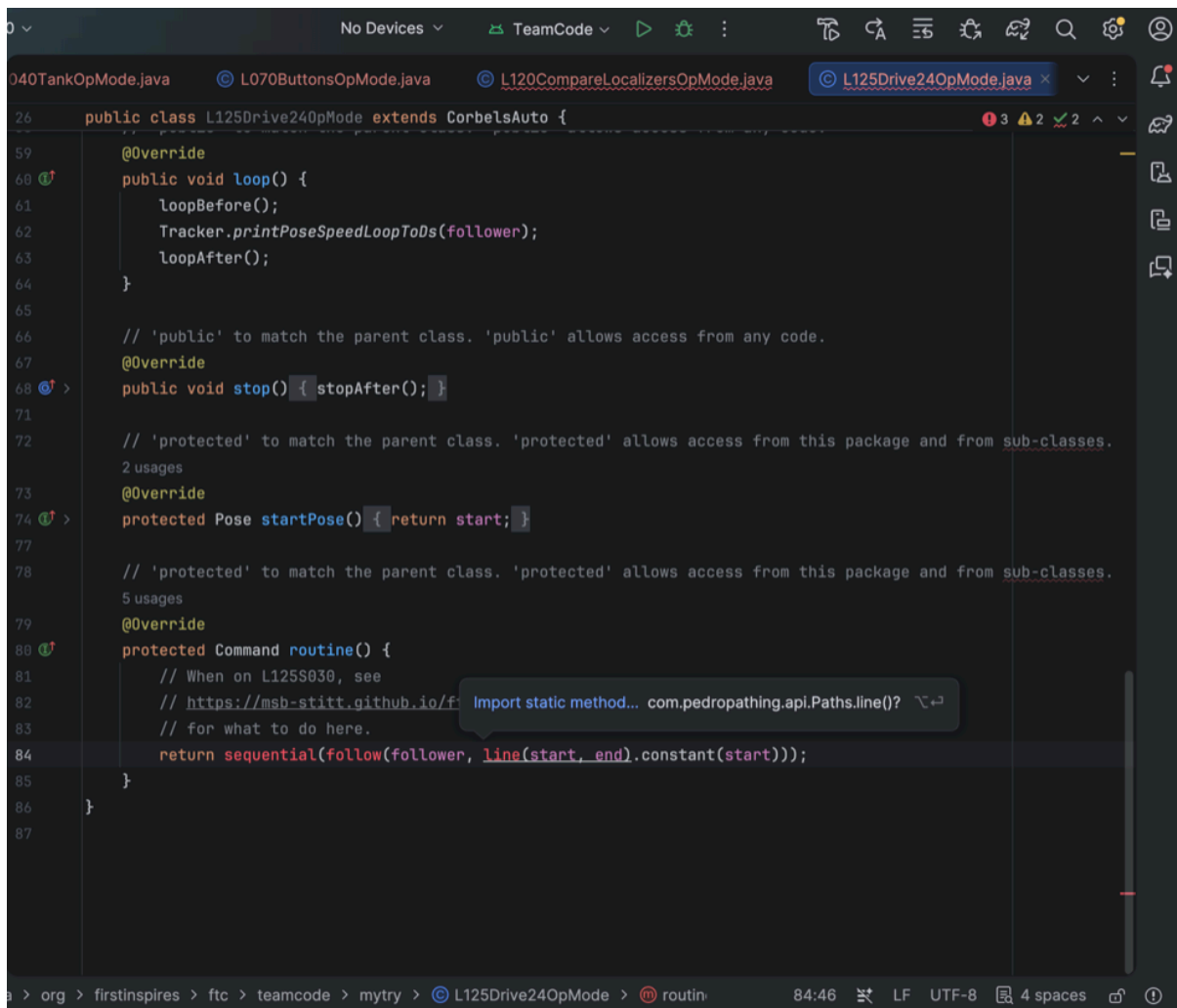


Figure 3.40. Android Studio with `sequential`, `follow` and `line` underlined in red and the popup offering **Import static method... `com.pedropathapi.Paths.line()`?** with `⌘↵`.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L125Drive24OpMode"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `Localizer` folder, then `pinPoint`.

Check.

- The robot has driven up the field and stopped.
- `y_in` reads 34.5, which is 24 more than the 10.5 it started at.
- `x_in` stays at 72 and `heading_deg` at 90.

L125S040: drive it on the floor

Robot. Stop the simulator. Put the robot on the floor with a piece of tape at its front edge. Send the code, and run `L125 Drive 24` from the Driver Station. It is in the autonomous list, not the `TeleOp` one. Keep your hands off the sticks.

Check.

- The robot drives forward, slows down, and stops, all by itself!

- Measure from the tape to the front edge. It is close to 24 inches.

If it didn't, and nothing moves, the routine still says `return Command.NOOP;`, the command that does nothing. `LessonsTest.l125_autoDrives24InchesForwardAndStops` fails with the robot never started following a path.

If it didn't, and nothing moves with the line typed, start and end are swapped. The robot is standing at that path's finish already, so it is done before it starts. The test fails with ends 24 inches further along y expected:<34.5> but was:<10.5>.

If it didn't, and the test fails with and does not wander in x expected:<72.0> but was:<70.81415129095723>, the heading is `.constant(90)`. The number form is radians, so 90 is not 90 degrees.

If it didn't, and on the floor it overshot or stopped short, that is the follower's tuning, not this line. L117 tunes it. If it curved, check the motor directions from L050S070 first.

What you just did. The follower is a library, used the way a team uses one: say where to go, and it works out the wheel powers. Nobody here wrote the path code.

`.constant()` holds one heading for the whole line, which is what this path wants. Pedro also has `.linear()`, which turns from one heading to another. Before Pedro 3.0.1 it turned the wrong way on a line path, reported in its issues #176 and #181. This project uses 3.0.1, which fixes it.

`sequential` takes a list, and this one has a single item in it. L130 puts a second item after it, and that is the whole of what an autonomous routine is: a list of things to do in order.

Where next. L130 adds a second leg.

3.16 L130: two legs, forward then sideways

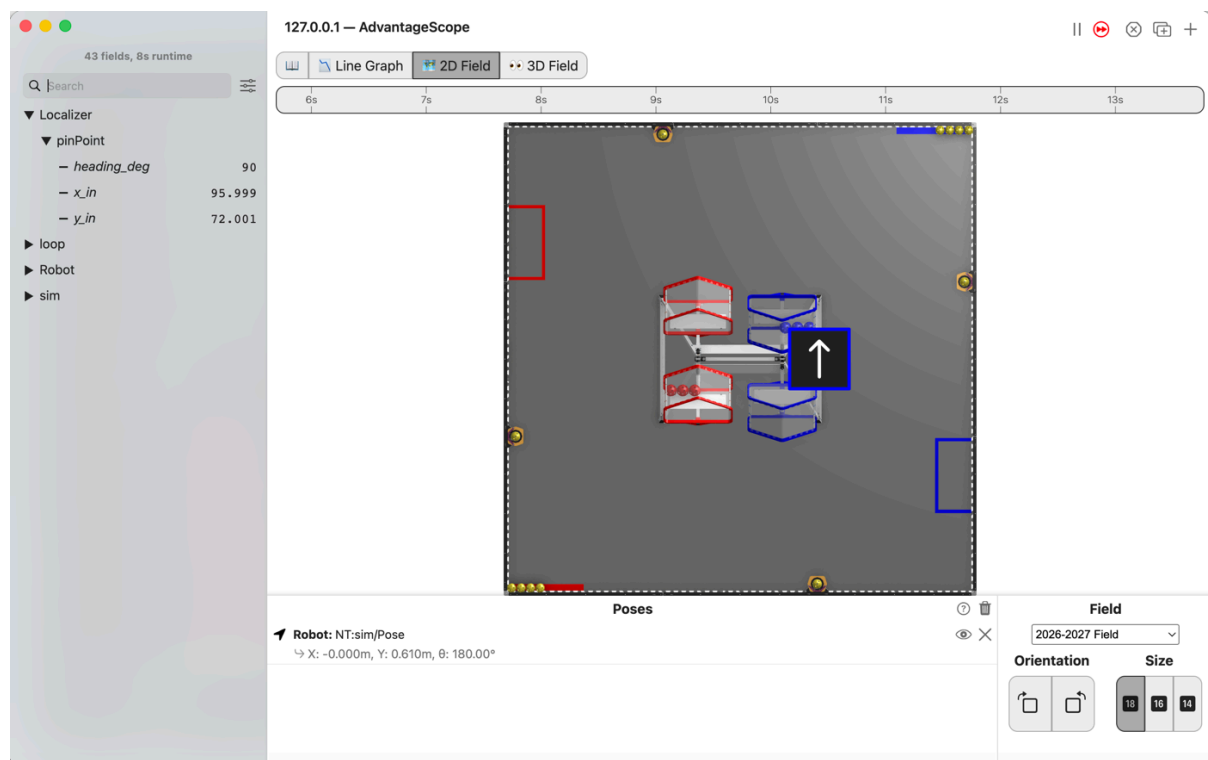


Figure 3.41. AdvantageScope's 2D Field with the robot stopped right of the middle of the field, its arrow still pointing up the field, and the list showing `pinPoint/x_in` at 95.999, `y_in` at 72.001 and `heading_deg` at 90.

The robot drives forward, then slides 24 inches to its right without turning, and holds where it ends

up. It is the first autonomous that does more than one thing.

Before you start.

- L125 is done.
- AdvantageScope is installed on the same laptop. No gamepad is needed.
- The robot, the Driver Station, a piece of tape, a tape measure, and about 1 m by 2 m of clear floor for the last step.

3.16.1 The steps

L130S010: copy the starter into mytry

Copy. In lessons, click `L130ForwardThenStrafeOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L130ForwardThenStrafeOpMode`, and the new file is open.

L130S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. `'public'` allows access from any code.

L130S030: add a second leg and a hold

A pose is a place on the field and the way the robot faces there. Near the top of the class are three poses: `start`, `corner` and `end`. All three face 90 degrees. `corner` is straight up the field from `start`, and `end` is 24 inches to the right of `corner`.

`routine()` already holds the first leg, from `start` to `corner`. It is L125's line, with one item on each line so there is room for more.

Change. In `routine()`, type a comma at the end of the first leg. The comment that names this step is under it. Under the comment, type:

```
return sequential(
    follow(follower, line(start, corner).constant(start)),
    // When on L130S030, see
    // https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
    // /l130.html#l130s030
    // for what to do here.
    follow(follower, line(corner, end).constant(start)),
    hold(follower, end)
);
```

Each item but the last ends with a comma. The second leg holds `start`'s heading too, so the robot never turns. It slides sideways instead, which is what mecanum wheels are for. `hold` keeps the robot at `end` once it gets there.

`hold` turns red. Click it and press Option+Enter, as in L125.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L130ForwardThenStrafeOpMode"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `Localizer` folder, then `pinPoint`.

Check.

- The robot has stopped right of the middle of the field.
- `x_in` reads 96 and `y_in` 72.
- `heading_deg` is still 90.

L130S040: drive it on the floor

Robot. Stop the simulator. The robot drives about 62 in forward, then 24 in to its right, so clear that much floor, plus room for the robot. Put the robot on the floor with a piece of tape at its front edge. Send the code, and run `L130 Forward Then Strafe` from the autonomous list.

Check.

- The robot drives straight forward and stops, then slides to its right and stops again.
- Its nose points the same way the whole time.

If it didn't, and the robot cuts across in one diagonal, the first leg goes to end instead of corner. `LessonsTest.l130_autoDrivesForwardThenStrafesSideways` fails with `drove straight up to the corner first expected:<0.0> but was:<20.307354640929788>`.

If it didn't, and the test fails with `never turned`, so it still faces +y `expected:<90.0> but was:<116.04530086607369>`, the second leg is `.constant(90)`. The number form is radians, as in L125.

If it didn't, and on the floor it slid left, check which way +x runs from where the robot stands. Facing +y, the robot's right hand points at +x.

`hold` is not checked: the test passes with it left out. On a floor it keeps the wheels pushing back if something bumps the robot once the path is over.

What you just did. An autonomous routine is a list. Two legs and a hold here, and a match autonomous is the same list with more items, some of them moving something other than the wheels.

A leg holds one heading from one end to the other. Hold the same heading on both legs and the robot never turns. Where it goes and which way it faces are two separate things, and L140 spends that.

Where next. L140 goes back to the sticks, where a stick means a direction on the field.

3.17 L140: make the sticks mean the field



Figure 3.42. AdvantageScope's 2D Field with the robot right of where it started and turned partway round, with the list showing arcade/forward at 0.160, arcade/turn at 0.490, and pinPoint/x_in at 107.911, y_in at 20.901 and heading_deg at 234.

Push the stick away from you and the robot goes away from you. It does that whichever way the robot happens to be facing, so a driver never has to think about which end of the robot is the front.

Before you start.

- L130 is done.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot, the Driver Station, and about 2 m by 2 m of clear floor for the last step.

Where the driver stands. "Away from the driver" is the field's +x. The robot starts at heading 90°, facing +y, so at the start a push away from the driver slides it to its right.

3.17.1 The steps

L140S010: copy the starter into mytry

Copy. In lessons, click L140FieldRelativeOpMode once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- mytry holds L140FieldRelativeOpMode, and the new file is open.

L140S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. 'public' allows access from any code.

L140S030: drive and turn at once, the L110 way

The starter drives the way L110 did. Forward on the stick is forward for the robot.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L140FieldRelativeOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `Localizer` folder, then `pinPoint`.

Check.

- Push the left stick a little forward and the right stick left, and hold both.
- The robot drives in a small circle, and `x_in` and `y_in` stay close to where they started.
- It goes where its nose points, and its nose keeps turning.

L140S040: turn the stick backwards through the heading

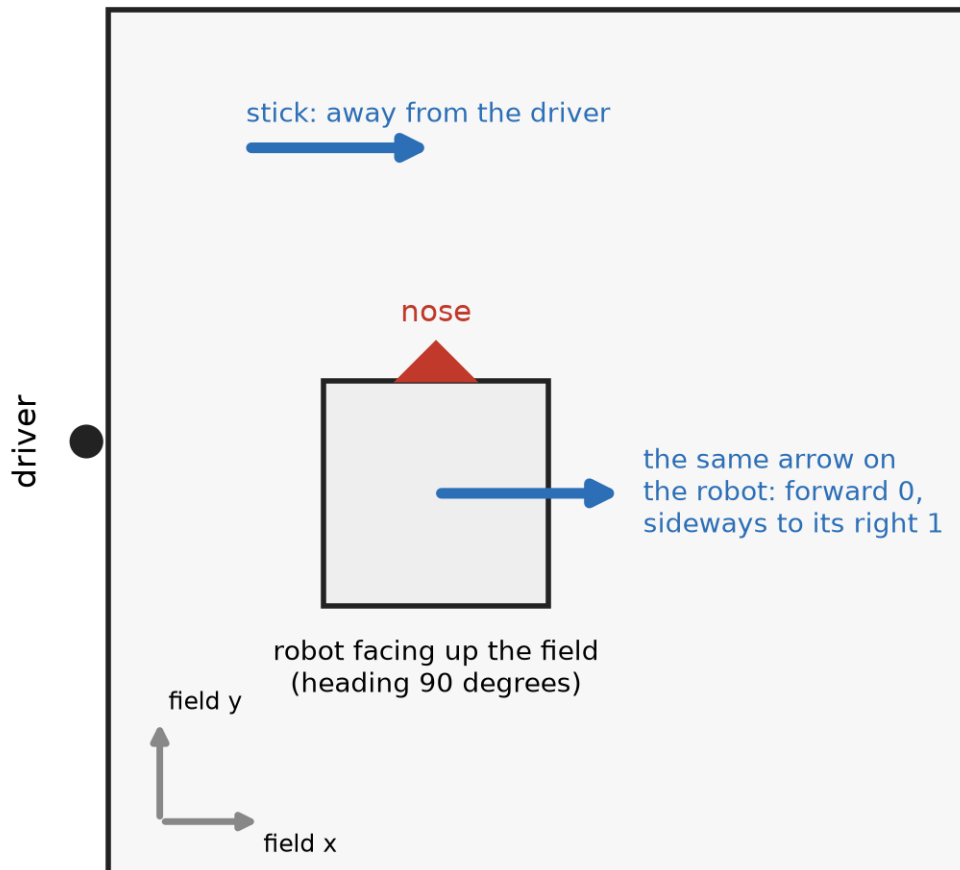


Figure 3.43. The field from above as AdvantageScope draws it, with the driver at the left edge, the robot facing up the field, the stick's direction as an arrow on the field, and the same arrow drawn again on the robot showing what forward and sideways it turns into.

From now on, *forward* on the stick means away from the driver, along the field's *x*. *strafe* means to the driver's left, along the field's *y*. On AdvantageScope's 2D Field, *+x* runs to the right and *+y* runs up.

The wheels only know the robot's own forward and sideways. L120 turned a step on the robot into a step on the field, through the heading. This turns the other way, from the field to the robot, so the signs on *sin* swap.

Change. Press Ctrl+C in the Terminal to stop the simulator. In `loop()`, find the comment that names this step. Delete the line `drivetrain.sticks(forward, strafe, turn);` under it, and type:

```
// When on L140S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l140.html#l140s040
// for what to do here.
double heading = follower.pose().heading();
double robotForward = forward * Math.cos(heading) + strafe *
Math.sin(heading);
double robotStrafe = -forward * Math.sin(heading) + strafe *
```

```
Math.cos(heading);  
drivetrain.sticks(robotForward, robotStrafe, turn);
```

The follower knows which way the robot faces. The drivetrain does not. So the OpMode asks the follower, and does the turn itself.

Simulator. Run the same command again, and connect AdvantageScope.

Check.

- Hold the same two sticks. The robot spins, and at the same time it travels to the right on the 2D Field, along +x.
- `x_in` climbs while `heading_deg` goes round and round.

L140S050: drive it facing you

Robot. Stop the simulator. Send the code, and run L140 Field Relative from the Driver Station. Stand off the robot's left side, so that away from you is the field's +x.

Check.

- Push the right stick until the robot's nose points at you.
- Then push the left stick away from you. The robot drives away from you while it faces you.
- Turn it a quarter turn and try again, and the stick still means the same way on the field.

No gamepad. `--args="mytry.L140FieldRelativeOpMode --left_stick_y=-0.4 --right_stick_x=-0.7"` holds both sticks, for the circle in L140S030 and the picture at the top. Before L140S040, the run was watched at `x_in` 70.812 and `y_in` 7.230 after about 8 s.

After L140S040, the robot drifts toward +y as it goes: one run went from `x` 72 in and `y` 10.5 in to `x` 179 in and `y` 43.5 in, about 17 degrees off a straight line. The simulator has no walls, so a run left going drives off the field.

If it didn't, and `LessonsTest.l140_fieldRelativeIgnoresWhichWayTheRobotFaces` fails with same stick, still moves away from the driver `expected:<0.0>` but `was:<1.0>`, the line still reads `drivetrain.sticks(forward, strafe, turn);`.

If it didn't, and it fails with `expected:<-1.0>` but `was:<1.0>`, the turn runs the wrong way. The minus sign is missing from `-forward`, or both signs on `sin` are the other way round.

If it didn't, and it fails with same stick, still moves away from the driver `expected:<0.0>` but `was:<-0.3338674754196965>`, the heading went in as degrees. `Math.cos` wants radians.

What you just did. Field relative is one turn of two numbers and nothing else. The same sticks, the same mix, the same scaling. All that changed is which pair of numbers goes in.

It also depends on the follower knowing which way the robot faces. If the heading drifts 20 degrees, the sticks are 20 degrees crooked. This is the first lesson where a mistake somewhere else in the robot shows up in the driver's hands.

Where next. L150 puts the robot's own directions back on a bumper, so a driver can have both.

3.18 L150: give the driver both

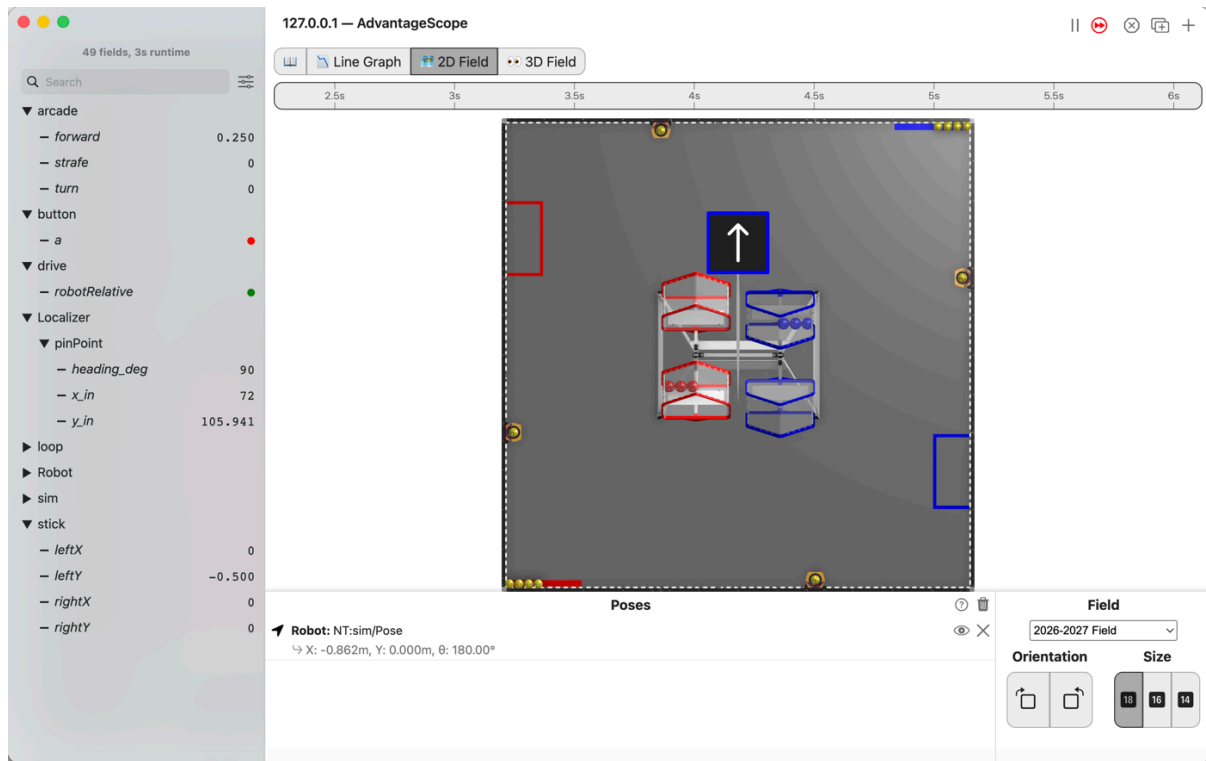


Figure 3.44. AdvantageScope's 2D Field with the robot above the middle of the field, its arrow pointing up the field, and the list showing drive/robotRelative green, arcade/forward at 0.250, and pinPoint /x_in at 72, y_in at 105.941 and heading_deg at 90.

The sticks mean the field, the way L140 left them, until the right bumper is held. While it is held they mean the robot. Let go and they mean the field again.

Before you start.

- L140 is done.
- AdvantageScope is installed on the same laptop, and a gamepad is plugged into the laptop.
- The robot, the Driver Station, and about 2 m by 2 m of clear floor for the last step.

3.18.1 The steps

L150S010: copy the starter into mytry

Copy. In `lessons`, click `L150RobotRelativeButtonOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L150RobotRelativeButtonOpMode`, and the new file is open.

L150S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. 'public' allows access from any code.

L150S030: choose with the bumper

Field relative is the easier way to drive most of the time. It is not the easier way to line up against something. Backing straight out of a corner, or nudging along a wall, is easier when forward means the robot's own forward.

So the driver gets both, and an if chooses. The starter already works out `robotForward` and `robotStrafe`, L140's turn. `forward` and `strafe` are the sticks as they are, the robot's own directions.

Change. In `loop()`, find the comment that names this step. Delete the line `drivetrain.sticks(robotForward, robotStrafe, turn);` under it, and type:

```
double heading = follower.pose().heading();
double robotForward = forward * Math.cos(heading) + strafe *
Math.sin(heading);
double robotStrafe = -forward * Math.sin(heading) + strafe *
Math.cos(heading);
// When on L150S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l150.html#l150s030
// for what to do here.
boolean robotRelative = gamepad1.right_bumper;
Tracker.publish("drive/robotRelative", robotRelative);
if (robotRelative) {
    drivetrain.sticks(forward, strafe, turn);
} else {
    drivetrain.sticks(robotForward, robotStrafe, turn);
}
```

When you type `{` and press Enter, Android Studio adds the `}` for you.

The bumper is read fresh every loop. Let go, and the next loop drives field relative again. The driver's thumb tells them which way the sticks work, so there is nothing to remember in a match.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L150RobotRelativeButtonOpMode --pad"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the drive folder, then Localizer and `pinPoint`.

Check.

- Hold the right bumper and push the left stick a little forward. `robotRelative` turns green, and the robot drives up the field, the way its nose points.
- `y_in` climbs and `x_in` stays at 72.
- Let go of the bumper and keep pushing. The robot slides to the right, along `+x`, and `x_in` climbs.

L150S040: drive both ways on the floor

Robot. Stop the simulator. Send the code, and run `L150 Robot Relative Button` from the Driver Station. Stand off the robot's left side, as in L140.

Check.

- Turn the robot so its nose points to one side, and push the left stick away from you.
- The robot drives away from you.

- Now hold the right bumper and push the same way. The robot drives along its own nose.
- Let go, and it drives away from you again.

No gamepad. This holds the bumper and the stick, for the picture at the top:

```
./gradlew :TeamCode:simRun --args="mytry.L150RobotRelativeButtonOpMode
--left_stick_y=-0.5 --right_bumper=true"
```

Without `--right_bumper=true`, the same run was watched at `x_in` 126.674 and `y_in` 10.5. The simulator has no walls, so a run left going drives off the field.

If it didn't, and `LessonsTest.l150_theBumperSwitchesToRobotRelative` fails with robot relative while held expected:<1.0> but was:<5.551115123125783E-17>, the line still reads `drivetrain.sticks(robotForward, robotStrafe, turn);`. That long number is 0 with a rounding error from the turn.

If it didn't, and it fails with field relative by default expected:<0.0> but was:<1.0>, the two branches are swapped. The bumper is the exception, so the `if` reads the bumper, not ! the bumper.

If it didn't, and it fails with and back again on release expected:<0.0> but was:<1.0>, the bumper flips a remembered flag instead of being read fresh every loop. That is a toggle, and this is the check that catches it.

No test watches the `drive/robotRelative` line. It is there for AdvantageScope: a driver who says "it went crooked" and a graph that says which way the sticks worked settle the question between them.

What you just did. This is the first lesson that is a driver's choice rather than a robot's. Nothing here is new arithmetic. Both ways of driving existed already, and what you added is who chooses and when.

That is most of what a competition teleop turns out to be: a few ways to drive, and a gamepad laid out so a driver can reach them without looking.

Where next. [L160](#) holds a heading for the driver, so the robot stops turning when nobody is turning it.

3.19 L160: hold the heading the driver left

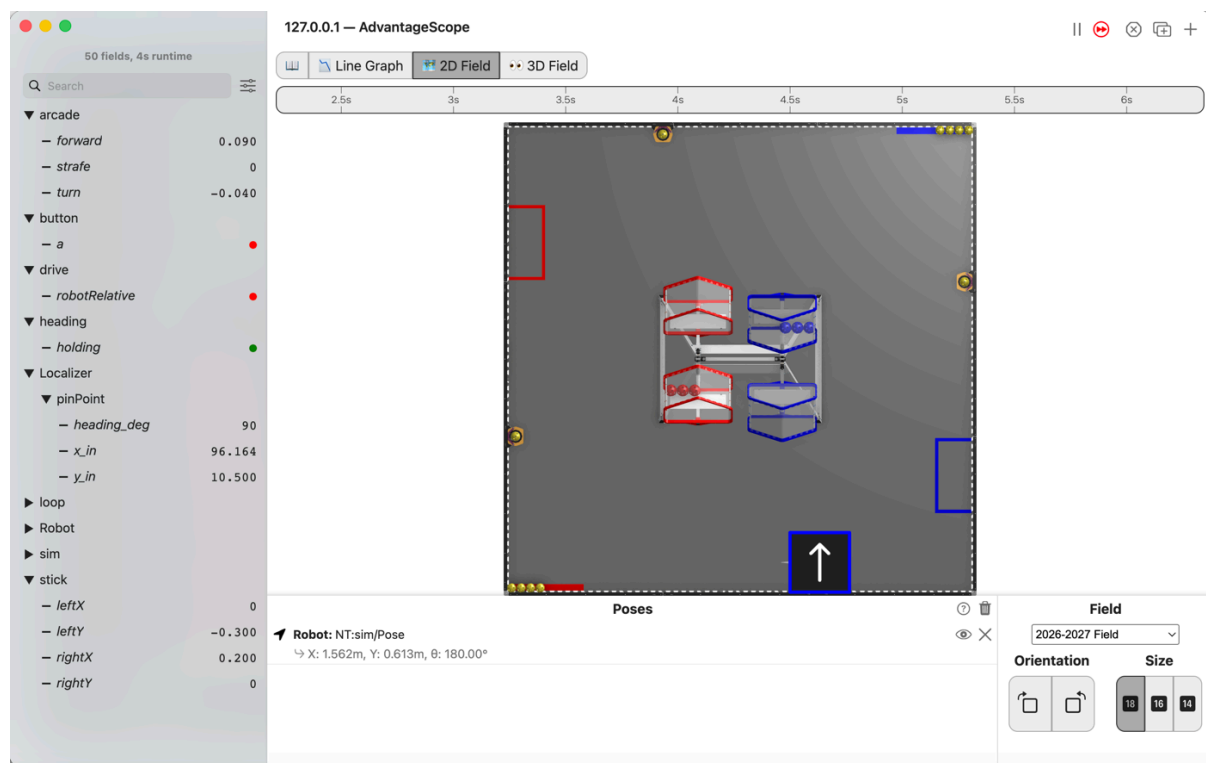


Figure 3.45. AdvantageScope's 2D Field with the robot near the bottom of the field, right of the middle, its arrow still pointing up the field, and the list showing arcade/turn at -0.040, heading/holding green, and pinPoint/heading_deg at 90 and x_in at 96.164.

Let go of the turn stick and the robot keeps pointing where it was left. Push the turn stick and the driver has it back at once.

Before you start.

- L150 is done.
- AdvantageScope is installed on the same laptop. No gamepad is needed until the last step.
- The robot, the Driver Station, a gamepad, and about 2 m by 2 m of clear floor for the last step.

3.19.1 The steps

L160S010: copy the starter into mytry

Copy. In `lessons`, click `L160HeadingHoldOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L160HeadingHoldOpMode`, and the new file is open.

L160S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this OpMode. 'public' allows access from any code.

L160S030: watch the nose creep

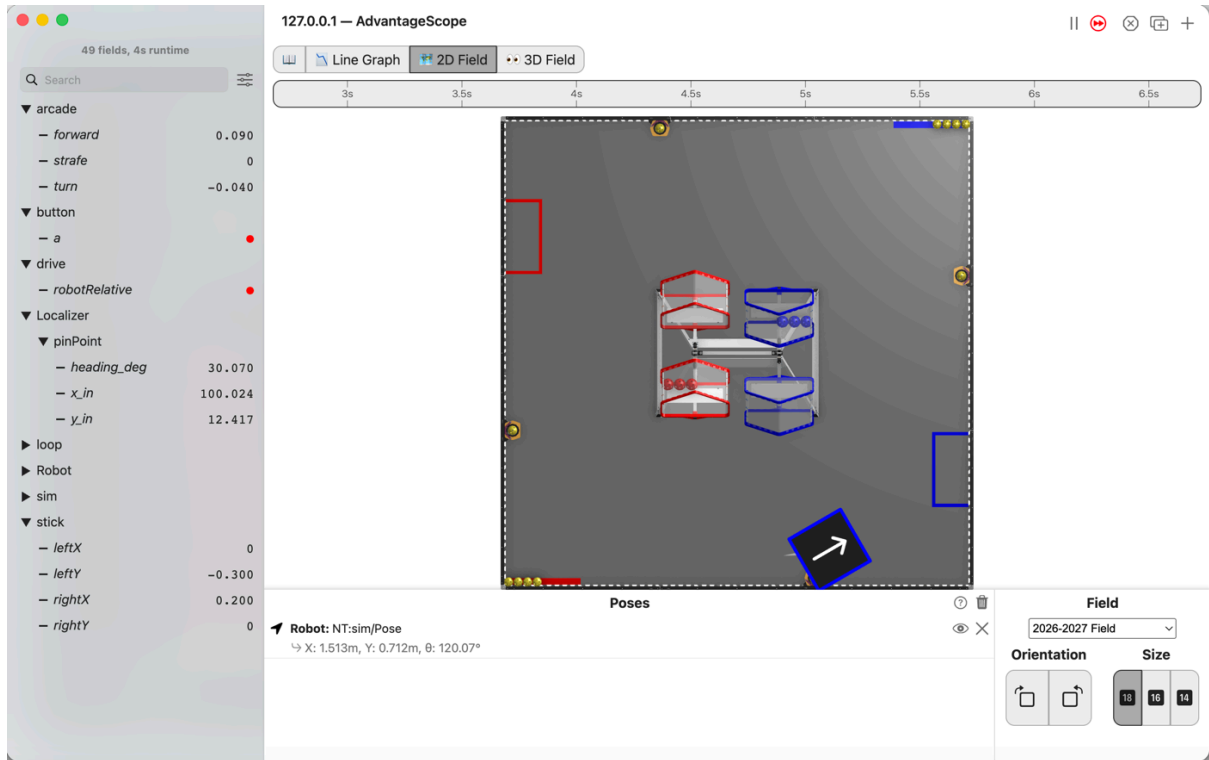


Figure 3.46. AdvantageScope's 2D Field with the robot near the bottom of the field, right of the middle, turned so its arrow points up and to the right, and the list showing `arcade/turn` at -0.040 and `pinPoint/heading_deg` at 30.070.

The starter is L150's finished OpMode. This run pretends the turn stick rests a little off centre, the way a worn stick does.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L160HeadingHoldOpMode --left_stick_y=-0.3
--right_stick_x=0.2"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the `arcade` folder, then `Localizer` and `pinPoint`.

Check.

- `turn` reads -0.040, which is small.
- The robot drives to the right, and it turns as it goes.
- `heading_deg` started at 90 and keeps dropping.

L160S040: hold the heading

`HeadingHold` is written for you, in base. While the turn stick is pushed, it hands the stick straight back and the driver steers. On the loop the stick is let go, it remembers the heading. After that it hands back the turn that steers the robot back to that heading.

Its own deadband is 0.05, so a stick that is nearly centred counts as let go. `turn` at -0.040 is inside it.

Change. Press `Ctrl+C` in the Terminal to stop the simulator. Find the line `CorbelsMecanum`

drivetrain; near the top of the class. Under it, type:

```
CorbelsMecanum drivetrain;
HeadingHold hold = new
HeadingHold(Constants.foresightConfig.headingFeedback.get());
```

Constants.foresightConfig holds the follower's own settings. headingFeedback is the part that steers the heading in autonomous. Tune the follower, and this hold gets the same tuning.

HeadingHold and Constants turn red. Click each one and press Option+Enter, as in L120. The popup for Constants names org.firstinspires.ftc.teamcode.pedro.Constants.

Change. In loop(), find the comment that names this step. Under it, type:

```
double turn = squared(deadband(-rightStickX, band));
Tracker.publish("arcade/turn", turn);
// When on L160S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /l160.html#l160s040
// for what to do here.
turn = hold.turn(follower, turn);
Tracker.publish("heading/holding", hold.target() != null);
```

The first new line swaps the stick's turn for the hold's. Both sticks calls below use turn, so both drive with it. Android Studio now underlines turn everywhere. That marks a variable that changes after it is made. It is not an error.

Simulator. Run the same command again, and connect AdvantageScope. Open the heading folder too.

Check.

- holding is green.
- The robot drives to the right, its nose still pointing up the field, and heading_deg stays at 90.

L160S050: drive it and then push it

Robot. Stop the simulator. Send the code, and run L160 Heading Hold from the Driver Station.

Check.

- Drive a long diagonal with the left stick, and do not touch the turn stick.
- The nose stays where it was.
- Then take hold of the robot and twist it a little. It turns back on its own.

No gamepad. The two runs above need none. With --pad, a driver can turn, let go, and watch the hold take over.

If it didn't, and LessonsTest.l160_theRobotHoldsItsHeadingWhenTheStickIsReleased fails with it steers back, nothing listens to the hold. Either the line is missing, or it reads hold.turn(follower, turn); without turn = in front, so its answer is thrown away.

If it didn't, and it fails with while steering, the stick wins expected:<-1.0> but was:<1.0>, the hold got rightStickX instead of turn. That skips the minus sign that makes a push to the right turn clockwise.

If it didn't, and on the floor the robot fights back and overshoots, that is the follower's tuning, not this lesson. L117 tunes it. If the nose drifts slowly anyway, the hold is working and the follower's heading is drifting with it.

The simulator turns the robot only when told to, so nothing there pushes it off its heading. The creep in L160S030 is a stick resting off centre, not a bump.

What you just did. Three things want to turn the robot, and they take turns in a fixed order. The driver's stick beats the hold. The hold beats doing nothing. The person with their hands on

the gamepad wins, and everything else fills in when they let go.

Where next. L170 gives a button a pose, and the robot drives there by itself while the driver watches.

3.20 L170: press Y and the robot drives itself there

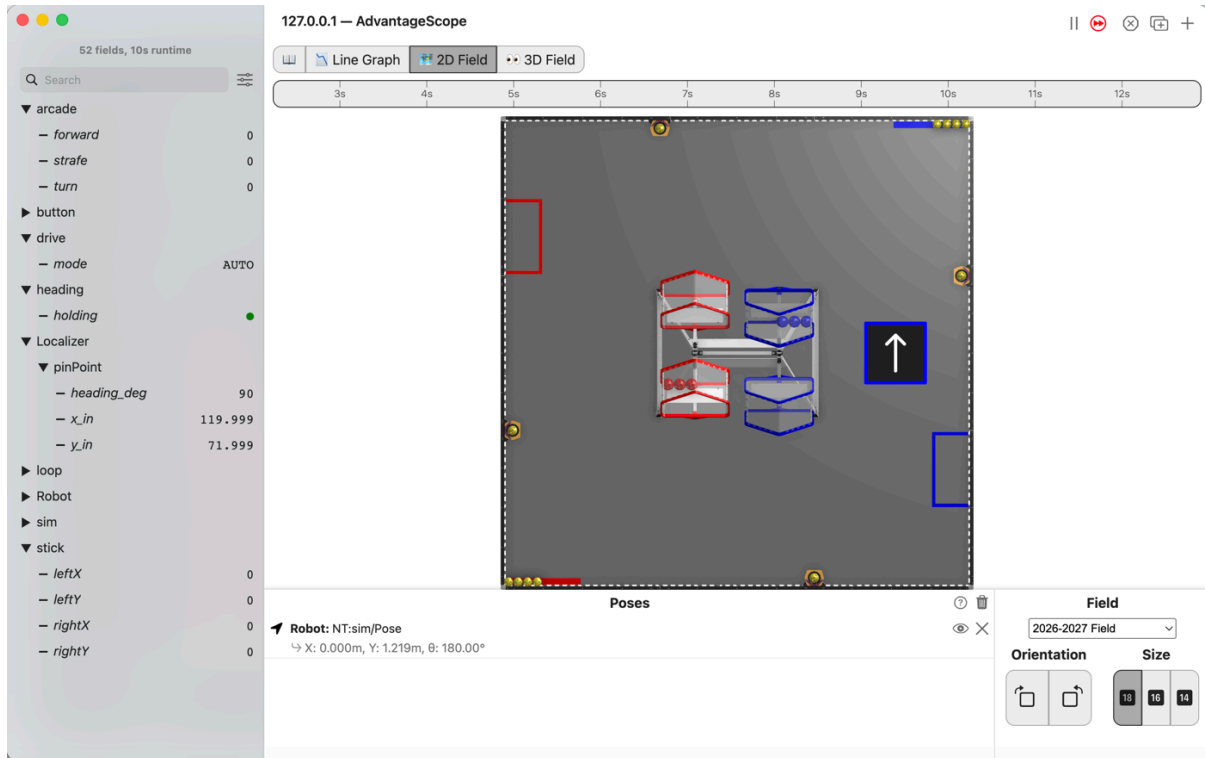


Figure 3.47. AdvantageScope's 2D Field with the robot right of the middle of the field, its arrow pointing up the field, and the list showing drive/mode at AUTO, every stick at 0, and pinPoint/x_in at 119.999, y_in at 71.999 and heading_deg at 90.

Press Y and the robot drives itself to a spot on the field. Push any stick and the driver has it back at once.

Before you start.

- L160 is done.
- AdvantageScope is installed on the same laptop. No gamepad is needed until the last step.
- The robot, the Driver Station, a gamepad, and about 2 m by 2 m of clear floor for the last step.

3.20.1 The steps

L170S010: copy the starter into mytry

Copy. In lessons, click L170DriveToPoseOpMode once and copy it. Click mytry once and paste. In the Copy Class box, click OK. If Android Studio asks whether to add the file to Git, click Add.

Check.

- mytry holds L170DriveToPoseOpMode, and the new file is open.

L170S020: switch your copy on

Change. Delete the line @Disabled, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this OpMode. 'public' allows access from any code.

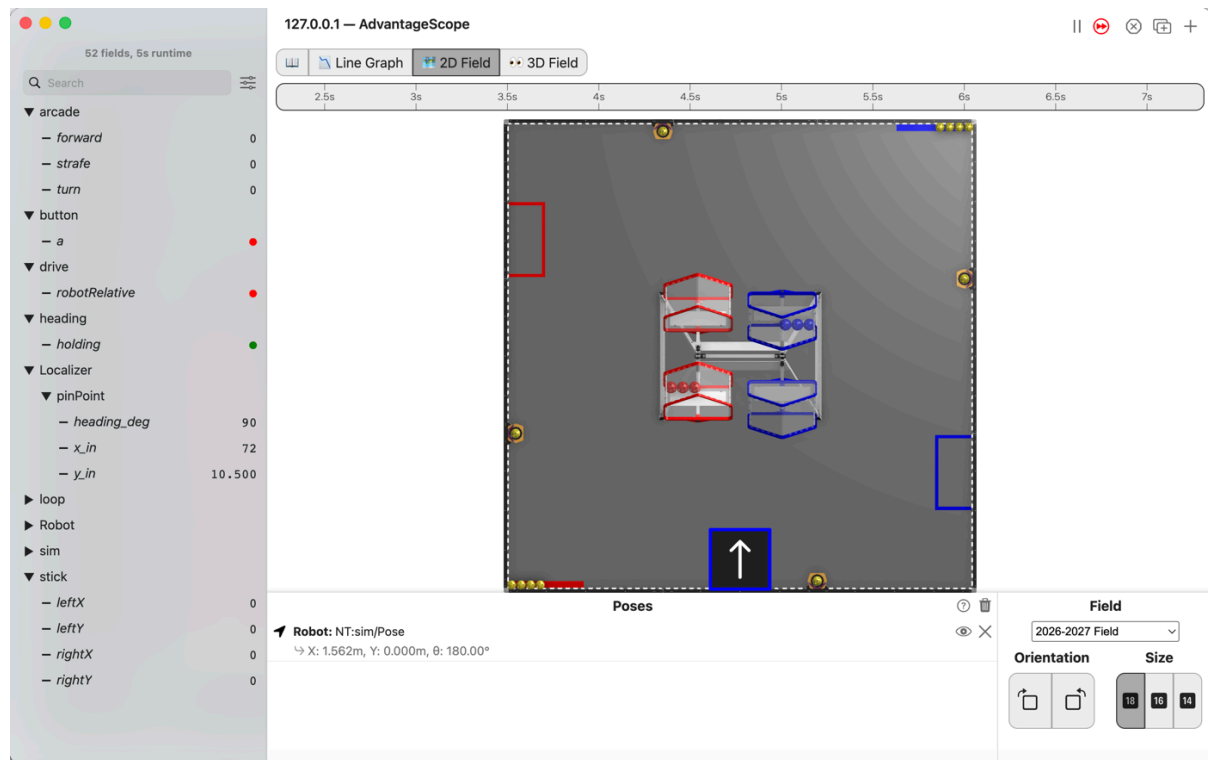
L170S030: give Y a pose

Figure 3.48. AdvantageScope's 2D Field with the robot at the bottom middle of the field, its arrow pointing up, and the list showing every stick at 0 and pinPoint/x_in at 72, y_in at 10.500 and heading_deg at 90.

The starter is L160's finished OpMode, moved around a little. All the driving now sits in `driveTheRobot()`, which `loop()` calls. Near the top are three new fields. `target` is the pose, at x 120 in, y 72 in and heading 90°. `drivingItself` remembers whether Y sent the robot there.

`bindings()` runs once, when PLAY is pressed. It is the place to say what each button does.

Change. In `bindings()`, find the comment that names this step. Under it, type:

```
// When on L170S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1170.html#l170s030
// for what to do here.
buttons.whenPressed(() -> gamepad1.y, Commands.instant(() -> {
    follower.hold(target);
    drivingItself = true;
}));
```

Android Studio closes the brackets for you, as `}})`. Type the `;` after them yourself.

`Commands` turns red. Click it and press `Option+Enter`, as in L120. The popup names `com.pedropathing.ivy.commands.Commands`.

`() ->` wraps code to run later, not now. `() -> gamepad1.y` is how `whenPressed` reads Y every loop, and the code inside `{ }` is what it does when Y goes down.

`whenPressed` runs its command once, the loop Y goes down. Holding Y does not run it again. The command tells the follower to drive to `target` and stay there.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L170DriveToPoseOpMode --y=true"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the Localizer folder, then `pinPoint`.

Check.

- The robot does not move.
- `x_in` stays at 72 and `y_in` at 10.5.

The follower wants to go, but every loop the sticks say stop, and since L110 the sticks win.

L170S040: let the follower drive until the driver takes over

`driverWantsControl` is in the starter. It is true when any stick is pushed past the deadband.

Change. Press Ctrl+C in the Terminal to stop the simulator. In `driveTheRobot()`, find the comment that names this step. Under it, type:

```
boolean driverWantsControl = Math.abs(leftStickY) > band
    || Math.abs(leftStickX) > band || Math.abs(rightStickX) > band;
// When on L170S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l170.html#l170s040
// for what to do here.
if (drivingItself) {
    if (!driverWantsControl) {
        Tracker.publish("drive/mode", "AUTO");
        drivetrain.releaseCommandedWheels();
        return;
    }
    drivingItself = false;
    follower.manual(0, 0, 0);
}
Tracker.publish("drive/mode", "DRIVER");
```

`||` means “or”, so `driverWantsControl` is true when any one stick is pushed. `!` means “not”, so `!driverWantsControl` is true when no stick is pushed.

While the robot drives itself and no stick is pushed, `releaseCommandedWheels()` lets go of the wheels, and `return` leaves `driveTheRobot()` before the sticks are read. The follower is left driving. Once a stick is pushed, `drivingItself` goes back to false and `follower.manual` stops the follower. The sticks drive again, from this loop on.

Simulator. Run the same command again, and connect AdvantageScope. Open the `drive` folder too.

Check.

- mode reads AUTO.
- The robot drives right and up the field and stops at `x_in` 120 and `y_in` 72, its nose still pointing up the field. It may show 119.999 and 71.999.

L170S050: send it and then take it back

Robot. Stop the simulator. Send the code, and run L170 Drive To Pose from the Driver Station.

Check.

- Press Y once and let go. The robot drives itself toward the middle of the field, right of where it started.
- Press Y again and push the left stick while it is on its way. It stops going there, and drives the way the stick says.

No gamepad. Both runs above hold Y with `--y=true` and need none. With `--pad`, a driver can press Y and take over with a stick.

If it didn't, and `LessonsTest.l170_pressingYDrivesToAPoseAndTheDriverCanTakeOver` fails with the command is driving, not the sticks expected:<HOLD> but was:<MANUAL>, the sticks are still driving after Y. The binding is missing, the return is missing, or the if reads `driverWantsControl` without the `!`.

If it didn't, and it fails with manual control returns expected:<MANUAL> but was:<HOLD>, the follower never lets go. Either `follower.manual(0, 0, 0);` is missing, or the binding leaves out `drivingItself = true;`, so the sticks never take the follower back.

Two lines can go missing and the test still passes. Neither has been run; both are read from the code.

- Without `drivingItself = false;`, a stick takes over as it should. Let go, and mode reads AUTO again while the follower, told `manual(0, 0, 0)`, keeps the robot still.
- Without `releaseCommandedWheels()`, the sticks' last command stays on the wheels, and the follower is ignored. The `--y=true` run still drove there, because Y was down from the first loop, before the sticks had set the wheels. After the driver has driven, Y would do nothing.

What you just did. The robot has two drivers now, the person and the follower, and one rule decides between them. The follower drives only when asked, and only until the person touches a stick. That is the same rule as L160, a step bigger: the person on the gamepad wins.

Where next. L180 puts every piece so far into one OpMode a driver could take to a match.

3.21 L180: one OpMode a driver could take to a match

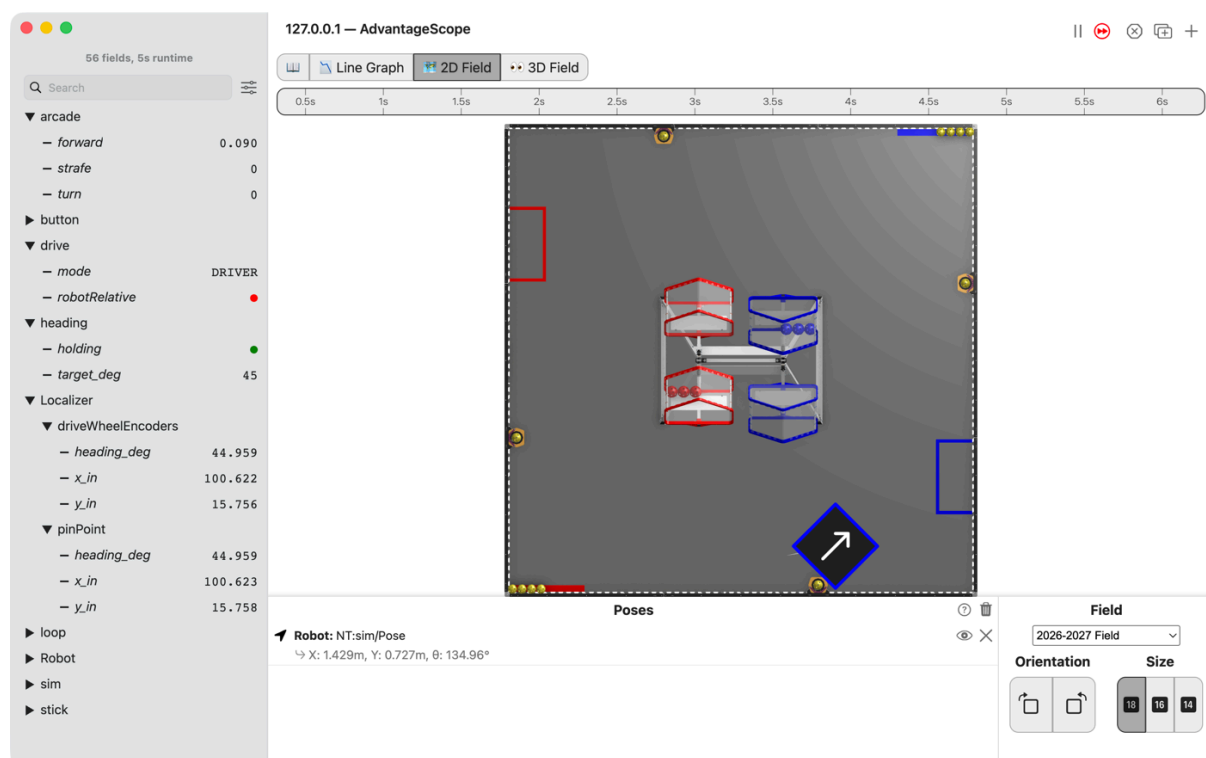


Figure 3.49. AdvantageScope's 2D Field with the robot near the bottom right of the field, its arrow pointing up and to the right, and the list showing drive/mode at DRIVER, heading/holding green and target_deg at 45, driveWheelEncoders at x 100.622, y 15.756 and heading 44.959, and pinPoint at x 100.623, y 15.758 and heading 44.959.

One teleop with all of it in. The sticks drive field relative and hold the heading they were left at. The right bumper means the robot's own directions. A points the robot at 45 degrees while the driver drives. Y sends the robot to its spot until a stick moves. And your own localizer from L120 runs beside the robot's.

Before you start.

- L170 is done, and so is L120, whose `L120EncoderLocalizer` this runs.
- AdvantageScope is installed on the same laptop. No gamepad is needed until the last step.
- The robot, the Driver Station, a gamepad, and about 3 m by 3 m of clear floor for the last step.

3.21.1 The steps

L180S010: copy the starter into mytry

Copy. In `lessons`, click `L180CombinedOpMode` once and copy it. Click `mytry` once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**.

Check.

- `mytry` holds `L180CombinedOpMode`, and the new file is open.

L180S020: switch your copy on

Change. Delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. `'public'` allows access from any code.

L180S030: run your localizer again

Change. Find `shadowLocalizers()` and the comment that names this step. Under it, type the same two lines as in L120:

```
// When on L180S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
// /l180.html#l180s030
// for what to do here.
shadowLocalizers.add("driveWheelEncoders",
    new L120EncoderLocalizer(new HardwareWheelSource(hardware)));
```

`HardwareWheelSource` turns red. Click it and press `Option+Enter`, as in L120. `L120EncoderLocalizer` does not turn red. It is your own, from L120, and it sits in `mytry` too.

Keep the name `driveWheelEncoders`. AdvantageScope shows each localizer under its name, so a run of this `OpMode` and a run of L120 line up.

Check.

- Nothing in the file is red.

L180S040: read what you are about to drive

Nothing else in the starter is new. Read `bindings()` and `driveTheRobot()`, and find the lesson each part came from.

- Both drive sticks go through deadband and then squared, from L060.
- `forward` and `strafe` are turned through the heading so they mean the field, from L140.
- The right bumper picks the robot's own directions, from L150.
- `hold.turn` keeps the heading the turn stick was left at, from L160.

- Y sends the robot to `target`, and any stick takes it back, from L170.

A is the one part that is new here, and it is `HeadingHold` again. `hold.aimAt` gives the hold a heading the driver did not leave, 45 degrees. The hold turns the robot there, and the left stick still drives. Push the turn stick and the driver has the turn back.

When a stick takes the robot back from Y, `hold.release()` forgets the heading the hold had before Y. Read the code without it: the hold would turn the robot back to that old heading.

Simulator. In the Terminal, run this. It holds A and pushes the left stick a little forward:

```
./gradlew :TeamCode:simRun --args="mytry.L180CombinedOpMode --a=true
--left_stick_y=-0.3"
```

AdvantageScope. Connect to the simulator as in L020, and click **2D Field**. Open the heading folder. Then open `Localizer` and both folders in it.

Check.

- `target_deg` reads 45.
- The robot turns until its arrow points up and to the right, and `pinPoint/heading_deg` comes to about 45.
- All the while it drives to the right, and `pinPoint/x_in` climbs.
- `driveWheelEncoders` is there, and it follows `pinPoint`: the same heading, and x and y within a tenth of an inch.

L180S050: drive all four

Robot. Stop the simulator. Send the code, and run L180 Combined from the Driver Station.

Check.

- Try them in this order: field relative, the bumper held, A, and Y.
- Each one does what its own lesson did, and switching between them needs nothing but a thumb.
- On the robot, `driveWheelEncoders` follows `pinPoint` as the robot drives.

No gamepad. The run above needs none. With `--pad`, a driver can try all four in turn.

If it didn't, and `LessonsTest.l180_everythingTogether` fails with the shadow localizer is still running, nothing was published under `Localizer/driveWheelEncoders/`. The two lines are missing, or the name is spelled differently. With the name typed as `encoders`, this was the message, and the other two tests passed.

If it didn't, and one of the four misbehaves on its own, go back to that lesson's page. This page adds no arithmetic, so a fault here is a fault there.

The other two tests, `l180_aPointsAt45DegreesWhileTheDriverKeepsDriving` and `l180_theTurnStickTakesAimingBack`, check the given A code. They pass on the starter.

What you just did. You wrote two lines and got a competition teleop. Every part lives in one place, so putting them together is choosing an order rather than writing code.

The order is the design. The driver's turn stick beats A's aim, and A's aim takes the place of the held heading. The follower drives only when nobody else has asked. The driver beats everything, and the robot fills in what the driver leaves.

Where next. L190 stops asking the wheels for power and starts asking them for a speed.

3.22 L190: ask the wheels for a speed

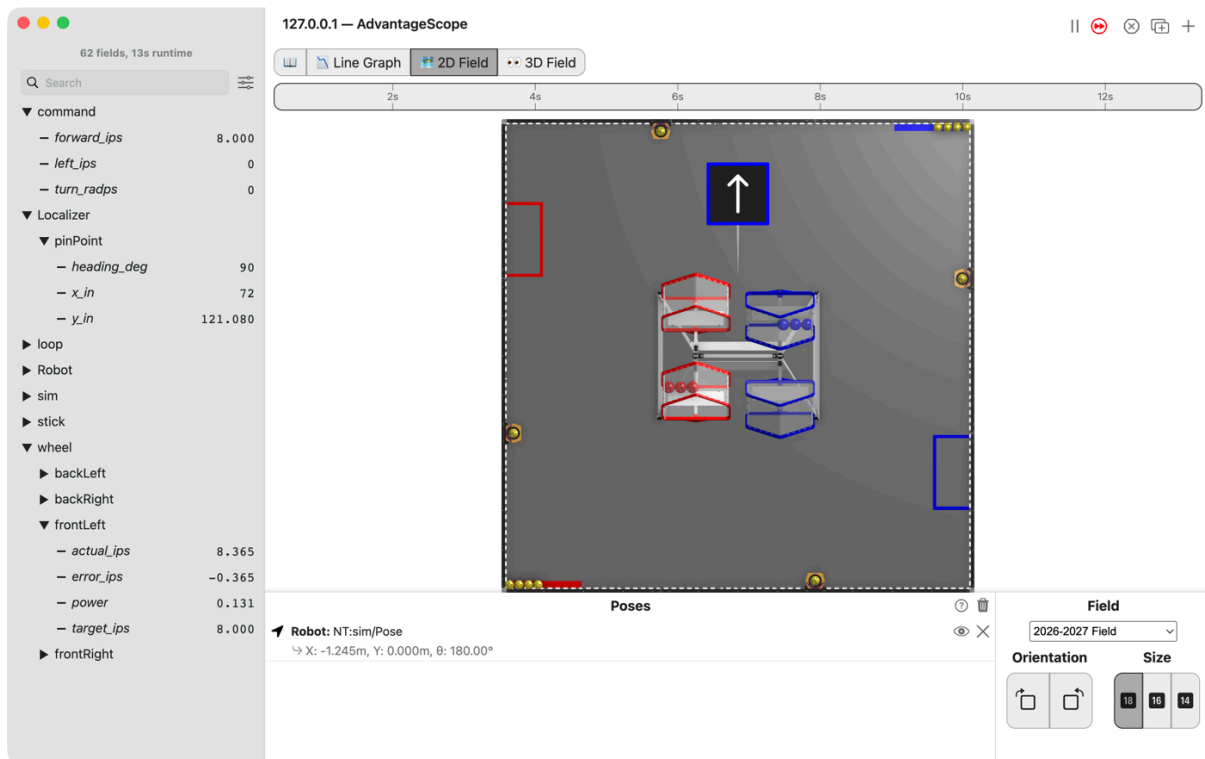


Figure 3.50. AdvantageScope's 2D Field with the robot near the top of the field, its arrow pointing up, and the list showing command/forward_ips at 8, pinPoint/y_in at 121.080, and wheel/frontLeft with target_ips at 8, actual_ips at 8.365, error_ips at -0.365 and power at 0.131.

Full stick means 40 inches per second, not full power. The drivetrain measures how fast each wheel turns and makes up the difference, on a fresh battery or a flat one, on the flat or up a slope.

Before you start.

- L180 is done.
- AdvantageScope is installed on the same laptop. No gamepad is needed until the last step.
- The robot, the Driver Station, a gamepad, about 3 m of clear floor, and something gentle to drive up, like a folded mat, for the last step.

3.22.1 The steps

L190S010: copy the two starters into mytry

This lesson has two files, as L110 did. One is the OpMode. The other is the drivetrain, which turns a speed into a power.

Copy. In lessons, click L190SpeedDriveTrain once and copy it. Click mytry once and paste. In the **Copy Class** box, click **OK**. If Android Studio asks whether to add the file to Git, click **Add**. Do the same for L190VelocityDriveOpMode.

Check.

- mytry holds L190SpeedDriveTrain and L190VelocityDriveOpMode.

L190S020: switch your copy on

Change. In `L190VelocityDriveOpMode`, delete the line `@Disabled`, under the comment that names this step.

Check.

- Under the step's comment, the next line reads `// 'public'` so the robot can run this `OpMode`. 'public' allows access from any code.

L190S030: make room for the drivetrain

Change. Near the top of the class, find the comment that names this step. Under it, type:

```
// When on L190S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1190.html#l190s030
// for what to do here.
L190SpeedDriveTrain drivetrain;
```

Check.

- The line has no red underline.
- `L190SpeedDriveTrain` is in `mytry` too, so it needs no import.

L190S040: hand the drivetrain to the follower

Change. In `init()`, find the comment that names this step. Under it, type the first line below. Then change `initAfter();` to `initAfter(drivetrain);`:

```
// When on L190S040, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1190.html#l190s040
// for what to do here.
drivetrain = new L190SpeedDriveTrain(hardware);

initAfter(drivetrain);
```

This is `L110S050` again, with the new drivetrain.

Check.

- Neither line has a red underline.

L190S050: turn the sticks into speeds

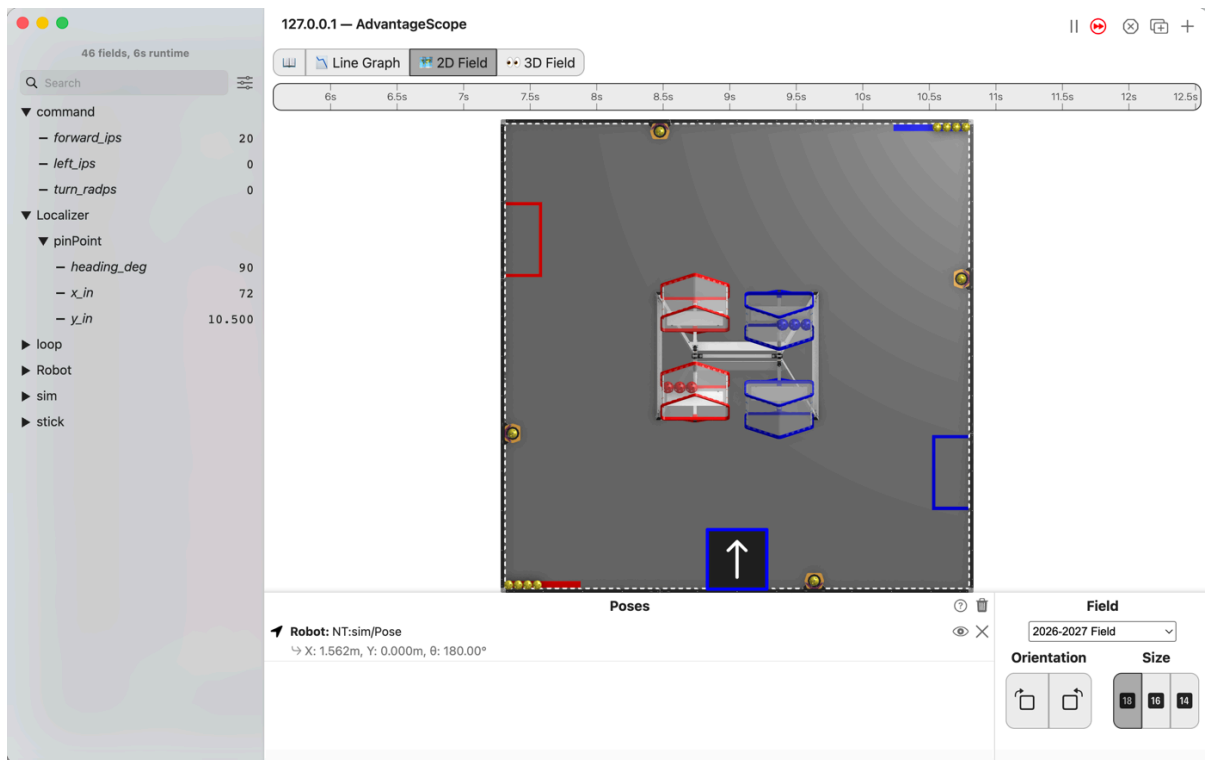


Figure 3.51. AdvantageScope's 2D Field with the robot at the bottom middle of the field, its arrow pointing up, and the list showing command/forward_ips at 20, left_ips and turn_radps at 0, and pinPoint at x 72, y 10.500 and heading 90, with no wheel folder.

MAX_IPS is 40, in inches per second. MAX_TURN_RADPS is pi radians per second, half a turn each second. Both are near the top of the class.

Change. In `loop()`, find the comment that names this step. Under it are three lines that end in `= 0;`. Change each 0 as below. Then, under the three publish lines, type the last two lines:

```
// When on L190S050, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//1190.html#l190s050
// for what to do here.
double forwardInPerS = deadband(-leftStickY, band) * MAX_IPS;
double strafeInPerS = deadband(-leftStickX, band) * MAX_IPS;
double turnRadPerS = deadband(-rightStickX, band) * MAX_TURN_RADPS;
Tracker.publish("command/forward_ips", forwardInPerS);
Tracker.publish("command/left_ips", strafeInPerS);
Tracker.publish("command/turn_radps", turnRadPerS);
double[] target = WheelTargets.forMecanum(forwardInPerS, strafeInPerS,
turnRadPerS, Constants.turnRadiusInches);
drivetrain.setCommandedWheelSpeeds(target[0], target[1], target[2],
target[3]);
```

The minus signs are the ones from L030. A stick pushed up reads less than 0.

`WheelTargets.forMecanum` works out how fast each wheel has to turn for the robot to move like that. It hands back an array, front left first, then front right, back left and back right. `turnRadiusInches` is how far each wheel sits from the middle of the robot.

`WheelTargets` and `Constants` turn red. Hold the mouse over each one until a box offers **Import class**, and click it. The box for `Constants` names `org.firstinspires.ftc.teamcode.pedro.Constants`.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L190VelocityDriveOpMode
--left_stick_y=-0.5"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the command folder. Then open Localizer and pinPoint.

Check.

- forward_ips reads 20, half of 40.
- The robot does not move, and y_in stays at 10.5.

The OpMode asks for a speed, but setCommandedWheelSpeeds in your drivetrain is still empty.

L190S060: turn a speed into a power

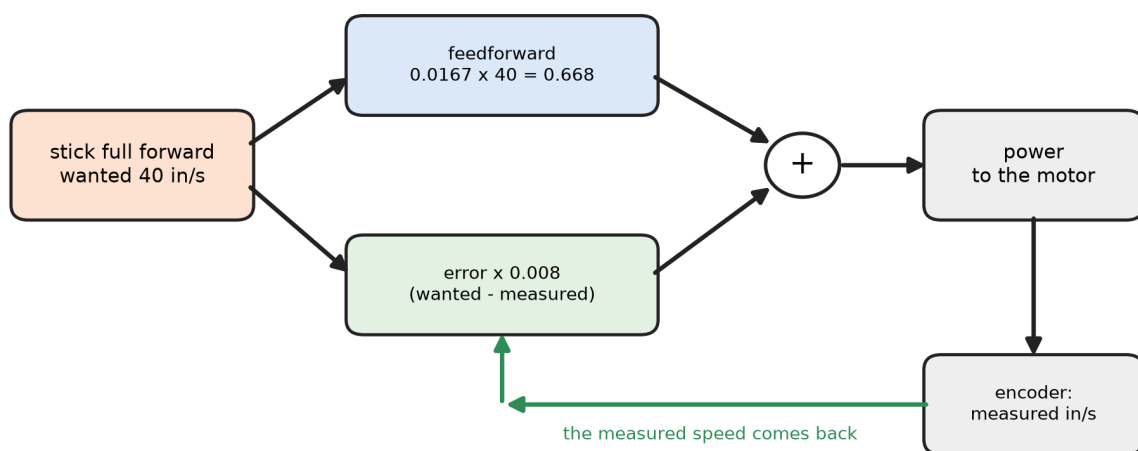


Figure 3.52. The loop drawn once round: the stick asking for 40 in/s, the feedforward guess at the power, the encoder's measured speed coming back, and the error being added in.

Open L190SpeedDriveTrain. It extends CorbelsMecanum, so it has everything L180's drivetrain had. setCommandedWheelSpeeds puts the four speeds into the array wanted. Each wheel gets its power from two parts added together.

- **Feedforward** is a guess, made before anything is measured. Constants.powerPerInchPerSecond is the power one inch per second needs, so the guess gets most of the way there at once.
- **Feedback** is the correction. measuredSpeeds.all() reads how fast each wheel really turns. The difference between wanted and measured, times VELOCITY_KP, is added on.

The guess is always a little off, and the correction only grows once the wheel is off its speed. Together they get there and stay there.

A for loop runs the lines inside its braces once for each wheel. i starts at 0 and goes up by 1 each time, while it is less than 4. So wanted[i] is wanted[FL] the first time round, then FR, BL and BR.

Change. Press Ctrl+C in the Terminal to stop the simulator. In setCommandedWheelSpeeds, find the comment that names this step. Under it, type:

```
// When on L190S060, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l190.html#l190s060
// for what to do here.
```

```

double[] measured = measuredSpeeds.all();
double[] powers = new double[4];
for (int i = 0; i < 4; i++) {
    double feedforward = Constants.powerPerInchPerSecond * wanted[i];
    double feedback = VELOCITY_KP * (wanted[i] - measured[i]);
    powers[i] = clampToPower(feedforward + feedback);
}
setCommandedWheels(powers[FL], powers[FR], powers[BL], powers[BR]);
publishWheelSpeeds(wanted, measured, powers);

```

Android Studio types the `}` of the `for` loop for you, when you press Enter after the `{`.

`clampToPower` keeps each power between -1 and 1. `setCommandedWheels` sends the four powers to the wheels, as in L110. `publishWheelSpeeds` puts each wheel's numbers in the `wheel` folder.

`Constants` turns red. Import it as in L190S050.

Simulator. Run the command again, with the stick pushed less so the robot stays on the field longer:

```

./gradlew :TeamCode:simRun --args="mytry.L190VelocityDriveOpMode
--left_stick_y=-0.2"

```

AdvantageScope. Connect, and open `wheel`, then `frontLeft`.

Check.

- `forward_ips` and `target_ips` read 8.
- The robot drives up the field.
- On the first loop `power` reads 0.198: the guess, 0.134, plus the correction for a wheel not yet turning.

`actual_ips` climbs to about 8 in under half a second and settles a little above 8, so `error_ips` settles a little below 0. `power` drops to about 0.13 and stays there. The guess was a little too much for the simulated robot, and the correction takes off what it needs.

L190S070: drive it up something

Robot. Stop the simulator. Send the code, run `L190 Velocity Drive` from the Driver Station, and watch it in `AdvantageScope`.

Check.

- Drive on the flat first.
- Then drive straight up the mat, with the stick held in the same place. The robot slows a little at the foot of the slope and then picks back up.
- In `AdvantageScope`, `actual_ips` dips below `target_ips`, then climbs back, and `power` rises to do it.

No gamepad. The runs above need none. With `--pad`, a driver can push the stick and watch the numbers.

If it didn't, and every `LessonsTest.L190_test` fails with this, nothing was published under `wheel` /:

```

NullPointerException: Cannot invoke "java.lang.Double.doubleValue()" because
the return value of "java.util.Map.get(Object)" is null

```

L190S060 is not typed yet, or `publishWheelSpeeds` is missing. The test asks the log for a number nobody wrote. The same message comes when the `drivetrain.setCommandedWheelSpeeds` line in L190S050 is missing.

If it didn't, and a test fails with one of these, the line named is the one to look at. Each was planted and run.

- full stick asks for a speed, in inches per second expected:<40.0> but was:<0.0>: the three speeds are still 0.
- expected:<40.0> but was:<1.0>: * MAX_IPS is missing. A full stick is 1, and 1 inch per second is very slow.
- expected:<40.0> but was:<-40.0>: the minus sign in front of leftStickY is missing. l190_whenTheWheelsAreUpToSpeedOnlyTheFeedforwardRemains fails with it, saying expected:<0.0> but was:<-80.0>.
- turning counter-clockwise drives the left side backwards, on its own: the minus sign in front of rightStickX is missing.
- expected:<0.9878391425450086> but was:<0.6678391425450086>: the feedback is not added. The difference is 0.008 * 40.
- expected:<0.9878391425450086> but was:<0.32>: the feedforward is not added. A wheel already at speed then gets no power at all.
- expected:<0.9878391425450086> but was:<0.34783914254500864>: the error is the wrong way round, measured[i] - wanted[i].
- measured speed matches the command expected:<40.0> but was:<0.0>: measured is not read from measuredSpeeds.all().

Two mistakes pass every test. Both were run in the simulator with `--left_stick_y=-0.2`, and in both the wheel numbers were right while the robot stood still at x 72 in, y 10.5 in.

- `setCommandedWheels` is missing, so the powers are worked out and logged but never sent.
- `initAfter()` ; was left as it was, so the follower holds a stand-in and never sends the powers on, as in L110.

A number in the log is what the code decided, not what the robot did.

If it didn't, and on the robot it is slower than `target_ips` everywhere, `powerPerInchPerSecond` is off for this robot. L117 measures it. If turning is the wrong speed while driving straight is right, `turnRadiusInches` is 8 in, and this robot may not be. L195 measures it.

What you just did. A power is an instruction, and a speed is a result. Every lesson before this one sent instructions. This one states the result it wants, measures, and corrects.

Where next. L195 measures the numbers this lesson trusted: how far a wheel goes for each encoder count, and the turn radius.

3.23 L195: measure the two numbers L190 trusted

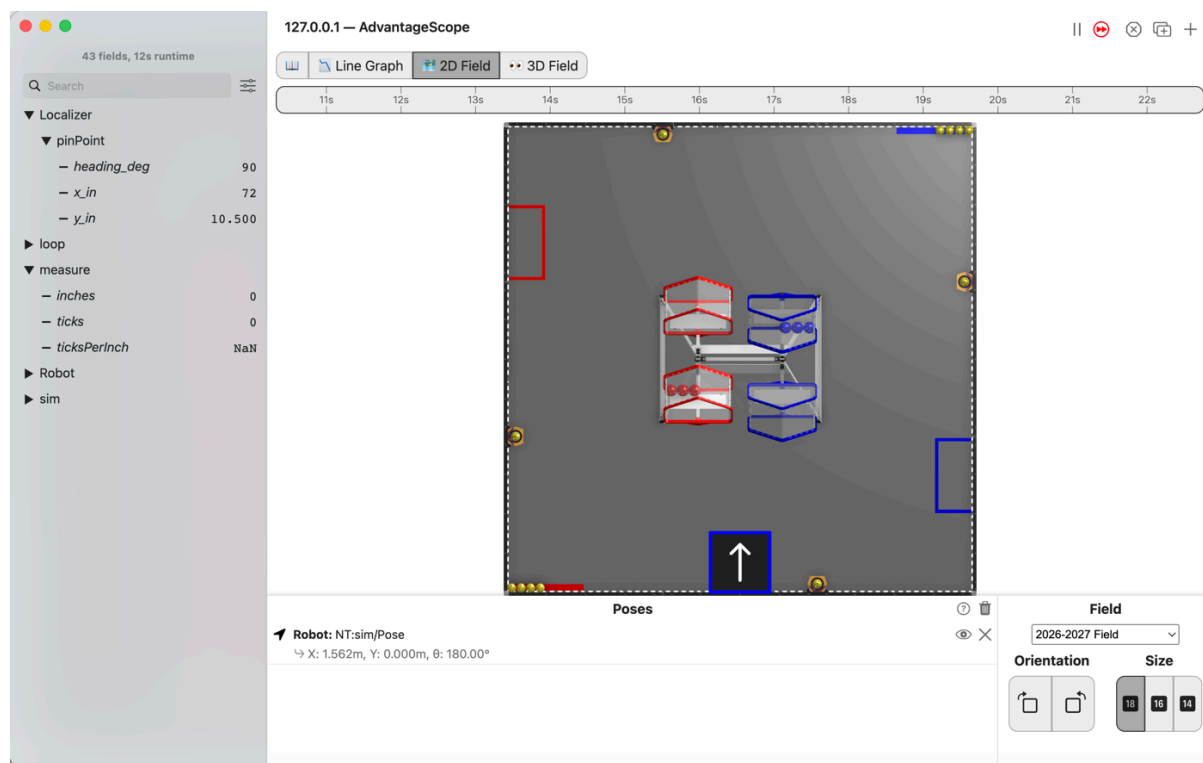


Figure 3.53. AdvantageScope's 2D Field with the robot at the bottom middle of the field, its arrow pointing up, and the list showing pinPoint at x 72, y 10.500 and heading 90, and measure with inches and ticks at 0 and ticksPerInch at NaN.

Two numbers in `Constants.java` that came off this robot instead of somebody else's: how many encoder ticks make an inch, and how far a wheel sits from the middle. The robot measures them while you push it and spin it by hand.

Before you start.

- L190 is done. It is the lesson that uses both numbers.
- The Pinpoint is mounted, as in L115, and its offsets set, as in L117, because this measurement believes it.
- AdvantageScope is installed on the same laptop.
- The robot, the Driver Station, about 1.5 m of clear straight floor for the push, and room to spin the robot on the spot.

3.23.1 The steps

L195S010: copy the three starters into mytry

This lesson has three files. Two are OpModes, one for each measurement. The third is the drivetrain, which tells them what each wheel's encoder has counted.

Copy. In `lessons`, click `L195MeasureTicksPerInchOpMode` once. Hold Shift and click `L195TicksDriveTrain`, so all three are picked. Copy them, click `mytry` once, and paste. In the **Copy** box, click **OK**. If Android Studio asks whether to add the files to Git, click **Add**.

Copy all three at once. Copied one at a time, the OpModes still point at the drivetrain in lessons, and turn red.

Check.

- mytry holds L195MeasureTicksPerInchOpMode, L195MeasureTurnRadiusOpMode and L195TicksDriveTrain, and none of them shows red.

L195S020: switch both OpModes on

Change. In L195MeasureTicksPerInchOpMode, delete the line @Disabled, under the comment that names this step. Do the same in L195MeasureTurnRadiusOpMode.

Check.

- In both, under the step's comment, the next line reads // 'public' so the robot can run this OpMode. 'public' allows access from any code.

L195S030: open a door to the encoders

Open L195TicksDriveTrain. A drivetrain sends four powers out to the motors. wheelTicks() is the same shape pointing the other way: it brings four numbers back.

Change. In wheelTicks(), find the comment that names this step. Under it, type:

```
// When on L195S030, see
// https://msb-stitt.github.io/ftc31459-26-27-biobuzz/guide/tasks
//l195.html#l195s030
// for what to do here.
ticks[FL] = hardware.frontLeft.getCurrentPosition();
ticks[FR] = hardware.frontRight.getCurrentPosition();
ticks[BL] = hardware.backLeft.getCurrentPosition();
ticks[BR] = hardware.backRight.getCurrentPosition();
```

getCurrentPosition() is how many ticks a motor's encoder has counted since the robot started. FL, FR, BL and BR name the slots, as in L190.

Simulator. In the Terminal, run:

```
./gradlew :TeamCode:simRun --args="mytry.L195MeasureTicksPerInchOpMode"
```

AdvantageScope. Connect to the simulator as in L020, click **2D Field**, and open the measure folder.

Check.

- inches and ticks read 0.
- ticksPerInch reads NaN, which means "not a number".

Nobody can push a robot in the simulator, so it stays at x 72 in and y 10.5 in. And 0 divided by 0 is not a number. The measuring is for the robot.

L195S040: push the robot and measure an inch

An encoder counts small steps, called **ticks**, as its motor turns. Many ticks make one turn of the wheel.

The robot keeps two accounts of the same trip. The Pinpoint says how far it went in inches, since its own wheels measure the floor. The drive encoders say how many ticks they counted. One divided by the other is the number.

Robot. Stop the simulator. Send the code, and run L195 Measure ticks per inch from the Driver Station.

Push. Push the robot slowly straight forward, at least 36 in, keeping it pointing the same way. Push slowly: a wheel that slips travels without turning, and the answer comes out short.

Check.

- The Driver Station shows `ticksPerInch` with a number after it.

Do it three times. Three answers a tick or two apart are a measurement.

Change. Open `Constants`, in the `pedro` folder, and put the number in `ticksPerInch`, which is 45.0 until you do. Send the code again and run it once more to check.

L195S050: spin the robot and measure the radius

A wheel r inches from the middle travels r inches for every radian the robot turns. So the radius is the wheel's inches divided by the radians. This `OpMode` turns ticks into inches with `ticksPerInch`, so L195S040 comes first.

Robot. Run `L195 Measure turn radius` from the Driver Station.

Spin. Spin the robot on the spot by hand, at least two full turns, keeping it in about the same place.

Check.

- The Driver Station shows a turn radius.
- Measure from the middle of the robot to the middle of one wheel, once across the robot and once along it. Add those two lengths. The sum is close to the turn radius the Driver Station shows.

Change. Put the number in `turnRadiusInches` in `Constants.java`, which is 8.0 until you do.

No gamepad. None of the runs above needs one. Pushing a robot by hand is not something the simulator does, so L195S040 and L195S050 are robot steps only.

If it didn't, and `LessonsTest.l195_theTicksDoorReportsWhatEachEncoderCounted` fails, the test puts 10, 20, 30 and 40 on the four encoders, so the numbers in the message name the wheel they came from. Each of these was planted and run.

- `front left, front right, back left, back right`: arrays first differed at element `[0]`; expected:`<10>` but was:`<0>`: L195S030 is not typed yet. Every measurement would say the robot never moved.
- arrays first differed at element `[1]`; expected:`<20>` but was:`<30>`: the front right and back left went into each other's slots. 30 is the back left's count, in the front right's place.

`getVelocity()` in place of `getCurrentPosition()` does not build. Android Studio underlines it red, and the test run stops with:

```
error: incompatible types: possible lossy conversion from double to int
```

`getVelocity()` is how fast a wheel turns, as a decimal. `getCurrentPosition()` is the one that counts.

With `@Disabled` left in, every test passes. The `OpMode` is missing from the Driver Station's list.

If it didn't, on the robot, and it says `Encoders counted 0 ticks` while the inches climb, L195S030 is not on the robot you sent to.

If it didn't, and the turn radius is about half what the tape says, the robot was swung round the room rather than spun on the spot. The Pinpoint counts the turning, and driving in a circle adds turning the wheels never made.

The wheels have to roll. Both `OpModes` make the wheels coast in `start()` and put braking back in `stop()`. It is given code, and the other two `l195_` tests watch it. They pass on the starter. Each of these was planted in `L195MeasureTicksPerInchOpMode` and run.

- `allowConfiguredBrakeMode()` added at the end of `loop()`: the coast lasts one loop, and the test says `L195MeasureTicksPerInchOpMode` still coasts five loops in. The wheels would brake while the student was still pushing.

- `allowConfiguredBrakeMode()` taken out of `stop()`: two messages, `L195MeasureTicksPerInchOpMode` brakes again afterwards and braking is back although no loop ran. The second is a run where STOP came before the first loop, which leaves the next `OpMode` coasting.

What you just did. Two constants became measurements. `Constants.java` came with 45.0 and 8.0 in it, and those are somebody else's robot. Nothing in L190 could tell they were wrong, because code cannot check a number it has no other way of knowing. Have the robot measure the same thing two ways, with two instruments that do not talk to each other, and compare.

Where next. L195 is the last lesson. [L117](#) tunes the numbers that need the robot driving itself rather than being pushed.

3.24 Pedro routes: a cheat sheet

Every way to say where the robot should go, and which ones to avoid. This project resolves `com.pedropathing:core:3.0.1`. Where the documentation and the jar disagree, the jar is what is written here.

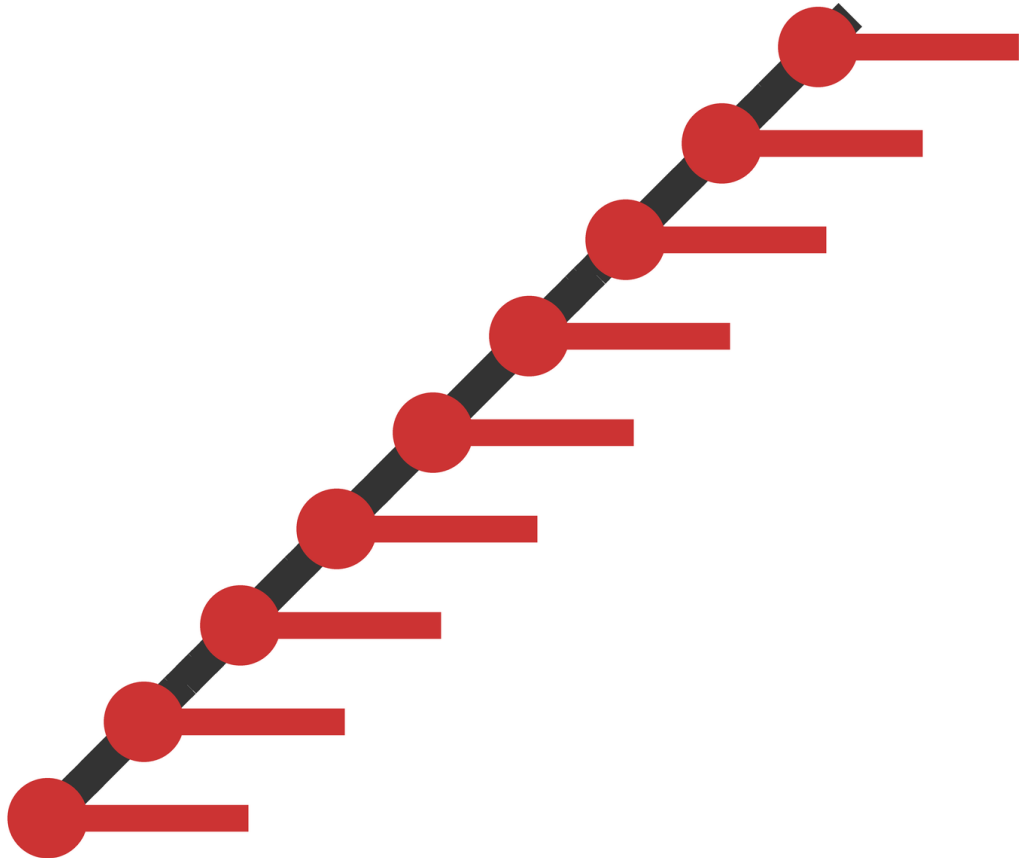
In the pictures the black line is where the robot drives and each red pin is its nose, sampled every eighth of the way along. Both come from `Path.get(t)` and `Path.heading(t)`, so they are the library's own output rather than a drawing of it.

3.24.1 The shapes

A route is a shape plus a heading rule. There are four shapes.

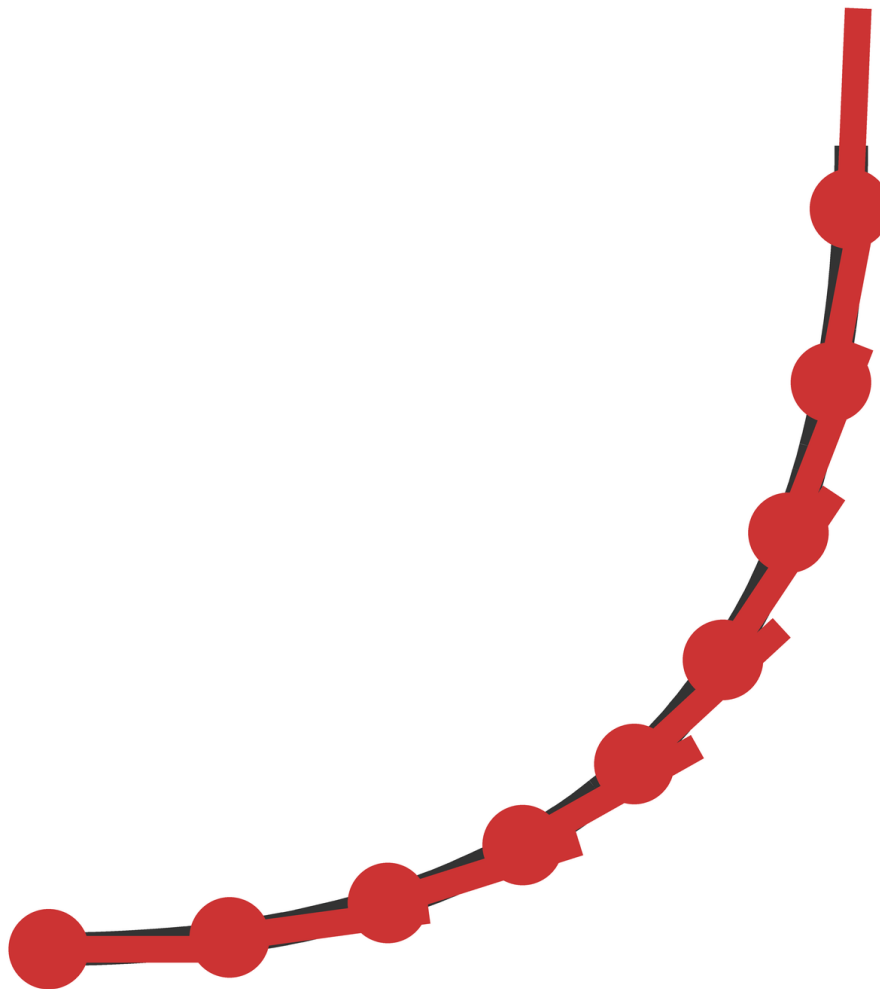
Shape	What it is
<code>Paths.line(start, end)</code>	A straight line between two poses.
<code>Paths.curve(start, control..., end)</code>	A Bézier curve. The first and last poses are the ends; every pose between them pulls the curve without being touched.
<code>Paths.through(poses...)</code>	A curve that passes through every pose given.
<code>Paths.path(paths...)</code>	Several paths run as one, in order.

`line`



Two poses, nothing between them. `line(start, end).constant(start)` is the plainest route there is and the one to reach for first.

curve



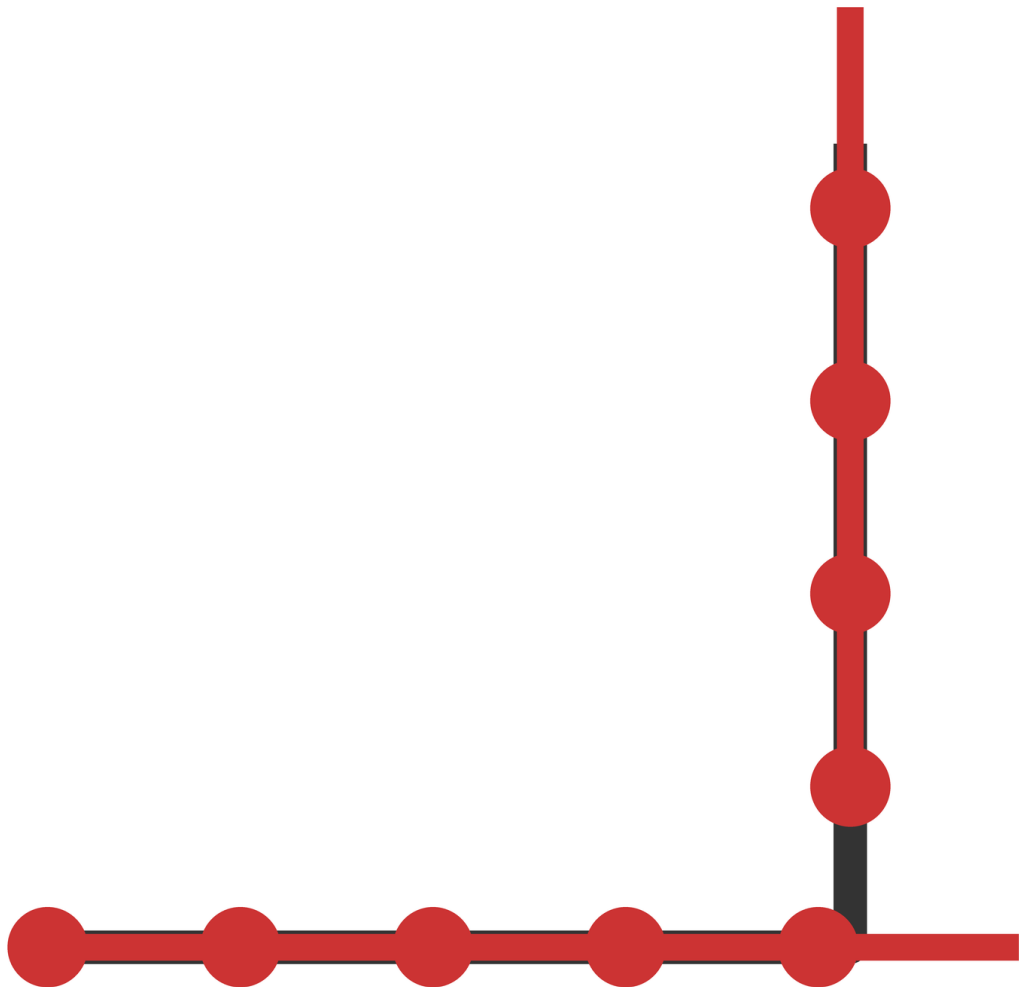
The control pose is a magnet, not a waypoint: the curve bends toward it and never reaches it. Its heading is ignored.

through



Every pose is on the route. The cost is at the ends: with `tangent()` the nose starts at -18.4 degrees and finishes at 108.4 degrees on the same three poses where `curve` runs 0 to 90, because the spline arrives at an angle. Check the end heading rather than assuming it.

path



Two legs of a route, joined. The nose snaps at the corner rather than easing round it: measured 0 degrees for the first leg and 90 for the second, with nothing in between.

3.24.2 The heading rules

Rule	What it does	Use it?
<code>constant (pose)</code>	Holds that pose's heading the whole way.	Yes.
<code>constant (radians)</code>	The same, from a number.	Careful: radians.
<code>tangent ()</code>	Nose follows the direction of travel.	Yes.
<code>reverseTangent ()</code>	Tangent plus 180 degrees, for driving backwards.	Yes.
<code>facingPoint (pose)</code>	Nose always aimed at one spot on the field.	Yes, with one edge.
<code>linear (a, b)</code>	Sweeps from one heading to another along the route.	Yes.

<code>longLinear(a, b)</code>	The same, taking the long way round.	Yes, undocumented.
<code>Interpolator.pieceswise()</code>	A different rule on each stretch of the route.	Not covered here.

There is no default

The documentation says `tangent()` “is the default interpolation type when you create a path”. It is not. A path with no heading rule throws as soon as anything asks for a heading:

```
java.lang.UnsupportedOperationException: No heading interpolator set.
```

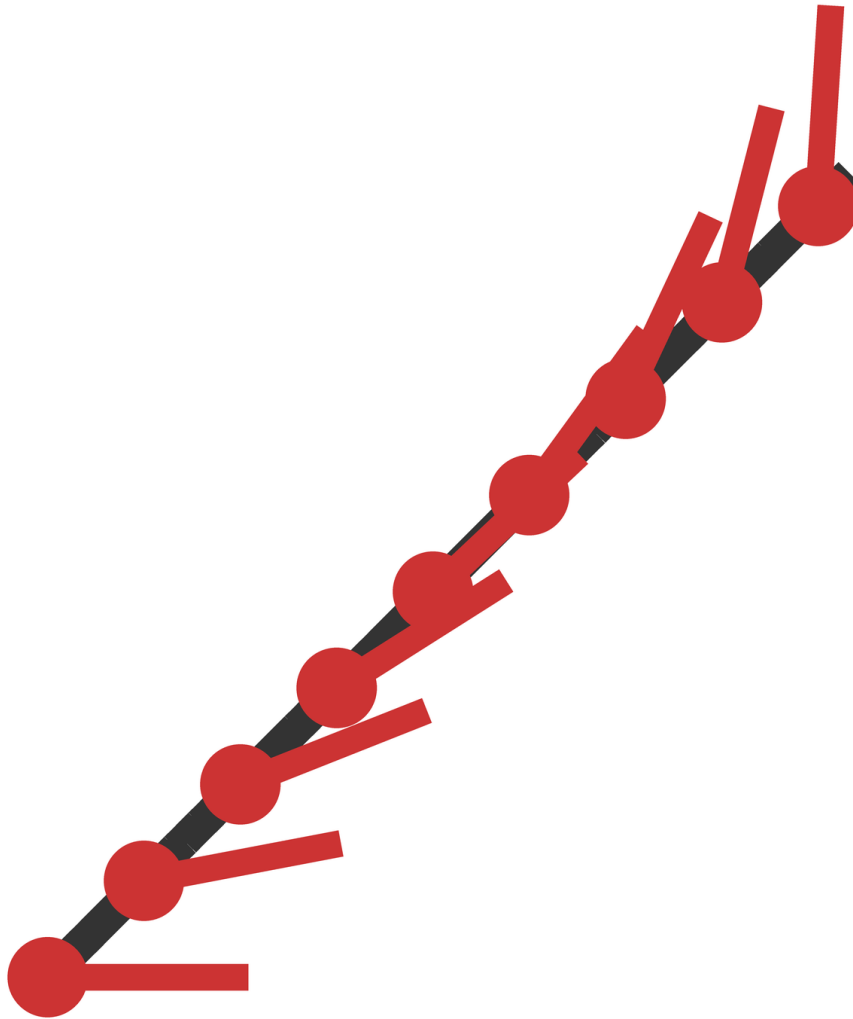
So every route needs a rule. It is not optional.

The number form is radians

`constant(90)` is not 90 degrees. It is 90 radians, which wraps to 116.6 degrees. `constant(0)` is safe only because zero is the same in both.

Two ways to avoid it. Pass a pose, `constant(start)`, and the heading comes from the pose. Or convert, `constant(Math.toRadians(90))`. A pose built by `PoseFactory.degrees()` converts for you; `constant` does not.

3.24.3 `linear()` on a line was reversed before 3.0.1



Asked to sweep from 0 to 90 degrees along a line, 3.0.1 does, and `endPose().heading()` reports the end heading. 3.0.0 swept from 90 to 0 and reported the start heading. On both:

on <code>line(s, e)</code>	version	t=0	t=0.5	t=1
<code>.linear(0, toRadians(90))</code>	3.0.1	0.0	45.0	90.0
<code>.linear(0, toRadians(90))</code>	3.0.0	90.0	45.0	0.0
<code>.linear(e, s), swapped</code>	3.0.1	90.0	45.0	0.0

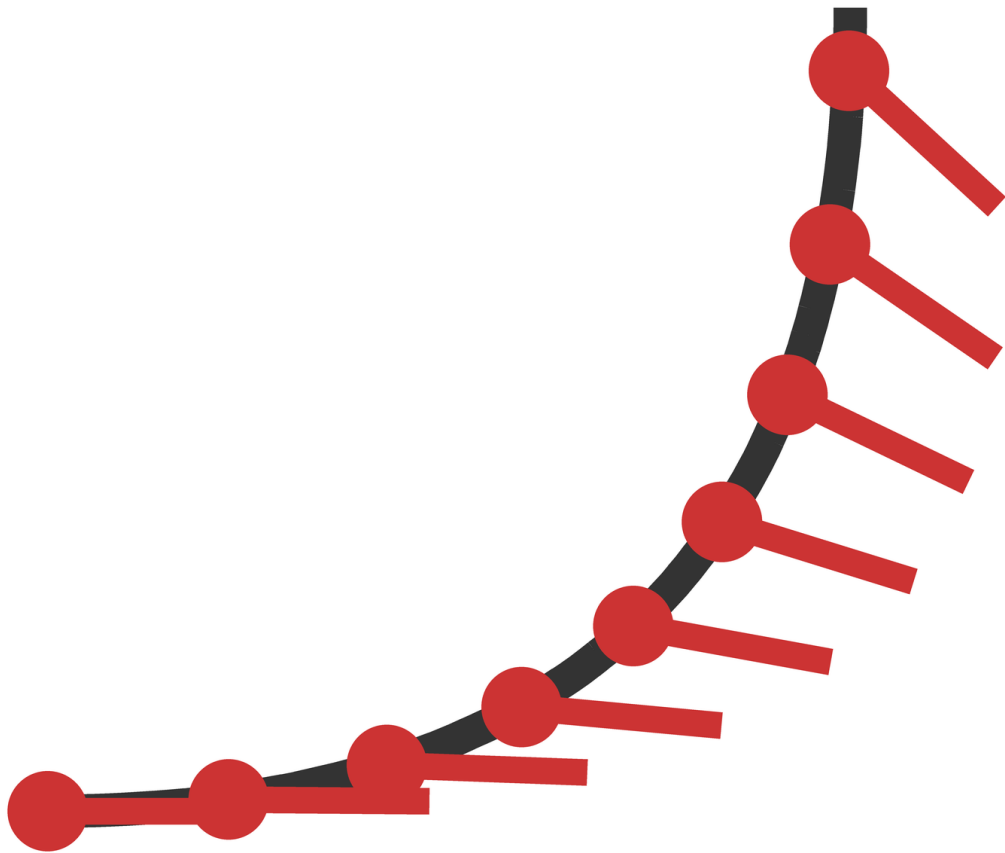
This was [issue 176](#), with [180](#) and [181](#) reporting the same thing. The cause given there is `Curve.pathCompletion()` returning the fraction of the path remaining rather than the fraction completed.

The workaround in those issues was to swap the arguments. On 3.0.1 the swap turns the robot the wrong way, so a swapped `linear()` in code written for 3.0.0 is now a bug. On a curve, `linear()` is the same in both versions: `curve(s, m, e).linear(0, toRadians(90))` runs 0, 24.6, 45, 90.

3.24.4 `longLinear()` takes the long way round

`longLinear(0, toRadians(90))` runs 0, 292.5, 225, then 90 at the end: clockwise, the long way from 0 to 90. On 3.0.0 it was reversed the same way as `linear()`, running 90, 157.5, 225, then 0. It is undocumented: the documentation page lists `tangent`, `linear`, `constant`, `facingPoint`, `piecewise` and `custom`, and not this.

3.24.5 `facingPoint()` has one edge



It works: the nose tracks the point as the robot moves. The edge is a route that starts *at* the point it faces, where there is no direction to face. Measured on a line leaving the point, the heading reads 0 at $t=0$ and the correct -135 degrees from the next sample on, so the first instant is a discontinuity. [Issue 69](#) is about this interpolator; it is closed.

3.24.6 What this page does not cover

- `Interpolator.piecewise()`, which nothing here has run.
- Custom interpolators.

- Deceleration, path and end constraints, and foresight, each of which has its own reference page and changes how a route is driven rather than what it is.
- Whether any of this behaves on a floor. Everything above is the library's own arithmetic, read out of the jar. When the simulator and the robot disagree, the robot is right.

3.25 Glossary

Every word here is one the task pages use, and each entry names the page that introduces it. Where the guide could have invented a friendlier word and did not, the reason is that the code, Pedro's documentation and the Driver Station all use this one.

OpMode

One program the Driver Station can run. Each lesson is an OpMode, and its name in the list is the one in its `@TeleOp` or `@Autonomous` line. [L020](#)

Simulator

A robot that exists only in the laptop. It runs the same OpMode, so code can be tried before the real robot is near. [L020](#)

AdvantageScope

The program that shows what the code published. It can show it live or from a flight log, as numbers, as graphs, or as a robot on the field. [The cheat sheet](#) lists every name the lessons publish. [L020](#)

Flight log

The file a run writes. It holds every number the code published, so you can look at the run later. [L020](#)

TeleOp and Autonomous

The two kinds of OpMode. A TeleOp reads the gamepads; an autonomous runs on its own. [L125](#) and [L130](#) are the two autonomous lessons. [L125](#)

Driver Station

The phone or tablet that holds the gamepads and picks the OpMode. Lines printed with `Tracker.printToDs` show up there. [L050](#)

Drivetrain

The four motors and the code that decides what power each one gets. [L040](#) sets the four powers itself. From [L110](#) on, a lesson's drivetrain is a class of its own. [L090](#)

Deadband

A band of stick readings near the middle that count as let go. Without one the robot creeps when nobody is touching it. [L060](#)

Method

A named piece of code you can use again. It takes numbers in and can give one back. [L060](#)

Squaring a stick

Making a small push mean a smaller power, by multiplying the stick by itself and putting the sign back. [L060](#)

Scaling the powers

Bringing two or four wheel powers down together when the biggest one is over 1. The robot still goes where it was asked. It only goes slower. [L080](#)

Strafe

Driving sideways without turning. [L090](#)

Mixing

Turning forward, sideways and turn into four wheel powers with four sums. [L090](#)

Mecanum

A wheel with rollers set at 45 degrees on its rim, which lets the robot drive sideways. The four

wheels' rollers make an X seen from above. [L090](#)

Commanded wheels

Four powers a lesson has set itself. The path follower cannot overrule them. It gets the wheels back when the lesson hands them back. [L110](#)

Array

Several numbers in a row, in one variable. `wheels[FL]` is one of them. [L110](#)

Follower

Pedro's code that drives the robot along a path, or holds it at a pose. From L110 on it holds the drivetrain and the lessons reach past it. [L110](#)

Pose

Where the robot is and which way it faces: x, y and a heading. A pose is a place on the field, not a distance from the robot. [L040](#)

Heading

Which way the robot is facing, as an angle. Counter-clockwise counts up, and the code keeps it in radians even where a page says degrees. [L090](#)

Localizer

Whatever works out the pose. A robot can run more than one at a time. That is what L120 is for. [L120](#)

Shadow localizer

A second localizer that is read and logged but never steers anything. It is how two answers get compared on one run. [L120](#)

Encoder

The counter inside a drive motor that counts how far the motor has turned. [L120](#)

Dead reckoning

Working out where you are by adding up every step you took, with nothing to check it against. The drive encoders do it; the Pinpoint does it better. [L120](#)

Pinpoint

The odometry computer. It has its own two measuring wheels, and it works out the pose. The follower believes this one. L195 measures against it. [L117](#)

Path

A line or curve the follower drives along, with a heading to hold while it does. [L125](#)

Hold

Telling the follower to sit at one pose and stay there. If something pushes the robot, the follower puts it back. [L130](#)

Field relative

Sticks that mean the field. Push away from you and the robot moves away from you, whichever way its nose points. [L140](#)

Robot relative

Sticks that mean the robot's own directions. Better for lining up against something. [L150](#)

Heading hold

Keeping the nose where the driver left it, and giving it straight back the moment the turn stick moves. [L160](#)

Instant command

A command that does its work and finishes in the same breath, rather than running for a while. A button press is usually one. [L170](#)

Feedforward

A guess at the power a wanted speed needs, made before anything is measured. Gets most of the way there at once. [L190](#)

Feedback

A correction worked out from two speeds: the one asked for and the one measured. It cleans up

what the guess got wrong. [L190](#)

Ticks

What an encoder counts as a motor turns. A tick is a fixed fraction of a turn, so ticks become inches once you know how many make one. [L195](#)

Ticks per inch

How many ticks a wheel counts for an inch of travel. Measured, not looked up. [L195](#)

Turn radius

How far a wheel rolls each time the robot spins one radian. Measure from the middle of the robot to a wheel, once across and once along, and add the two. It turns a spin speed into a wheel speed. [L195](#)

Coast and brake

What a motor does at zero power. Braking holds the robot still; coasting lets the wheels roll, which is what a measurement you push by hand needs. [L195](#)

3.26 Cheat sheet

Every table here is read straight out of the lesson code, so it says what the code says.

3.26.1 What to run

One OpMode name for the Driver Station, one command for the laptop. Each command runs only that lesson's own tests, which is what makes a red one easy to read. A dash means no test names that lesson: its check is on the floor.

Lesson	On the robot	On the laptop
L020	L020 Sticks (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1020*'</code>
L030	L030 Speeds (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1030*'</code>
L040	L040 Tank (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1040*'</code>
L050	L050 Blocks (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1050*'</code>
L060	L060 Shaping (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1060*'</code>
L070	L070 Buttons (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1070*'</code>
L080	L080 Arcade (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1080*'</code>
L090	L090 Mecanum (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1090*'</code>
L110	L110 Follower (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.1110*'</code>

L120	L120 Compare Localizers (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l120_theEncoderLocalizerRunsAlongsideAndIsLogged'</code> <code>--tests '*L120EncoderLocalizerTest*'</code>
L125	L125 Drive 24 (Autonomous)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l125*'</code>
L130	L130 Forward Then Strafe (Autonomous)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l130*'</code>
L140	L140 Field Relative (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l140*'</code>
L150	L150 Robot Relative Button (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l150*'</code>
L160	L160 Heading Hold (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l160*'</code>
L170	L170 Drive To Pose (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l170*'</code>
L180	L180 Combined (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l180*'</code>
L190	L190 Velocity Drive (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l190*'</code>
L195	L195 Measure ticks per inch (TeleOp) L195 Measure turn radius (TeleOp)	<code>./gradlew :TeamCode:testDebugUnitTest --tests '*LessonsTest.l195*'</code>

3.26.2 What shows up in AdvantageScope

A key ending in . . . has a name added to the end of it, one per wheel or per localizer.

Log key	Published by
Localizer/driveWheelEncoders /...	L120CompareLocalizersOpMode, L180CombinedOpMode
arcade/forward	L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
arcade/strafe	L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
arcade/turn	L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
button/a	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
button/event	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
command/forward_ips	L190VelocityDriveOpMode
command/left_ips	L190VelocityDriveOpMode
command/turn_radps	L190VelocityDriveOpMode
drive/mode	L170DriveToPoseOpMode, L180CombinedOpMode
drive/robotRelative	L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
heading/holding	L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode
heading/target_deg	L180CombinedOpMode
measure/inches	L195MeasureTicksPerInchOpMode
measure/radians	L195MeasureTurnRadiusOpMode
measure/ticks	L195MeasureTicksPerInchOpMode
measure/ticksPerInch	L195MeasureTicksPerInchOpMode
measure/turnRadius_in	L195MeasureTurnRadiusOpMode
measure/wheel_inches	L195MeasureTurnRadiusOpMode

power/backLeft	L090MecanumOpMode
power/backRight	L090MecanumOpMode
power/frontLeft	L090MecanumOpMode
power/frontRight	L090MecanumOpMode
power/left	L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode
power/right	L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode
speed/left	L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode
speed/right	L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode
stick/leftX	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode, L190VelocityDriveOpMode
stick/leftY	L020SticksOpMode, L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode, L190VelocityDriveOpMode
stick/rightX	L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode, L190VelocityDriveOpMode
stick/rightY	L020SticksOpMode, L030SpeedsOpMode, L040TankOpMode, L050BlocksOpMode, L060ShapingOpMode, L070ButtonsOpMode, L080ArcadeOpMode, L090MecanumOpMode, L110FollowerOpMode, L120CompareLocalizersOpMode, L140FieldRelativeOpMode, L150RobotRelativeButtonOpMode, L160HeadingHoldOpMode, L170DriveToPoseOpMode, L180CombinedOpMode

Generated by applying solutions/ on solutions-1010.1, every lesson in order.

3.27 Where this came from

3.27.1 The guide's own terms

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You may share these pages and build on them. Credit Catholic Central Spires Robotics, link the license, and do not sell what you make.

The repository's own `LICENSE` is FIRST's notice over the FTC SDK. It covers the code, not these pages.

3.27.2 Game Manual 0

[Game Manual 0](#) is licensed [CC BY-NC 2.0 Generic](#). It is a general FTC reference, written by FTC teams. This guide is not trying to replace it. These pages are about one robot and one set of lessons.

One page leans on it. [L090](#) explains why the mecanum rollers make an X, and links GM0's page on holonomic drivetrains, which shows it. No page here copies GM0's words. The guide's license is chosen to keep every restriction GM0's asks for anyway.

3.27.3 Pedro Pathing

[Pedro Pathing](#) is the path following library the lessons hand the drivetrain to. It is licensed BSD 3-Clause, and its name and logo are its own. This guide links Pedro's documentation rather than restating it, and the code the lessons write is ours.

3.27.4 The FTC SDK

The **FTC SDK** is FIRST's, under the BSD-style notice in the repository's root `LICENSE`. The Driver Station, the Control Hub and every `@TeleOp` in this guide are its.

3.27.5 AdvantageScope

AdvantageScope, from the Mechanical Advantage team, is what opens a flight log after a run.

3.27.6 Not affiliated with FIRST

FIRST and FIRST Tech Challenge are trademarks of For Inspiration and Recognition of Science and Technology.

This guide is written by one team for its own students. It is not produced by FIRST, endorsed by FIRST, or affiliated with FIRST.

3.28 Answers

What each lesson changes, file by file, out of the solutions line. Use it when you are stuck, not instead of being stuck.

3.28.1 L020SticksOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying `solutions/` on `solutions-l010.1`:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry  
/L020SticksOpMode.java
```

L020*A change*

Before L020:

```
(nothing)
```

After L020:

```
import org.firstinspires.ftc.teamcode.base.Tracker;
```

A change

Before L020:

```
(nothing)
```

After L020:

```
double leftStickY = gamepad1.left_stick_y;
```

A change

Before L020:

```
(nothing)
```

After L020:

```
Tracker.publish("stick/leftY", leftStickY);
```

A change

Before L020:

```
(nothing)
```

After L020:

```
double rightStickY = gamepad1.right_stick_y;
Tracker.publish("stick/rightY", rightStickY);
```

3.28.2 L030SpeedsOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L030SpeedsOpMode.java
```

L030*A change*

Before L030:

```
(nothing)
```

After L030:

```
double leftSpeed = -leftStickY;
Tracker.publish("speed/left", leftSpeed);
```

A change

Before L030:

```
(nothing)
```

After L030:

```
double rightSpeed = -rightStickY;  
Tracker.publish("speed/right", rightSpeed);
```

3.28.3 L040TankOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010.1:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry  
/L040TankOpMode.java
```

L040

A change

Before L040:

```
(nothing)
```

After L040:

```
hardware.frontLeft.setPower(leftSpeed);  
hardware.backLeft.setPower(leftSpeed);
```

A change

Before L040:

```
(nothing)
```

After L040:

```
hardware.frontRight.setPower(rightSpeed);  
hardware.backRight.setPower(rightSpeed);
```

3.28.4 L050BlocksOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010.1:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry  
/L050BlocksOpMode.java
```

L050

A change

Before L050:

```
(nothing)
```

After L050:

```
hardware.frontLeft.setDirection(hardware.mecanumConfig.frontLeftDirection.get());  
hardware.frontRight.setDirection(hardware.mecanumConfig.frontRightDirection.get());  
hardware.backLeft.setDirection(hardware.mecanumConfig.backLeftDirection.get());  
hardware.backRight.setDirection(hardware.mecanumConfig.backRightDirection.get());
```

3.28.5 L060ShapingOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/

on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L060ShapingOpMode.java

L060

A change

Before L060:

```
(nothing)
```

After L060:

```
double band = 0.05;
double leftPower = squared(deadband(leftSpeed, band));
Tracker.publish("power/left", leftPower);
double rightPower = squared(deadband(rightSpeed, band));
Tracker.publish("power/right", rightPower);
```

A change

Before L060:

```
hardware.frontLeft.setPower(leftSpeed);
hardware.backLeft.setPower(leftSpeed);
```

After L060:

```
hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);
```

A change

Before L060:

```
hardware.frontRight.setPower(rightSpeed);
hardware.backRight.setPower(rightSpeed);
```

After L060:

```
hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

A change

Before L060:

```
(nothing)
```

After L060:

```
double deadband(double value, double band) {
    if (Math.abs(value) < band) {
        return 0;
    }
    return value;
}
```

A change

Before L060:

```
(nothing)
```

After L060:

```
double squared(double value) {
```

```
        return value * Math.abs(value);  
    }
```

3.28.6 L070ButtonsOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L070ButtonsOpMode.java

L070

A change

Before L070:

```
(nothing)
```

After L070:

```
boolean previousButtonA;
```

A change

Before L070:

```
(nothing)
```

After L070:

```
double leftStickX = gamepad1.left_stick_x;  
Tracker.publish("stick/leftX", leftStickX);  
double rightStickX = gamepad1.right_stick_x;  
Tracker.publish("stick/rightX", rightStickX);
```

A change

Before L070:

```
(nothing)
```

After L070:

```
boolean buttonA = gamepad1.a;  
Tracker.publish("button/a", buttonA);
```

A change

Before L070:

```
(nothing)
```

After L070:

```
if (buttonA != previousButtonA) {  
    if (buttonA) {  
        Tracker.publish("button/event", "A pressed");  
    } else {  
        Tracker.publish("button/event", "A released");  
    }  
}  
previousButtonA = buttonA;
```

3.28.7 L080ArcadeOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L080ArcadeOpMode.java

L080

A change

Before L080:

```
(nothing)
```

After L080:

```
double band = 0.05;
double forward = squared(deadband(-leftStickY, band));
Tracker.publish("arcade/forward", forward);
double turn = squared(deadband(-rightStickX, band));
Tracker.publish("arcade/turn", turn);
```

A change

Before L080:

```
(nothing)
```

After L080:

```
double leftPower = forward - turn;
double rightPower = forward + turn;
double biggest = Math.max(1, Math.max(Math.abs(leftPower),
Math.abs(rightPower)));
leftPower = leftPower / biggest;
rightPower = rightPower / biggest;
Tracker.publish("power/left", leftPower);
Tracker.publish("power/right", rightPower);

hardware.frontLeft.setPower(leftPower);
hardware.backLeft.setPower(leftPower);
hardware.frontRight.setPower(rightPower);
hardware.backRight.setPower(rightPower);
```

3.28.8 L090MecanumOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L090MecanumOpMode.java

L090

A change

Before L090:

```
(nothing)
```

After L090:

```
double strafe = squared(deadband(-leftStickX, band));
Tracker.publish("arcade/strafe", strafe);
```

A change

Before L090:

(nothing)

After L090:

```
double frontLeftPower = forward - strafe - turn;
double frontRightPower = forward + strafe + turn;
double backLeftPower = forward + strafe - turn;
double backRightPower = forward - strafe + turn;
double biggest = Math.max(1, Math.max(Math.max(Math.abs(frontLeftPower),
Math.abs(frontRightPower)),
Math.max(Math.abs(backLeftPower), Math.abs(backRightPower))));
frontLeftPower = frontLeftPower / biggest;
frontRightPower = frontRightPower / biggest;
backLeftPower = backLeftPower / biggest;
backRightPower = backRightPower / biggest;
Tracker.publish("power/frontLeft", frontLeftPower);
Tracker.publish("power/frontRight", frontRightPower);
Tracker.publish("power/backLeft", backLeftPower);
Tracker.publish("power/backRight", backRightPower);

hardware.frontLeft.setPower(frontLeftPower);
hardware.frontRight.setPower(frontRightPower);
hardware.backLeft.setPower(backLeftPower);
hardware.backRight.setPower(backRightPower);
```

3.28.9 L110FollowerDriveTrain

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L110FollowerDriveTrain.java

L110*A change*

Before L110:

(nothing)

After L110:

```
wheels[FL] = forward - strafe - turn;
wheels[FR] = forward + strafe + turn;
wheels[BL] = forward + strafe - turn;
wheels[BR] = forward - strafe + turn;
```

3.28.10 L110FollowerOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L110FollowerOpMode.java

L110*A change*

Before L110:

```
(nothing)
```

After L110:

```
L110FollowerDriveTrain drivetrain;
```

A change

Before L110:

```
(nothing)
```

After L110:

```
drivetrain = new L110FollowerDriveTrain(hardware);
```

A change

Before L110:

```
initAfter();
```

After L110:

```
initAfter(drivetrain);
```

A change

Before L110:

```
(nothing)
```

After L110:

```
drivetrain.sticks(forward, strafe, turn);
```

3.28.11 L120CompareLocalizersOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

```
TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry  
/L120CompareLocalizersOpMode.java
```

L120

A change

Before L120:

```
(nothing)
```

After L120:

```
import org.firstinspires.ftc.teamcode.base.odometry.HardwareWheelSource;
```

A change

Before L120:

```
(nothing)
```

After L120:

```
shadowLocalizers.add("driveWheelEncoders",  
    new L120EncoderLocalizer(new HardwareWheelSource(hardware)));
```

3.28.12 L120EncoderLocalizer

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L120EncoderLocalizer.java

L120

A change

Before L120:

```
(nothing)
```

After L120:

```
if (lastWheels == null) {
    lastWheels = wheels.clone();
    lastRawHeading = rawHeading;
    return;
}

double frontLeft = wheels[0] - lastWheels[0];
double frontRight = wheels[1] - lastWheels[1];
double backLeft = wheels[2] - lastWheels[2];
double backRight = wheels[3] - lastWheels[3];

double forward = (frontLeft + frontRight + backLeft + backRight) / 4;
double strafe = (-frontLeft + frontRight + backLeft - backRight) / 4;
```

A change

Before L120:

```
(nothing)
```

After L120:

```
double heading = rawHeading + headingOffset;
x += forward * Math.cos(heading) - strafe * Math.sin(heading);
y += forward * Math.sin(heading) + strafe * Math.cos(heading);
```

3.28.13 L125Drive24OpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L125Drive24OpMode.java

L125

A change

Before L125:

```
return Command.NOOP;
```

After L125:

```
return sequential(follow(follower, line(start, end).constant(start)));
```

3.28.14 L130ForwardThenStrafeOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L130ForwardThenStrafeOpMode.java

L130

A change

Before L130:

```
(nothing)
```

After L130:

```
import static com.pedropathing.ivy.pedro.PedroCommands.hold;
```

A change

Before L130:

```
follow(follower, line(start, corner).constant(start))
```

After L130:

```
follow(follower, line(start, corner).constant(start)),
```

A change

Before L130:

```
(nothing)
```

After L130:

```
follow(follower, line(corner, end).constant(start)),  
hold(follower, end)
```

3.28.15 L140FieldRelativeOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L140FieldRelativeOpMode.java

L140

A change

Before L140:

```
drivetrain.sticks(forward, strafe, turn);
```

After L140:

```
double heading = follower.pose().heading();  
double robotForward = forward * Math.cos(heading) + strafe *  
Math.sin(heading);  
double robotStrafe = -forward * Math.sin(heading) + strafe *  
Math.cos(heading);  
drivetrain.sticks(robotForward, robotStrafe, turn);
```

3.28.16 L150RobotRelativeButtonOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L150RobotRelativeButtonOpMode.java

L150

A change

Before L150:

```
drivetrain.sticks(robotForward, robotStrafe, turn);
```

After L150:

```
boolean robotRelative = gamepad1.right_bumper;
Tracker.publish("drive/robotRelative", robotRelative);
if (robotRelative) {
    drivetrain.sticks(forward, strafe, turn);
} else {
    drivetrain.sticks(robotForward, robotStrafe, turn);
}
```

3.28.17 L160HeadingHoldOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L160HeadingHoldOpMode.java

L160

A change

Before L160:

```
(nothing)
```

After L160:

```
import org.firstinspires.ftc.teamcode.base.HeadingHold;
```

A change

Before L160:

```
(nothing)
```

After L160:

```
import org.firstinspires.ftc.teamcode.pedro.Constants;
```

A change

Before L160:

```
(nothing)
```

After L160:

```
HeadingHold hold = new
HeadingHold(Constants.foresightConfig.headingFeedback.get());
```

A change

Before L160:

(nothing)

After L160:

```
turn = hold.turn(follower, turn);
Tracker.publish("heading/holding", hold.target() != null);
```

3.28.18 L170DriveToPoseOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L170DriveToPoseOpMode.java

L170*A change*

Before L170:

(nothing)

After L170:

```
import com.pedropathing.ivy.commands.Commands;
```

A change

Before L170:

(nothing)

After L170:

```
buttons.whenPressed(() -> gamepad1.y, Commands.instant(() -> {
    follower.hold(target);
    drivingItself = true;
}));
```

A change

Before L170:

(nothing)

After L170:

```
if (drivingItself) {
    if (!driverWantsControl) {
        Tracker.publish("drive/mode", "AUTO");
        drivetrain.releaseCommandedWheels();
        return;
    }
    drivingItself = false;
    follower.manual(0, 0, 0);
}
Tracker.publish("drive/mode", "DRIVER");
```

3.28.19 L180CombinedOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/ on solutions-1010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L180CombinedOpMode.java

L180

A change

Before L180:

```
(nothing)
```

After L180:

```
import org.firstinspires.ftc.teamcode.base.odometry.HardwareWheelSource;
```

A change

Before L180:

```
(nothing)
```

After L180:

```
shadowLocalizers.add("driveWheelEncoders",  
    new L120EncoderLocalizer(new HardwareWheelSource(hardware)));
```

3.28.20 L190SpeedDriveTrain

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/
on solutions-1010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L190SpeedDriveTrain.java

L190

A change

Before L190:

```
(nothing)
```

After L190:

```
import org.firstinspires.ftc.teamcode.pedro.Constants;
```

A change

Before L190:

```
(nothing)
```

After L190:

```
double[] measured = measuredSpeeds.all();  
double[] powers = new double[4];  
for (int i = 0; i < 4; i++) {  
    double feedforward = Constants.powerPerInchPerSecond * wanted[i];  
    double feedback = VELOCITY_KP * (wanted[i] - measured[i]);  
    powers[i] = clampToPower(feedforward + feedback);  
}  
setCommandedWheels(powers[FL], powers[FR], powers[BL], powers[BR]);  
publishWheelSpeeds(wanted, measured, powers);
```

3.28.21 L190VelocityDriveOpMode

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/

on solutions-l010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L190VelocityDriveOpMode.java

L190

A change

Before L190:

```
(nothing)
```

After L190:

```
import org.firstinspires.ftc.teamcode.base.WheelTargets;
import org.firstinspires.ftc.teamcode.pedro.Constants;
```

A change

Before L190:

```
(nothing)
```

After L190:

```
L190SpeedDriveTrain drivetrain;
```

A change

Before L190:

```
(nothing)
```

After L190:

```
drivetrain = new L190SpeedDriveTrain(hardware);
```

A change

Before L190:

```
initAfter();
```

After L190:

```
initAfter(drivetrain);
```

A change

Before L190:

```
double forwardInPerS = 0;
double strafeInPerS = 0;
double turnRadPerS = 0;
```

After L190:

```
double forwardInPerS = deadband(-leftStickY, band) * MAX_IPS;
double strafeInPerS = deadband(-leftStickX, band) * MAX_IPS;
double turnRadPerS = deadband(-rightStickX, band) * MAX_TURN_RADPS;
```

A change

Before L190:

```
(nothing)
```

After L190:

```
double[] target = WheelTargets.forMecanum(forwardInPerS, strafeInPerS,
```

```
turnRadPerS, Constants.turnRadiusInches);  
    drivetrain.setCommandedWheelSpeeds(target[0], target[1], target[2],  
target[3]);
```

3.28.22 L195TicksDriveTrain

What each lesson's patch does to this file, in the order the lessons come, from applying solutions/
on solutions-1010.1:

TeamCode/src/main/java/org/firstinspires/ftc/teamcode/mytry
/L195TicksDriveTrain.java

L195

A change

Before L195:

```
(nothing)
```

After L195:

```
ticks[FL] = hardware.frontLeft.getCurrentPosition();  
ticks[FR] = hardware.frontRight.getCurrentPosition();  
ticks[BL] = hardware.backLeft.getCurrentPosition();  
ticks[BR] = hardware.backRight.getCurrentPosition();
```